# Notes

Qiutong Wang




*The bad news is that we have still a long way to go.*
*The good news is that we have still a long way to go.*

— Alain Fournier, 1998

# Contents


1. Tori, Pinnings, and Coherent Continuation 3
2. Regular Parameters and Outer Automorphisms 7
3. Young Diagrams and Discrete Series 9
4. Continuous complex characters 10
5. Associated Varieties of Casselman-Wallach Representations 10
6. Split reductive algebraic groups over arbitrary field 11
7. Geometric Pre-quantization 11
8. Tianyuan Problems [1] 13
9. Orbit Method and Wavefront Sets 14
10. Conormal Bundles and Characteristic Cycles 16
11. Springer Representations and Characteristic Cycles 19
12. Holonomic Systems and Characteristic Cycles 19
13. Double Covers and Genuine Representations 20
14. Integrable connections and local systems 21
15. Kashiwara's equivalence and its applications 22
16. Nilpotent orbits and degenerate Whittaker models 23
17. Spectral Sequences of Filtered Complexes 24
18. Motivation for Quotient Stacks 25
19. Unipotent Arthur Packets [ABV92] 26
20. Questions on the Unitary Dual 29
21. Annihilator Varieties via Cells 30
22. Annihilator and Whittaker Support Problems 31
23. Robinson-Schensted and Annihilator Varieties 31
24. Linear Double Covers and Genuine Representations 32
25. Geometry of the flag variety of $\mathrm{SL}_2(\mathbb{C})$ 32
26. Perverse sheaves on $\mathbb{C}$ 33
27. Enhanced flag variety and line bundles on $\mathcal{B}$ 34
28. Tangent Maps for Associated Bundles 35
29. Symplectic Geometry and Formal Quantization 36
30. Normality of Nilpotent Orbit Closures 40
31. Filtered Quantization of the Nilpotent Cone 40
32. Divisors and Picard Groups 42
33. Symplectic Leaves of Nilpotent Orbits 42
34. Richardson Orbits and Symplectic Resolutions 43
35. Equivariant Structures on the Structure Sheaf 44
36. Principal Nilpotent Elements 45
37. Conical Symplectic Singularities 46
38. Zariski's Main Theorem 49
39. Twisted Differential Operators 49
40. Quantization of Nonaffine Varieties 50
41. Harish-Chandra Bimodules 51
42. Principal Connections and Picard Groups 52
43. Formal quantization and Star product 53


[1] This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre.

44. Filtered Quantization and Graded Quantization 54
45. Quantum Slices Theorem 55
46. Symplectic Lie algebra and Weyl algebra 56
47. Koszul Duality Pattern in Category $\mathcal{O}$ 56
48. Examples of resolution of the nilpotent orbit closure 58
49. Oscillator Representations as Quantization 60
50. Structure of Cell Representations 61
51. Atlas Parameter 66
52. Calculation of Lusztig-Vogan Modules 67
53. Hamiltonian Reduction of Differential Operators 72
54. Orbits on Partial Flag Variety 73
55. Geometric Explanation of $\tau$-Invariants 74
56. Ample cones and Parabolic Subgroups 81
57. Computation of Birational Induction 82
58. Strict GK-Dimension Drop for Harish–Chandra Cells 90
References 102

# 1. Tori, Pinnings, and Coherent Continuation

2023-12-22

1.1. **Identification of co-character groups with Lie algebras.** Settings: $G$ complex connected reductive algebraic group. $Q \subseteq P \subseteq \mathfrak{h}^*$ are root lattice and weight group, respectively. $\check{Q} \subseteq \check{P} \subseteq \mathfrak{h}$ are coroot lattice and coweight group, respectively. $H$ is a maximal torus of $G$, $\mathfrak{h}$ is its Lie algebra.

There are identifications

$$\begin{aligned} I: \quad X_*(H) \otimes \mathbb{C} &\longrightarrow \mathfrak{h} \\ \phi &\longmapsto \mathrm{d}\phi\,(1) \\ J: \quad X^*(H) \otimes \mathbb{C} &\longrightarrow \mathfrak{h}^* \\ \varphi &\longmapsto \mathrm{d}\varphi \end{aligned}$$

Under these identifications we have

$$\begin{aligned} P_* &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \left\langle \alpha, \check{\lambda} \right\rangle \in \mathbb{Z}, \text{for all } \alpha \in Q \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \exp(2\pi\sqrt{-1}\left\langle \alpha, \check{\lambda} \right\rangle) = 1, \text{for all } \alpha \in Q \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \alpha(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{for all } \alpha \in Q \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \exp(2\pi\sqrt{-1}\check{\lambda}) \in Z(G) \right\}. \end{aligned}$$

And

$$\begin{aligned} X_*(H) &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \left\langle \phi, \check{\lambda} \right\rangle \in \mathbb{Z}, \text{for all } \phi \in X^*(H) \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \exp(2\pi\sqrt{-1}\left\langle \phi, \check{\lambda} \right\rangle) = 1, \text{for all } \phi \in X^*(H) \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \phi(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{for all } \phi \in X^*(H) \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \,\middle|\, \exp(2\pi\sqrt{-1}\check{\lambda}) = 1 \right\}. \end{aligned}$$

Similarly

$$P^* = \left\{ \lambda \in \mathfrak{h}^* \,\middle|\, \exp(2\pi\sqrt{-1}\lambda) \in Z(\check{G}) \right\}.$$

$$X^*(H) = \left\{ \lambda \in \mathfrak{h}^* \,\middle|\, \exp(2\pi\sqrt{-1}\lambda) = 1 \right\}.$$

Define $e : \mathfrak{h} \to H$ by $e(X) = \exp(2\pi\sqrt{-1}X)$, then $\ker(e) = X_*(H)$.

$$\begin{array}{ccc} \mathfrak{h} & \xrightarrow{e} & H \\ \uparrow & & \uparrow \\ P_* & \xrightarrow{e} & Z(G) \end{array}$$

$Z(G) \simeq P_*/X_*(H)$.

## 1.2. **Pinnings of algebraic groups.**

**Theorem 1.1** (see [AC09])**.** *Inner automorphism group* $\mathrm{Inn}(G)$ *of* $G$ *is equal to* $\mathrm{G/Z(G)}$, $\mathrm{Inn}(G)$ *acts freely and transitively on the set of pinnings.*

I know that there is an analogy in compact connected groups, and it has been proved.

**Theorem 1.2.** *If* G *is a compact connected group, then its inner automorphism group* $\mathrm{Inn}(G) = \mathrm{G/Z(G)}$ *acts freely and transitively on the set of pinnings. (where pinnings for it is a bit different from the complex case, that* $(T, B_{\mathbb{C}}, \mathcal{X})$ *is a pinning if* $T$ *is a maximal torus,* $B_{\mathbb{C}}$ *a Borel of* $G_{\mathbb{C}}$ *containing* $T$*,* $\mathcal{X}$ *is a set of real rays in simple root space.)*

*Proof.* Here's proof from mathoverflow LSpice's answer:

To prove this, I'll use a few pieces of structure theory:

1. All maximal tori in $G$ are $G^\circ$-conjugate.
2. All Borel subgroups of $G_{\mathbb{C}}$ are $G_{\mathbb{C}}^\circ$-conjugate.
3. For every maximal torus $T$ in $G$, the map $\mathbf{W}(G^\circ, T) \to \mathbf{W}(G_{\mathbb{C}}^\circ, T_{\mathbb{C}})$ is an isomorphism.
4. If $G_{\mathbf{sc}}$ and $(G_{\mathbb{C}})_{\mathbf{sc}}$ are the simply connected covers of the derived groups of $G^\circ$ and $G_{\mathbb{C}}^\circ$, then $(G_{\mathbf{sc}})_{\mathbb{C}}$ equals $(G_{\mathbb{C}})_{\mathbf{sc}}$.
5. Every compact Lie group has a finite subgroup that meets every component.

I only need (4) to prove that, for every maximal torus $T$ in $G$, the map from $T$ to the set of conjugation-fixed elements of $T/\mathbf{Z}(G^\circ)$ is surjective. This is probably a well known fact in its own right for real-group theorists.

Now consider triples $(T, B_{\mathbb{C}}, \mathcal{X})$ as follows: $T$ is a maximal torus in $G$; $B_{\mathbb{C}}$ is a Borel subgroup of $G_{\mathbb{C}}^\circ$ containing $T_{\mathbb{C}}$, with a resulting set of simple roots $\Delta(B_{\mathbb{C}}, T_{\mathbb{C}})$; and $\mathcal{X}$ is a set consisting of a real ray in each complex simple root space (i.e., the set of positive real multiples of some fixed non-$\mathbf{0}$ vector). (Sorry about the pair of modifiers "complex simple".) I will call these 'pinnings', although it doesn't agree with the usual terminology (where we pick individual root vectors, not rays). I claim that $G^\circ/\mathbf{Z}(G^\circ)$ acts simply transitively on the set of pinnings.

Once we have transitivity, freeness is clear: if $g \in G^\circ$ stabilises some pair $(T, B_{\mathbb{C}})$, then it lies in $T$, and so stabilises every complex root space; but then, for it to stabilise some choice of rays $\mathcal{X}$, it has to have the property that $\alpha(g)$ is positive and real for each simple root $\alpha$; but also $\alpha(g)$ is a norm-$1$ complex number, hence trivial, for each simple root $\alpha$, hence for each root $\alpha$, so that $g$ is central.

For transitivity, since (1) all maximal tori in $G$ are $G^\circ$-conjugate, so (2) for every maximal torus $T$ in $G$, the Weyl group $\mathbf{W}(G^\circ_{\mathbb{C}}, T_{\mathbb{C}})$ acts transitively on the Borel subgroups of $G^\circ_{\mathbb{C}}$ containing $T_{\mathbb{C}}$, and (3) $\mathbf{W}(G^\circ, T) \to \mathbf{W}(G^\circ_{\mathbb{C}}, T_{\mathbb{C}})$ is an isomorphism, it suffices to show that all possible sets $\mathcal{X}$ are conjugate. Here's the argument that I came up with to show that they are even $T$-conjugate; I think it can probably be made much less awkward. Fix a simple root $\alpha$, and two non-$0$ elements $X_\alpha$ and $X'_\alpha$ of the corresponding root space. Then there are a positive real number $r$ and a norm-$1$ complex number $z$ such that $X'_\alpha = rzX_\alpha$. Choose a norm-$1$ complex number $w$ such that $w^2 = z$. There is then a unique element $s_{\mathrm{ad}}$ of $T_{\mathbb{C}}/\mathbf{Z}(G^\circ_{\mathbb{C}})$ such that $\alpha(s_{\mathrm{ad}}) = w$, and $\beta(s_{\mathrm{ad}}) = 1$ for all simple roots $\beta \neq \alpha$. By (4), we can choose a lift $s_{\mathrm{sc}}$ of $s_{\mathrm{ad}}$ to $(G_{\mathrm{sc}})_{\mathbb{C}} = (G_{\mathbb{C}})_{\mathrm{sc}}$, which necessarily lies in the preimage $(T_{\mathbb{C}})_{\mathrm{sc}}$ of (the intersection with the derived subgroup of) $T$, and put $t_{\mathrm{sc}} = s_{\mathrm{sc}} \cdot \overline{s_{\mathrm{sc}}}$. Then

$$\alpha(t_{\mathrm{sc}}) = \alpha(s_{\mathrm{sc}})\overline{\alpha(s_{\mathrm{sc}})} = \alpha(s_{\mathrm{sc}})\overline{\alpha(s_{\mathrm{sc}})^{-1}} = w \cdot \overline{w^{-1}} = z,$$

and, similarly, $\beta(t_{\mathrm{sc}}) = 1$ for all simple roots $\beta \neq \alpha$. Now the image $t$ of $t_{\mathrm{sc}}$ in $G^\circ_{\mathbb{C}}$ lies in $T_{\mathbb{C}}$ and is fixed by conjugation, hence lies in $T$; and $\mathbf{Ad}(t)X_\alpha = zX_\alpha$ lies on the ray through $X'_\alpha$.

Since $G$ also acts on the set of pinnings, we have a well defined map $p : G \to G^\circ/\mathbf{Z}(G^\circ)$ that restricts to the natural projection on $G^\circ$. Now $\ker(p)$ meets every component, but it contains $\mathbf{Z}(G^\circ)$, so it need not be finite. Applying (5) to the Lie group $\ker(p)$ yields the desired subgroup $H$. Note that, as requested in your improved classification, conjugation by any element of $H$ fixes a pinning, hence, if inner, must be trivial.

□

**Remark 1.3.** The pinnings show that split reductive algebraic groups look like butterflies!

According to Milne, Grothendieck used the following analogy to talk about the structure theory of algebraic groups. An algebraic group is like a butterfly. The body of the butterfly is the maximal torus. The wings are two opposite Borel subgroups. And the pins are the pinning maps.



FIGURE 1. Pinning of butterfly

1.3. **Coherent continuation representations.** Until the end of today, $G$ denotes a real reductive group in Harish-Chandra class.

Let's recall some basic definitions of coherent continuation representations, in order for me to do calculations in the case of real classical groups.

Fix a connected reductive complex Lie group $G_{\mathbb{C}}$, together with a Lie group homomorphism $\iota : G \to G_{\mathbb{C}}$ such that its differential $\mathrm{d}\iota : \mathrm{Lie}(G) \to \mathrm{Lie}(G_{\mathbb{C}})$ has the following two properties:

(1) the kernel of $\mathrm{d}\iota$ is contained in the center of $\mathrm{Lie}(G)$;
(2) the image of $\mathrm{d}\iota$ is a real form of $\mathrm{Lie}(G)$.

The analytical weight lattice of $\mathrm{G}_{\mathbb{C}}$ is identified with a subgroup of ${}^a\mathfrak{h}^*$ via $\mathrm{d}\iota$. We write $Q_\iota \subset {}^a\mathfrak{h}^*$ for this subgroup. Denote root lattice by $Q_{\mathfrak{g}}$, weight lattice by $Q^{\mathfrak{g}}$, then

$$Q_{\mathfrak{g}} \subset Q_\iota \subset Q^{\mathfrak{g}}.$$

They are all $W$ stable subgroups. And we have some categories:

(1) $\mathrm{Rep}(\mathfrak{g}, Q_\iota)$ the category of all finite-dimensional representations of $\mathfrak{g}$ whose weight are contained in $Q_{\mathfrak{g}}$, it can be identified with $\mathrm{Rep}(Lie(G_{\mathbb{C}}), Q_\iota)$ and hence $\mathrm{Rep}_{finite}(G_{\mathbb{C}})$;
(2) $\mathrm{Rep}(G)$ the category of all Casselman-Wallach representations of $G$, and its full subcategories $\mathrm{Rep}_\mu(G)$, $\mathrm{Rep}_S(G)$.

To denote their Grothendieck groups, we replace Rep by $\mathcal{K}$.

Fix a $Q_\iota$-coset $\Lambda = \lambda + Q_\iota \subset {}^a\mathfrak{h}^*$.

**Definition 1.4** (coherent family)**.** A $\mathcal{K}(G)$-valued $\Lambda$ coherent family is a map

$$\Phi : \Lambda \to \mathcal{K}(G)$$

satisfies the following two conditions:

(1) for all $\nu \in \Lambda$, $\Phi(\nu) \in \mathcal{K}_\nu(G)$;

(2) for all representations $F \in \mathrm{Rep}(\mathfrak{g}, Q_\iota)$ and all $\nu \in \Lambda$,

$$F \cdot (\Phi(\nu)) = \sum_{\mu} \Phi(\nu + \mu),$$

where $\mu$ runs over all weights of F, counted with multiplicities.

The set of coherent families is denoted by $\mathrm{Coh}_\Lambda(\mathcal{K}(G))$

We have the integral Weyl group $W_\Lambda = \{w \in W \mid w\lambda - \lambda \in Q_\iota\}$ there is a representation of $W_\Lambda$ on $\mathrm{Coh}_\Lambda(\mathcal{K}(G))$ by

$$(w \cdot \Phi)(\nu) = \Phi(w^{-1}\nu).$$

This is called a coherent continuation representation.

## 2. Regular Parameters and Outer Automorphisms

2023-12-23

2.1. **Review of regular parameter.** Following yesterday's discussion of coherent continuation representations, we define two basis for it, use Langlands classification, follow [BMSZ22].

Suppose $H$ is a Cartan subgroup of $G$, by which we mean a subgroup that is the centralizer of a Cartan subalgebra of $\mathrm{Lie}(G)$ (In the case of linear groups, it's just a real algebraic torus.). $H$ has a unique maximal compact subgroup (since $G$ is in Harish-Chandra's class, adjoint representation of $H$ is trivial, $H_0$ lies in the center of $H$, as we know, all maximal subgroups of Lie groups are conjugate to each other, use Cartan decomposition of $H$, we win. In fact, $H$ has a unique maximal compact torus and a unique maximal split torus) We denote by $\Delta_\mathfrak{h} \subset \mathfrak{h}^*$ the root system of $\mathfrak{g}$. A root is called imaginary if $\check{\alpha} \in \mathfrak{t}$ (i.e. roots which take pure imaginary values on $\mathrm{Lie}(H)$), an imaginary root $\alpha$ is called compact if $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ is contained in the complexified Lie algebra of a common compact subgroup of $G$.(this is equivalent to $\mathfrak{g}_\alpha$ belong to a Lie subalgebra of a compact subgroup, since $\mathfrak{g}_\alpha$ lies in a Lie subalgebra of maximal compact subgroup is equivalent to $\mathfrak{g}_{-\alpha}$ Lies in the same subgroup)

There are two important facts about the representation theory of $H$:

(1) Every Casselman-Wallach representation of $H$ is finite-dimensional, it follows directly from that $H$ is an extension of an abelian group and a finite group;
(2) For every $\Gamma \in \mathrm{Irr}(H)$ differential of $\Gamma$ is a direct sum of one-dimensional representations attached to a unique $\mathrm{d}\Gamma \in \mathfrak{h}^*$. (it is true because $H$ is in Harish-Chandra's class).

For every Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ containing $\mathfrak{h}$, write

$$\xi_\mathfrak{b} : \mathfrak{h} \to {}^a\mathfrak{h},$$

for the linear isomorphism attached to $\mathfrak{b}$ defined by

$$\mathfrak{h} \hookrightarrow \mathfrak{b} \to \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}] = {}^a\mathfrak{h}.$$

The transpose inverse of this map is still denoted by $\xi_\mathfrak{b} : {}^a\mathfrak{h}^* \to \mathfrak{h}^*$.

Write

$$W({}^a\mathfrak{h}^*, \mathfrak{h}^*) := \{\xi_\mathfrak{b} : {}^a\mathfrak{h}^* \to \mathfrak{h}^* \mid \mathfrak{b} \text{ is a Borel subalgebra containing } \mathfrak{h}\}.$$

**Definition 2.1.** For every element $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$, put

$$\delta(\xi) := \frac{1}{2} \cdot \sum_{\alpha \text{ is an imaginary root in } \xi\Delta^+} \alpha - \sum_{\beta \text{ is an compact imaginary root in } \xi\Delta^+} \beta \in \mathfrak{h}^*.$$

Write $\mathcal{P}_\Lambda(G)$ for the set of all triples $\gamma = (H, \xi, \Gamma)$, where $H$ is a Cartan subgroup of $G$, $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$, and

$$\begin{array}{rccc} \Gamma\colon & \Lambda & \longrightarrow & \mathrm{Irr}(H) \\ & \nu & \longmapsto & \Gamma_\nu \end{array}$$

is a map with the following properties:

(1) $\Gamma_{\nu+\beta} = \Gamma_\nu \otimes \xi(\beta)$ for all $\beta \in Q_\iota$ and $\nu \in \Lambda$;
(2) $\mathrm{d}\Gamma_\nu = \xi(\nu) + \delta(\xi)$ for all $\nu \in \Lambda$.(note that Cartan subgroups not always have irr repn with this differential!)

Here $\xi(\beta)$ is naturally viewed as a character of $H$ by using the homomorphism $\iota : H \to H_{\mathbb{C}}$, and $H_{\mathbb{C}}$ is a Cartan subgroup of $G_{\mathbb{C}}$ containing $\iota(H)$.

## 2.2. **Outer automorphism of algebraic groups.**

**Lemma 2.2** (a simple discovery of abstract algebra)**.** *$G$ an abstract group, $H$ a normal subgroup of $G$, if $G$ acts transitively on a set $X$ such that the restriction of this action to $H$ is free and transitive, then the short exact sequence of groups*

$$1 \to H \to G \to G/H \to 1$$

*Proof.* Actually choose an element $x \in X$, we have

$$G \simeq \mathrm{Stab}_G(x) \ltimes H.$$

□

Using this lemma, we can see that for a complex connected reductive algebraic group $G$, $\mathrm{Aut}(G)$ acts on the set of pinnings transitively, with its normal subgroup the $\mathrm{Inn}(G)$ acts freely and transitively, so every pinning gives us a split of the short exact sequence

$$1 \to \mathrm{Inn}(G) \to \mathrm{Aut}(G) \to \mathrm{Out}(G) \to 1.$$

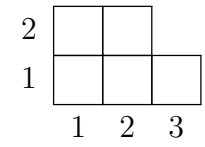
Some facts about split connected reductive algebraic groups over arbitrary field $k$:

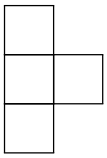
(1) $\{\text{split connected reductive algebraic groups}\}/_\backsimeq = \{\text{root data}\}/_\backsimeq$;
(2) For any two automorphisms of $G$, if they induce the same automorphism of root datum, then they differ by an inner automorphism;
(3) Fix a pinning, we get an isomorphism $\mathrm{Out}(G) \simeq \mathrm{Aut}(D_b)$.

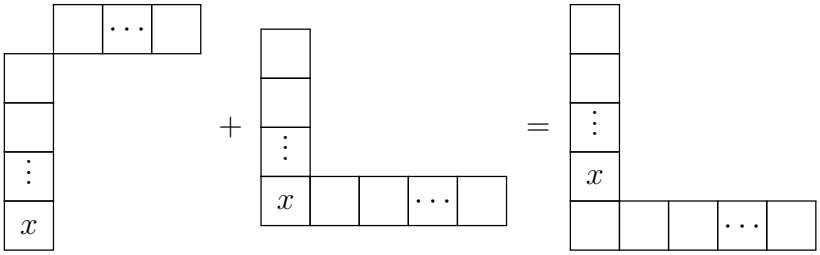
# 3. Young Diagrams and Discrete Series

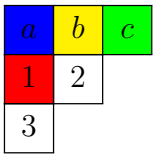
2024-02-27

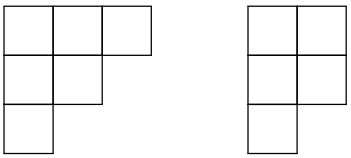
Some code examples of Young diagrams and tableaux using the `ytableau` package:

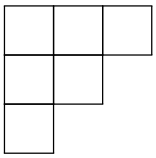


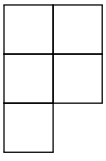



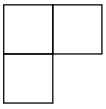


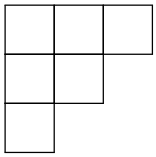



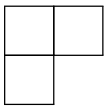




ce est chinois  english

$$a = \begin{array}{|c|c|}\hline \phantom{x} & \phantom{x} \\ \hline \phantom{x} \\ \cline{1-1} \end{array} + b$$

youngdiagram1: 

young2: 

For a compact connected Lie group $G$, choose a maximal torus $T$, there is a canonical 1-1 correspondence:

$$X^*(T)/W \longleftrightarrow (X^*(T)+\rho)/W \longleftrightarrow \mathrm{Irr}(G)$$

Moreover, if we choose a positive system $\Psi^+$, and then $\rho_{\Psi^+}$, the above sets also canonically isomorphic to the set of dominant analytically integral weights:

$$\{\lambda \in X^*(T) \,|\, \langle \lambda, \alpha \rangle \geq 0, \text{for all } \alpha \in \Psi^+\}$$

This can be generalized to parameter of discrete series representations for connected semisimple Lie groups $G$ with compact Cartan subgroup $T$ :

$$(X^*(T)+\rho)^{reg}/W_c$$

If a discrete series representation $V$ has minimal $K$ type which has maximal weight $\nu$, then it's parameter is $\lambda = \nu - \rho + 2\rho_c$.(not sure)

Note that for a fixed infinitesimal character, there are exactly $|W/W_c|$ different discrete series representations.

## 4. Continuous complex characters

2024-02-28

Some facts about continuous complex characters of some common abelian topological groups:

| field | character group | correspondence |
|---|---|---|
| $(\mathbb{R}_+^\times, \times)$ | $\mathbb{C}$ | $\xi \mapsto (t \mapsto t^\xi)$ |
| $(\mathbb{C}^\times, \times)$ | $\mathbb{C} \times \mathbb{Z}$ | $(\xi, n) \mapsto (z \mapsto \lvert z\rvert^\xi (\frac{z}{\lvert z\rvert})^n)$ |
| $(\mathbb{S}^1, \times)$ | $\mathbb{Z}$ | $n \mapsto (z \mapsto z^n)$ |
| $(\mathbb{Q}_p, +)$ | $\mathbb{Q}_p$ | $\xi \mapsto (x \mapsto \exp(2\pi i \xi \{x\}_p))$ |

**Remark 4.1.** Note that all continuous characters of $\mathbb{Q}_p$ are automatically locally constant (smooth), which is consistent with the case of real lie groups.

## 5. Associated Varieties of Casselman-Wallach Representations

2024-10-22

**Lemma 5.1.** *The associated variety of an irreducible Casselman-Wallach representation of a real reductive group in Harish-Chandra class is the closure of a nilpotent orbit in $\mathfrak{g}^*$.*

*Proof.* By the work of Borho, Brylinski [BB82], and Joseph [Jos85], the associated variety of any primitive ideal (annihilator of an irreducible $\mathfrak{g}$-module) is the closure of a single nilpotent orbit. So we only need to prove that $\mathrm{gr}(\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V))$ is a primitive ideal.

Since the $K$-finite part $V_K$ is dense in $V$, $\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V) = \mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V_K)$, we only need to consider the annihilator of the irreducible $(\mathfrak{g}, K)$-module $V_K$.

If $G$ is connected, then $K$ is also connected, so $V_K$ is irreducible as $\mathcal{U}(\mathfrak{g})$-module, in this case, $\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V)$ is a primitive ideal.

If $G$ is not connected, let $G_0$ be the identity component of $G$, $K_0 = G_0 \cap K$ is a maximal compact subgroup of $G_0$, then $V_K|_{G_0}$ is a finite length $(\mathfrak{g}, K_0)$-module, it has an irreducible quotient $V_K \to V_0$, then according to the theory of induction of $(\mathfrak{g}, K)$-modules [KV95, Chapter 2] we have an injective morphism of $(\mathfrak{g}, K_0)$-modules:

$$V_K|_{K_0} \hookrightarrow (\mathrm{induced}_{K_0}^{K}(V_0))|_{K_0} \cong \bigoplus_{\text{double cosets } K_0 k K_0} kV_0, \tag{5.1}$$

where $kV_0$ be the $(\mathfrak{g}, K_0)$-module whose underlying space is $V_0$, whose action by $X \in \mathfrak{g}$ is the usual action by $\mathrm{Ad}(k)X$, whose action by $g \in K_0$ is the usual action by $k^{-1}gk$, it has the same annihilator in $\mathcal{U}(\mathfrak{g})$ as $V_0$ since $G$ is in Harish-Chandra's class, and hence the annihilator of $V_K$ is the same as the annihilator of $V_0$, which is a primitive ideal.

□

## 6. Split reductive algebraic groups over arbitrary field

2024-11-28

Any split connected reductive group over arbitrary field can be uniquely defined and split over $\mathbb{Z}$. (Chevalley group)

Classification of split reductive group is the same over arbitrary non-empty schemes, which is equivalent to the classification of root data.(cf. [SGA3])

## 7. Geometric Pre-quantization

2024-12-17

**Geometric Pre-quantization**

Let $X$ be a smooth manifold. There are natural maps

$$\mathrm{Pic}(X) \longrightarrow \mathrm{H}^2_{\mathrm{dR}}(X;\mathbb{R}) \simeq \mathrm{H}^2_{\check{\mathrm{C}}\mathrm{ech}}(X;\mathbb{R}).$$

Construction of the first map: For a (complex) line bundle $\mathbb{L} \in \mathrm{Pic}^h(X)$, let $\langle , \rangle$ be a Hermitian metric on it and $\nabla$ be a unitary connection. (Every line bundle $\mathbb{L} \in \mathrm{Pic}(X)$ has a Hermitian metric and a unitary connection on it.)

- choose a locally trivialization $\{U_\alpha\}$ of $\mathbb{L}$, such that all $U_\alpha$ and $U_\alpha \cap U_\beta$ are contractible;
- choose Unitary local frames $\{s_\alpha\}$ for $\mathbb{L}$ on each $U_\alpha$, then $s \leftrightarrow (f_\alpha)$ and $\nabla \leftrightarrow (\theta_\alpha)$ (unitary implies that $\theta_\alpha$ are pure imaginary);
- define $\Omega := \mathrm{d}\theta_\alpha$, and $c_1(\mathbb{L}) := [\frac{1}{2\pi i}\Omega] \in \mathrm{H}^2_{dR}(X;\mathbb{R})$ ($\Omega = \mathrm{curv}\nabla$).

$c_1(\mathbb{L})$ is independent of the choice of $\langle , \rangle$ and $\nabla$ by Chern-Weil theory. And we can see that $\theta_\alpha - \theta_\beta = \frac{\mathrm{d}g_{\alpha,\beta}}{g_{\alpha,\beta}} = \mathrm{dlog}(g_{\alpha,\beta})$ on $U_\alpha \cap U_\beta$, where $g_{\alpha,\beta}$ is the translation function ($s_\beta = g_{\alpha,\beta}s_\alpha$).

Construction of the second map (classical de-Rham isomorphism), choose a 2-form $\omega \in \mathrm{H}^2_{dR}(X;\mathbb{R})$

- choose a cover $\{U_\alpha\}$ of $\mathbb{L}$, such that all $U_\alpha$ and $U_\alpha \cap U_\beta$ are contractible;
- on each $U_\alpha$, choose a 1-form $\theta_\alpha$ such that $\mathrm{d}\theta_\alpha = \omega|_{U_\alpha}$;
- since $\mathrm{d}\theta_\alpha = \mathrm{d}\theta_\beta$ on $U_\alpha \cap U_\beta$, we can choose smooth function $c_{\alpha,\beta}$ such that $\mathrm{d}c_{\alpha,\beta} = \theta_\alpha - \theta_\beta$;

- let $c_{\alpha,\beta,\gamma} = c_{\alpha,\beta} + c_{\beta,\gamma} + c_{\gamma,\alpha}$, which is a constant, and we get the corresponding 2-cocycle $(c_{\alpha,\beta,\gamma}) \in \mathrm{H}^2_{\check{\mathrm{C}}\mathrm{ech}}(X;\mathbb{R})$.

If $\omega$ come from the first map, we can choose $c_{\alpha,\beta} = \frac{1}{2\pi i}\log g_{\alpha,\beta}$, and hence $c_{\alpha,\beta,\gamma} = \frac{1}{2\pi i}(\log g_{\alpha,\beta} + \log g_{\beta,\gamma} + \log g_{\gamma,\alpha}) \in \mathbb{Z}$. So the morphism is in fact

$$\mathrm{Pic}(X) \longrightarrow \mathrm{H}^2_{dR}(X;\mathbb{Z}) \simeq \mathrm{H}^2_{\check{\mathrm{C}}\mathrm{ech}}(X;\mathbb{Z}).$$

In fact the first map is also an isomorphism. (cf. prequantization) For a 2-form $\omega$, choose $\theta_\alpha$ and $c_{\alpha,\beta}$ as above, let $g_{\alpha,\beta} = \exp(2\pi i c_{\alpha,\beta})$, these are transition functions of a complex line bundle $\mathbb{L}$.

**Corollary 7.1.** *Assume that $G$ is a connected Lie group. The following are equivalent:*

- $\Omega \subseteq \mathfrak{g}^*$ *is integral* ($[\sigma_\Omega] \in \mathrm{H}^2(X;\mathbb{Z})$)*;*
- *there exists a $G$-equivariant complex line bundle over $\Omega$ with a $G$-invariant Hermitian connection $\nabla$ such that*

$$\frac{1}{2\pi i}\mathrm{curv}\,(\nabla) = \sigma_\Omega.$$

**Theorem 2.6** (Weil)**.** *Let $M$ be a smooth manifold and $\omega$ a real, closed 2-from whose cohomology class $[c]$ is integral. Then there is a unique Hermitian line bundle $\mathbb{L}$ over $M$ with unitary connection $\nabla$ so that $c_1(\mathbb{L}) = [c]$.*

*Sketch of Proof.* Existence: Reverse the arguments above. Since $[c]$ is integral, $e^{2\pi c_{ijk}} = 1$. As a consequence, $g_{ij}g_{jk}g_{kl} = 1$. So $g_{ij}$'s are the transition function for some line bundle whose first Chern class is $[c]$.

Uniqueness: Suppose $c_1(\mathbb{L}) = c_1(\widetilde{\mathbb{L}})$. Let $h_{ij} = \frac{1}{2\pi i}g_{ij}$ and define $\tilde{h}_{ij}$ similarly. Then the functions $\hat{h}_{ij} = h_{ij} - \tilde{h}_{ij}$ satisfies the relation

$$\hat{h}_{ij} + \hat{h}_{jk} + \hat{h}_{ki} = 0.$$

We take a partition of unity $\rho_k$ and let $\lambda_i = e^{2\pi i \sum \hat{h}_{ki}\rho_k}$. Then

$$\lambda_i g_{ij} \lambda_j^{-1} = e^{2\pi i(h_{ij} + \sum(\hat{h}_{ki} - \hat{h}_{kj})\rho_k)} = e^{2\pi i(h_ij - \hat{h}_{ij})} = \tilde{g}_{ij}.$$

So as line bundles $\mathbb{L} \simeq \widetilde{\mathbb{L}}$. □

FIGURE 2. Proof

## 8. Tianyuan Problems [2]

2025-08-14

Let $G$ be a finite central extension of an open subgroup of $\mathbb{R}$-points of a connected reductive algebraic group (reductive Nash group [Sun15]) with Lie algebra $\mathfrak{g}$. Fix an invariant nondegenerate bilinear form $B(-,-)$ on $\mathfrak{g}$.

For an irreducible Casselman-Wallach representation $\pi$ of $G$, let $\mathrm{Wh}_{\mathcal{O}}(\pi)$ be the space of generalized Whittaker models of $\pi$ associated to the nilpotent orbit $\mathcal{O}$. Let

$$\mathrm{WO}(\pi) := \{\mathcal{O} \,|\, \mathcal{O} \text{ is a nilpotent orbit such that } \mathrm{Wh}_{\mathcal{O}}(\pi) \neq 0\},$$

and the *Whittaker support* $\mathrm{WS}(\pi)$ of $\pi$ be the set of all maximal orbits in $\mathrm{WO}(\pi)$ with respect to closure inclusion:

$$\mathrm{WS}(\pi) := \{\mathcal{O} \,|\, \mathcal{O} \text{ is a maximal orbit in } \mathrm{WO}(\pi)\}.$$

Another nilpotent invariant is the wavefront cycle:

$$\mathcal{WF}(\pi) = \sum_i m_i[\mathcal{O}_i].$$

With the corresponding wavefront set :

$$\mathrm{WF}(\pi) = \bigcup_{m_i \neq 0} \overline{\mathcal{O}_i}.$$

Mœglin-Waldspurger [MW87] proved that $\mathrm{WF}(\pi) = \mathrm{WS}(\pi)$ when $F$ is a $p$-adic field. We want to test this conjecture for arbitrary reductive almost linear Nash groups.

**Conjecture 8.1** (Main conjecture)**.** *For any reductive almost linear Nash group $G$ and any irreducible Casselman-Wallach representation $\pi$:*

$$\mathrm{WF}(\pi) = \mathrm{WS}(\pi).$$

Take a nilpotent orbit $\mathcal{O} \subseteq \mathfrak{g}$ (or $\mathfrak{sl}_2$-triple $\gamma = \{Y, H, X\}$ in $\mathfrak{g}$). $\mathfrak{g}/\mathfrak{g}^Y$ is a symplectic space with symplectic form $\langle -,- \rangle := B(Y, [-,-])$. Set $G^Y := \mathrm{Stab}_G(Y)$, there is a natural homomorphism $G^Y \to \mathrm{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -,- \rangle)$. Let $\widetilde{G^Y}$ be the double cover of $G^Y$ induced by the metaplectic double cover of $\mathrm{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -,- \rangle)$.

**Definition 8.2.** The nilpotent $\mathcal{O}$ is called admissible if the double cover $\widetilde{G^Y}$ is split over the identity component of $(G^Y)_0$. It is called quasi-admissible if $\widetilde{G^Y}$ has a finite-dimensional genuine representation.

Result of [GGS21] says that $\mathrm{WS}(\pi)$ consists of quasi-admissible orbits. But they do not know whether the non-admissible quasi-admissible orbits appear in Whittaker supports of representations. We may ask the same question for the wavefront set:

**Problem 1**: Are all the nilpotent orbits in the wavefront cycle admissible?

For the case when $G$ is general linear, it is well known that the wavefront set $\mathrm{WF}(\pi)$ of an irreducible Casselman-Wallach representation $\pi$ of $G$ consists of a single nilpotent adjoint orbit. Together with some counting results, it maybe possible to construct all irreducible representations of general linear group via theta correspondence and coherent continuation.

[2] This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre.

**Problem 2**: Can we construct all irreducible representations via theta correspondence, coherent continuation, and twist of characters?

Assuming the construction steps above are done, it then remains to study the Whittaker support $\mathrm{WS}(\pi)$ of $\pi$. We focus on two types of interactions with Whittaker models - theta correspondence and coherent continuation.

On one hand, Gomez-Zhu [GZ14] proved for type I reductive (symplectic-orthogonal) dual pairs in the stable range that the big theta lift preserves the structure of Whittaker models:

$$\mathrm{Wh}_{\Theta(\mathcal{O})}\left(\Theta(\pi)\right) = \mathrm{Wh}_{\mathcal{O}}(\pi^\vee).$$

Noting that $(G, G') = (\mathrm{GL}_n(F), \mathrm{GL}_m(F))$ is a (reductive) dual pair for any $n, m \in \mathbb{N}$, one observes that if one can prove an analogue of Gomez-Zhu's result for type II dual pairs, one can use it to infer properties of the structure of the Whittaker support $\mathrm{WS}(\pi)$. Hence we propose the following subconjecture:

**Conjecture 8.3.** *If $(G, G') = (\mathrm{GL}_n(F), \mathrm{GL}_m(F))$ is a type II reductive dual pair over $F$ in the stable range (with $n \leq m$), and $\pi$ is a genuine Casselman-Wallach representation of $G$, then*

$$\mathrm{Wh}_{\Theta(\mathcal{O})}\left(\Theta(\pi)\right) = \mathrm{Wh}_{\mathcal{O}}(\pi^\vee),$$

*where $\pi^\vee$ denotes the smooth contragredient of $\pi$. Furthermore, the above will imply that*

$$\mathrm{WS}\left(\Theta(\pi)\right) = \mathrm{WS}(\pi^\vee).$$

On the other hand, Barbash-Vogan [BV80] proved that under coherent continuation, the multiplicity of wavefront cycle changes in the form of a Goldie rank polynomial, i.e.

$$\mathcal{WF}(\pi(\lambda)) = p(\lambda) \cdot (\sum_i c_i[\mathcal{O}_i]),$$

where $p(\lambda)$ is the goldie rank polynomial, $\lambda$ is dominant. We expect some similar result for the multiplicity of the generalized Whittaker model.

**Problem 3**: For a fixed nilpotent orbit $\mathcal{O}$, does there exist a constant $c_{\mathcal{O}}$ such that $\dim \mathrm{Wh}_{\mathcal{O}}(\pi(\lambda)) = c_{\mathcal{O}} \cdot p(\lambda)$?

## 9. Orbit Method and Wavefront Sets

2025-10-01

Orbit method for nilpotent Lie groups.

Let $G$ be a connected, simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Let $\mathfrak{g}^*$ be the dual space of $\mathfrak{g}$.

We have Fourier transform:

$$\mathcal{F} : \mathcal{S}(\mathfrak{g})\mathrm{d}X \to \mathcal{S}(\sqrt{-1}\mathfrak{g}^*), \quad f\mathrm{d}X \mapsto \widehat{f\mathrm{d}X}, \quad \widehat{f\mathrm{d}X}(l) = \int_{\mathfrak{g}} f(X)e^{\langle l, X\rangle}\mathrm{d}X.$$

Take its dual:

$$\mathcal{S}'(\sqrt{-1}\mathfrak{g}^*) \to (\mathcal{S}(\mathfrak{g})\mathrm{d}X)', \quad T \mapsto \check{T}, \quad \langle \check{T}, f\mathrm{d}X\rangle = \langle T, \widehat{f\mathrm{d}X}\rangle.$$

For any irreducible unitary representation $\pi$ of $G$, we have its character $\Theta_\pi \in C^{-\infty}(G)$ defined by

$$\Theta_\pi(f) = \mathrm{Tr}\left(\pi\left(f\mathrm{d}g\right)\right), \quad f\mathrm{d}g \in C_c^\infty(G)\mathrm{d}g.$$

Using the exponential map $\exp : \mathfrak{g} \to G$ (isomorphism in this case), we can view $\Theta_\pi$ as a distribution on $\mathfrak{g}$:

$$\theta_\pi(f\mathrm{dX}) = \Theta_\pi(f \circ \exp), \quad f\mathrm{dX} \in C_c^\infty(\mathfrak{g})\mathrm{dX}.$$

In fact, $\theta_\pi$ is a tempered generalized function on $\mathfrak{g}$ (it can continuously extends to $\mathcal{S}(\mathfrak{g})\mathrm{dX}$). By the work of Howe, Kirillov, and others, we have the following theorem:

**Theorem 9.1.** *There exists a unique coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$ such that $\check{\theta}_\pi = \mu_\mathcal{O}$, where $\mu_\mathcal{O}$ is the canonical measure on $\mathcal{O}$.*

This defines a bijection between $\widehat{G}$ (the unitary dual of $G$) and the set of coadjoint orbits in $\sqrt{-1}\mathfrak{g}^*$.

Conversely, for any coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$, we can construct an irreducible unitary representation $\pi_\mathcal{O}$ of $G$ such that $\check{\theta}_{\pi_\mathcal{O}} = \mu_\mathcal{O}$. This is the more interesting part, use the method of geometric quantization.

Take any element $\lambda \in \mathcal{O}$, we have an identification $\mathcal{O} = G/G_\lambda$, *lambda* : $\mathfrak{g}_\lambda \to \mathbb{C}$ is a Lie algebra homomorphism, and hence $\chi_\lambda : G_\lambda \to \mathbb{C}^\times$ defined by $\chi_\lambda(\exp X) = e^{\langle \lambda, X\rangle}$ is a unitary character of $G_\lambda$. Then we can defined a Hermitian line bundle $\mathcal{L}_\lambda = G \times^{G_\lambda} \mathbb{C}$ over $\mathcal{O}$, with a natural $G$-invariant connection $\nabla$ such that $\mathrm{curv}(\nabla) = \omega_\mathcal{O}$, where $\omega_\mathcal{O}$ is the Kirillov-Kostant-Souriau symplectic form on $\mathcal{O}$.

Then we can define the Hilbert space of square integrable sections of $\mathcal{L}_\lambda$, but this space is too large. We need to choose a polarization $\mathfrak{h} \subseteq \mathfrak{g}_\mathbb{C}$ (maximal isotropic subalgebra with respect to the bilinear form $(X, Y) \mapsto \langle \lambda, [X, Y]\rangle$) and consider the space of square integrable holomorphic sections of $\mathcal{L}_\lambda$ which are covariant constant along $\mathfrak{h}$.

In fact the polarization $\mathfrak{h}$ corresponds to a $G$-invariant integrable distribution on $\mathcal{O}$, its a subbundle $P$ of $T\mathcal{O}$ each fiber is a lagrangian subspace. Then we can define the space of polarized sections:

$$\Gamma_P(\mathcal{O}, \mathcal{L}_\lambda) = \{s \in \Gamma^\infty(\mathcal{O}, \mathcal{L}_\lambda \otimes |\omega_{\mathcal{O}/\mathcal{P}}|^{\frac{1}{2}}) \,|\, \nabla_X s = 0, \forall X \in P\}.$$

On this space, we have a unitary $G$ action and get a unitary representation $\pi_\mathcal{O}$ of $G$, it turns out that $\pi_\mathcal{O}$ is irreducible and independent of the choice of the polarization $\mathfrak{h}$.

Kirillov's philosophy works well for nilpotent Lie groups, but for more general Lie groups, it is not true anymore.

Now assume $G$ is a real reductive group, with Lie algebra $\mathfrak{g}_0$. For any irreducible Casselman-Wallach representation $\pi$ of $G$, we still have its generalized function character $\Theta_\pi \in C^{-\infty}(G)$, and hence a tempered generalized function $\theta_\pi$ on $\mathfrak{g}$. Here $\theta_\pi \in C^{-\infty}(\mathfrak{g}_0)$ is defined as follows:

$$\theta_\pi(f\mathrm{dX}) = \Theta_\pi(\exp_*(\sqrt{\det(\exp)}f\mathrm{dX})), \quad f\mathrm{dX} \in C_c^\infty(\mathfrak{g}_0)\mathrm{dX}.$$

The generalized function $\Theta_\pi$ has the following properties:

- $\Theta_\pi$ is invariant under conjugation of $G$;
- $\Theta_\pi$ is an eigendistribution of $\mathcal{U}(\mathfrak{g})^G$;
- $\Theta_\pi$ is locally integrable, and real analytic on the regular semisimple locus $G^{\mathrm{rs}}$.

By the work of Rossmann, we have the following theorem:

**Theorem 9.2.** *If $\pi$ is tempered, then*

$$\theta_\pi|_{\textit{nbhd of 0}} = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}|_{\textit{nbhd of 0}}.$$

*Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$, is the orbital integral along $\mathcal{O}_i$.*

For an arbitrary generalized function $\Theta \in C^{-\infty}(M)$, we can define its wavefront set $\mathrm{WF}(\Theta) \subseteq \sqrt{-1}T^*M$ as follows:

**Definition 9.3.** For any $(x,\xi) \in \sqrt{-1}T^*M$, $(x,\xi) \notin \mathrm{WF}(\Theta)$ if there exists a compactly supported smooth function $\varphi$ on $M$ such that $\varphi(x) \neq 0$, and an open conic neighborhood $\Gamma$ of $\xi$ such that for any $N \in \mathbb{N}$, there exists a constant $C_N > 0$ such that

$$|\widehat{\varphi\Theta}(t\eta)| \leq C_N t^{-N}, \quad \forall t > 0, \eta \in \Gamma.$$

$\mathrm{WF}(\Theta)$ is a closed conic subset of $\sqrt{-1}T^*M$ (singular direction of $\Theta$).

In particular, we can define the wavefront set $\mathrm{WF}(\pi)$ of an irreducible Casselman-Wallach representation $\pi$ of $G$ as $\mathrm{WF}(\Theta_\pi) \cap (\sqrt{-1}T_1^*(G))$, it is a closed conic $G$-invariant subset of $\sqrt{-1}\mathfrak{g}_0^*$. The GK-dimension of $\pi$ is defined as

$$\mathrm{Dim}(\pi) = \frac{1}{2}\dim(\mathrm{WF}(\pi)).$$

$\theta_\pi$ is a real analytical function on $\mathfrak{g}_0^{\mathrm{rs}}$, and

$$\theta_\pi^*(X) := \lim_{t\to 0} t^{\mathrm{Dim}(\pi)}\theta_\pi(tX)$$

is still a real analytical function on $\mathfrak{g}_0^{\mathrm{rs}}$, where $\mathcal{B}$ is the flag variety of $G$. We can consider it as in $C^{-\infty}(\mathfrak{g}_0)$.

**Theorem 9.4.** *(Barbasch-Vogan) The asymptotic of $\theta_\pi$ defined above is a non-negative linear combination of Fourier transforms of nilpotent orbital integrals:*

$$\theta_\pi^* = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}.$$

*Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$, is the orbital integral along $\mathcal{O}_i$.*

$\mathrm{WF}(\pi) = \bigcup_{c_i \neq 0} \overline{\mathcal{O}_i}$.

## 10. Conormal Bundles and Characteristic Cycles

2025-10-02

10.1. **Conormal bundle of singular subvariety.** Let $\mathcal{B}$ be the flag variety of $G$. $\overline{\mathcal{B}_y} \subsetneq \overline{\mathcal{B}_w}$ two schubert varieties, what is the relation between their conormal bundles $T^*_{\mathcal{B}_y}\mathcal{B}$ and $T^*_{\mathcal{B}_w}\mathcal{B}$?

Let $X$ be a smooth algebraic variety over $\mathbb{C}$, $Z \subseteq X$ be a closed subvariety, define the conormal bundle $T^*_Z X$ of $Z$ in $X$ as follows:

$$T^*_Z X = \overline{T^*_{Z^{\mathrm{reg}}} X}.$$

For a singular point $z \in Z$ what is the fiber of $T^*_Z X$ at $z$?

Here is a simple example:

**Example 10.1.** Let $X = \mathbb{C}^2$ and

$$Z = \{xy = 0\} \subset X$$

be the union of the coordinate axes.

- The regular locus is

$$Z^{\mathrm{reg}} = (x\text{-axis} \setminus \{0\}) \cup (y\text{-axis} \setminus \{0\}).$$

- The singular locus is
$$S := Z \setminus Z^{\mathrm{reg}} = \{0\}.$$

Now, the fibre of the conormal bundle at the origin is
$$(T^*_Z X)_0 = \mathrm{Ann}(T_{x\text{-axis}}) \cup \mathrm{Ann}(T_{y\text{-axis}}) = \langle dy \rangle \cup \langle dx \rangle,$$
which is a union of two lines in $T^*_0\mathbb{C}^2$.

On the other hand,
$$(T^*_S X)_0 = (T^*_{\{0\}} X)_0 = T^*_0 X,$$
the entire cotangent plane.

Thus,
$$(T^*_Z X)_0 \subsetneq (T^*_{Z-Z^{\mathrm{reg}}} X)_0,$$

10.2. **Summary of paper [KT84, The characteristic cycles of holonomic systems on a flag manifold].** Let $G$ be a semisimple simply connected complex algebraic group. Define Steinberg variety:

$$Z = \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, x \in \mathrm{Lie}(B') \cap \mathrm{Lie}(B'')\} (= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, \}).$$

$(= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, x \in \mathfrak{n}'^{\perp} \cap \mathfrak{n}''^{\perp}\}$ more naturally.)

Here $\mathcal{N} \subseteq \mathfrak{g}$ is the nilpotent cone of $\mathfrak{g}$, $\mathcal{B}$ is the flag variety of $G$. It can be identify with the union of conormal bundles of all $G$-orbits in $\mathcal{B} \times \mathcal{B}$:
$$Z = \bigcup_{w \in W} T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B}),$$
which is a closed conic lagrangian subvariety of $T^*(\mathcal{B} \times \mathcal{B})$.

We also have the moment map:
$$\mu : Z \to \mathcal{N}, \quad (x, B', B'') \mapsto x.$$

By taking characteristic cycles, we get a map from the (complex) Grothendieck group of $G$-equivariant holonomic $D$-modules on $\mathcal{B} \times \mathcal{B}$ to the top Borel-Moore homology of $Z$:
$$\mathrm{CC} : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}})) \to H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}).$$
We also have isomorphisms:
$$h : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}})) \to \mathbb{C}[W], \quad \mathcal{M}_w \mapsto w, \quad \mathcal{L}_w \mapsto a(w)$$

By Kazhdan-Lusztig, we have:
$$a(w) = \sum_{y \le w} (-1)^{l(y)+l(w)} P_{y,w}(1) y.$$

There is a convolution product on $K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}}))$ which makes $h$ an isomorphism of algebras.

On
$$H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{w \in W} H^{\mathrm{BM}}_{\mathrm{top}}(\overline{T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B})}, \mathbb{C}) \simeq \bigoplus_{w \in W} \mathbb{C}[\overline{T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B})}],$$
In the paper of Kazhdan-Lusztig [KL80], they use topological method to construct a $W \times W$ action on $H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C})$.

Moreover there is a $W \times W$ module decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}),$$

where $\mathcal{O}$ runs over all nilpotent orbits in $\mathfrak{g}^*$, $Z_{\mathcal{O}} = \mu^{-1}(\mathcal{O})$. There is also an isomorphism of $W \times W$-modules:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq (H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C}) \otimes H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C}))_{A_G(x)},$$

($A_G(x)$ coinvariants) where $x \in \mathcal{O}$, $\mathcal{B}_x = \{B' \in \mathcal{B} \,|\, x \in \mathrm{Lie}(B')\}$ (the Springer fiber), $A_G(x) = \mathrm{Stab}_G(x)/\mathrm{Stab}_G(x)^0$.

By classical Springer theory, $H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C})$ is a representation of $W \times A_G(x)$, and we have the following decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \rho.$$

Here $\tau(x, \rho)$ is an irreducible representation of $W$ or zero.

So we have a decomposition of $W \times W$-modules:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \tau(x, \rho).$$

Combine the above together, we finally have the following decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \tau(x, \rho), \quad H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}).$$

For any $w \in W$, there is a unique nilpotent orbit $\mathrm{St}(w)$ such that $\mu(\overline{T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B})}) = \overline{\mathrm{St}(w)}$. The map $w \mapsto \mathrm{St}(w)$ defines a surjective(?) map from $W$ to the set of nilpotent orbits in $\mathfrak{g}^*$ (this map can be computed via Robinson-Schensted algorithm).

**Corollary 10.2.** *For any nilpotent orbit $\mathcal{O}$, there is an isomorphism of $W \times W$-modules:*

$$\bigoplus_{w \in W, \mathrm{St}(w) = \mathcal{O}} \mathbb{C}[\overline{T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B})}] \simeq \bigoplus_{\rho \in \widehat{A_G(\mathcal{O})}} \tau(\mathcal{O}, \rho) \otimes \tau(\mathcal{O}, \rho).$$

**Theorem 10.3.** *(Main theorem of [KT84]) The map*

$$\mathrm{CC} : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \to H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}),$$

*is an isomorphism and $W \times W$-equivariant.*

Define $b(w) = h \circ \mathrm{CC}^{-1}([\overline{T^*_{\mathcal{Y}_w} \mathcal{B} \times \mathcal{B}}])$. Now we have three basis of $\mathbb{C}[W]$:

$$\{w\}_{w \in W}, \quad \{a(w)\}_{w \in W}, \quad \{b(w)\}_{w \in W}.$$

It was conjectured by Kashiwara and Tanisaki that: $a(w) = b(w)$ when $G = \mathrm{SL}_n(\mathbb{C})$ which means the characteristic cycle of $\mathcal{L}_w$ is irreducible, but unfortunately it is not true. Counterexample exists when $G = \mathrm{SL}_8(\mathbb{C})$, where Kashiwara-Saito singularities appears in some Schubert varieties.

## 11. Springer Representations and Characteristic Cycles

2025-10-05

**The Springer representation**

Let $G$ be a connected complex reductive group, $K$ a symmetric subgroup of $G$. We can define the Springer representation of $W$ on $H_c^\bullet(T_K^*\mathcal{B},\mathbb{Q})$.

Recall that we have the following diagram:

$$\begin{array}{ccccc} \widetilde{\mathcal{N}} & \hookrightarrow & \widetilde{\mathfrak{g}} & \hookleftarrow & \widetilde{\mathfrak{g}^{\mathrm{rs}}} \\ \downarrow{\scriptstyle p_{\mathcal{N}}} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p_{\mathrm{rs}}} \\ \mathcal{N} & \hookrightarrow & \mathfrak{g} & \hookleftarrow & \mathfrak{g}^{\mathrm{rs}} \end{array}$$

In this diagram $p_{\mathrm{rs}}$ is a $W$-torsor. So we have a natural $W$-action on $p_{\mathrm{rs},*}\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}}$ and hence on $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}},\mathcal{L})$, where $\mathcal{L}=(p_{\mathrm{rs}})_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}}$. There is a canonical isomorphism of perverse sheaves $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}},\mathcal{L})\simeq \mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[\dim(\mathfrak{g})]$, so there is a natural $W$-action on $\mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}$. By proper base change, we have

$$\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}\simeq \mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}|_{\mathcal{N}}. \quad (\mathrm{Spr}=\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[\dim\mathcal{N}]\text{called the Springer sheaf})$$

So there is a natural $W$-action on $\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}$ (actually $\mathrm{End}_{\mathrm{Perv}(\mathcal{N})}(\mathrm{Spr})=\mathbb{Q}[W]$) and hence on $H_c^\bullet(T^*\mathcal{B},\mathbb{Q})\simeq \mathbb{R}\Gamma_c(\mathcal{N},\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$ and $H_c^\bullet(\mathcal{B}_x,\mathbb{Q})\simeq \mathbb{R}\Gamma_c(\{x\},\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$. (action on Springer fiber)

Since $T_K^*\mathcal{B}=p_{\mathcal{N}}^{-1}(\mathfrak{p})$, we also have a natural $W$-action on

$$H_c^\bullet(T_K^*\mathcal{B},\mathbb{Q})\simeq \mathbb{R}\Gamma_c(\mathfrak{p}\cap\mathcal{N},(\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})|_{\mathfrak{p}\cap\mathcal{N}}).$$

Finally, there is a $W$-action on $(H_c^{2d}(T_K^*\mathcal{B},\mathbb{Q}))^*=\bigoplus_{\mathcal{O}\in K\backslash\mathcal{B}}\mathbb{Q}[\overline{T_{\mathcal{O}}^*\mathcal{B}}]$.

**Theorem 11.1** (Tanisaki)**.** *The map:*

$$\mathrm{CC}: K_0(\mathrm{Mod}_K^{\mathrm{Coh}}(\mathcal{D}_{\mathcal{B}}))\to (H_c^{2d}(T_K^*\mathcal{B},\mathbb{Q}))^*=H_{2d}^{\mathrm{BM}}(T_K^*\mathcal{B},\mathbb{Q}),$$

*is surjective and $W$-equivariant.*

## 12. Holonomic Systems and Characteristic Cycles

2025-10-09

$X=\mathrm{Spec}(R)$ smooth affine with coordinate $x_1,\dots,x_n$, $M$ a finitely generated $\mathcal{D}_X$-module.

**Definition 12.1.** $M$ is called holonomic if $\dim(\mathrm{Ch}(M))=n$.

**Corollary 12.2.** *If $M$ is holonomic, then* $\mathrm{Ch}(M)$ *is lagrangian in $T^*X$, and* $\mathrm{CC}(M)=\sum_{E_i\subseteq X,irr}n_i[\overline{T_{E_i}^*X}]$.

The de-Rham complex and analytification:

$$\mathcal{K}(M;\partial_1)=[\,M\xrightarrow{\partial_1}M\,],$$

$$\mathcal{K}(M;\partial_1,\partial_2)=[\,M\xrightarrow{(\partial_1,\partial_2)}M\oplus M\xrightarrow{(\partial_2,-\partial_1)}M\,].$$

We can define $\mathcal{K}(M;\partial_1,\dots,\partial_n)$ inductively.

There is another way to define the de-Rham complex, independent of the choice of coordinates:

$$\mathrm{DR}(M) = [M \xrightarrow{\nabla} \Omega^1_X \otimes M \xrightarrow{\nabla} \cdots \xrightarrow{\nabla} \Omega^n_X \otimes M].$$

**Example 12.3.** Let $X = \mathbb{C}$, $M = \mathbb{C}[x, x^{-1}] \cdot x^\alpha$, $\alpha \in \mathbb{C}$, $\partial_x x^\alpha = \alpha \frac{1}{x} x^\alpha$. Then $M_\alpha = \mathcal{D}_X / \mathcal{D}_X(x\partial_x - (\alpha - k))$. To get the *multiplicities* (the characteristic cycle), we examine the primary decomposition of the defining principal-symbol ideal:

$$(x\xi) = (x) \cap (\xi).$$

Each factor appears with exponent 1, and the associated graded module localized at a generic point of either component has length 1. Concretely:

- Localize at the generic point of $V(\xi)$ (i.e. at the prime $(\xi)$). In that localization, $\xi$ is nonunitally small and $x$ is a unit, so
$$(\mathrm{gr}\, M_\alpha)_{(\xi)} \cong \mathbb{C}[x, \xi]_{(\xi)}/(\xi)$$
which is a copy of $\mathbb{C}[x]_{(0)}$ of length 1 over the local ring — so the multiplicity is 1 on the zero section.
- Localize at the generic point of $V(x)$ (i.e. at $(x)$). There $\xi$ is a unit and
$$(\mathrm{gr}\, M_\alpha)_{(x)} \cong \mathbb{C}[x, \xi]_{(x)}/(x)$$
which is a copy of $\mathbb{C}[\xi]_{\mathrm{gen}}$ of length 1 — so the multiplicity is 1 on the conormal $T^*_0 X$.

Hence both components appear with multiplicity 1.

Therefore the characteristic cycle is

$$\boxed{\mathrm{CC}(M_\alpha) \;=\; [T^*_X X] \;+\; [T^*_0 X].}$$

This calculation is independent of $\alpha$, because the parameter $\alpha$ contributes only lower-order terms that do not affect the principal symbol.

Now consider the de-Rham complex of $M_\alpha$:

$$\mathrm{DR}(M_\alpha) = [M_\alpha \xrightarrow{\partial_x} M_\alpha].$$

It is exact, this is bad. To fix this, we need to analytify $M_\alpha$:

$$M^{\mathrm{an}}_\alpha = \mathcal{O}_{X^{\mathrm{an}}} \otimes_{\mathbb{C}[x]} M_\alpha = \mathcal{O}_{X^{\mathrm{an}}}[x^{-1}] \cdot x^\alpha.$$

**Example 12.4.** $N = \mathbb{C}[\delta]$ where $\delta$ is the delta function at 0, $x \cdot \delta = 0$. Then $N = \mathcal{D}_X / \mathcal{D}_X x$. Characteristic cycle is $[T^*_0 X]$ with multiplicity 1.

**Remark 12.5.** The two examples above consists of all regular singularities in some sence.

## 13. Double Covers and Genuine Representations

2025-10-13

**Lemma 13.1.** *Let $G$ be a connected Lie group, $\widetilde{G}$ is a double cover. Then the following are equivalent:*

- *$\widetilde{G}$ split;*
- *$\widetilde{G}$ is disconnected;*
- *there exists a genuine representation of $\widetilde{G}$ which is trivial on the identity component $\widetilde{G}_0$.*

*Proof.* Only need to notice that $\widetilde{G}_0 \to G$ is a covering map of degree 1 or 2. □

There is a same statement for p-adic groups in [Nev99].

**Lemma 13.2.** *Let $G$ be an arbitrary Lie group, $\pi : \widetilde{G} \to G$ a double cover. Then the following are equivalent:*

- *there exists a genuine finite dimensional representation of $\widetilde{G}$ which is trivial on the identity component $\widetilde{G}_0$;*
- *$z \notin \widetilde{G}_0$;*
- *$z \notin (\pi^{-1}(G_0))_0$;*
- *there exist a genuine finite dimensional representation of $\pi^{-1}(G_0)$ which is trivial on the identity component $(\pi^{-1}(G_0))_0$;*
- *$\pi^{-1}(G_0)$ is disconnected;*
- *$\pi^{-1}(G_0)$ is split.*

*Here $z$ is the nontrivial element in $\ker(\pi)$.*

*If in addition the double cover $\widetilde{G}$ is induced by a square root of a character $\gamma : G \to \mathbb{C}^\times$, i.e. $\widetilde{G} = \{(g, \epsilon) \in G \times \mathbb{C}^\times \mid \epsilon^2 = \gamma(g)\}$, these are also equivalent to:*

- *$\gamma$ has a square root.*

*Proof.* Only need to notice that $(\pi^{-1}(G_0))_0 = \widetilde{G}_0$, together with results in [Sch87]. □

## 14. Integrable connections and local systems

2025-10-14

$X = \mathrm{Spec}(R)$ smooth affine with coordinate $x_1, \dots, x_n$, $M$ is an integrable connection.

**Theorem 14.1.** *$M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$ for some local system $L$ on $X^{an}$.*

*Proof.* We know that $M$ is locally free, suppose $M|_U$ is freely generated by $m_1, \cdots, m_n$, suppose $\partial_n \underline{m} = \chi \underline{m}$ where $\chi$ is a matrix of holomorphic functions on $U$. Locally we can expand $\chi$ with respect to $x_n$:

$$\chi = \sum_{i=0}^{\infty} \chi_i(x_1, \dots, x_{n-1}) x_n^i.$$

Take $\Psi = \sum_{i=1}^{\infty} \frac{1}{i+1} \chi_i(x_1, \dots, x_{n-1}) x_n^i$, so $\partial_n \Psi = \chi$. Take $\Phi = \exp(-\Psi)$ which is invertible, and set $\underline{s} = \Phi \underline{m}$.

So

$$\partial_n s = \partial_n(\Phi) \cdot \underline{m} + \Phi \cdot (\partial_n \underline{m}) = 0.$$

There is a small open subset $V \subset U$ such that $M^{an}|_V = \bigoplus_{i=1}^r \mathcal{O}_V s_i$, take

$$\begin{aligned} M_1 &= \ker(M^{an}|_V \xrightarrow{\partial_n} M^{an}|_V) \\ &= \left\{ \sum_{i=1}^r f_i s_i \in M^{an}|_V \,\middle|\, \partial_n(\sum_{i=1}^r f_i s_i) = 0, \forall i \right\} \\ &= \left\{ \sum_{i=1}^r f_i s_i \in M^{an}|_V \,\middle|\, \partial_n f_i = 0, \forall i \right\} \end{aligned}$$

$$= \bigoplus_{i=1}^{r} \mathcal{O}_{V_1} s_i, \quad V_1 = V \cap \{x_n = 0\}.$$

Continue this process, we can finally get a subsheaf

$$M_n|_V = \bigoplus_{i=1}^{n} \mathbb{C} s_i.$$

of $M^{an}$, where $s_i$ are horizontal sections. So $M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$ where $L$ is the local system generated by $s_i$.

□

**Remark 14.2.** The proof actually implies that integrable connections are locally generated by horizontal sections, and the sheaf of horizontal sections is a local system.

**Corollary 14.3.** $\mathrm{DR}(M) = L$

*Proof.* $H^0(\mathrm{DR}(M)) \simeq \ker(M \xrightarrow{\nabla} \Omega^1_X \otimes M) = L$. All the other cohomologies vanish because of Poincaré lemma, since this complex is locally de-Rham complex. □

**Remark 14.4.** This means that de Rham complex of $M$ gives a locally free resolution of $L$. So the hypercohomology of $\mathrm{DR}(M)$ computes the cohomology of $L$.

## 15. Kashiwara's equivalence and its applications

2025-10-16

**Kashiwara's equivalence**

$Y$ smooth affine $/\mathbb{C}$ with coordinate $x_1, \cdots, x_n, y_1, \cdots, y_l$, $X = \{y_1 = \cdots = y_l = 0\} \subset Y$, $X$ is also smooth. $I = (y_1, \ldots, y_l)$, $\mathcal{D}_Y$ the ring of differential operators on $Y$, $\mathcal{D}_X$ the ring of differential operators on $X$. $\mathrm{Mod}_{qc}(\mathcal{D}_X)$: the category of quasi-coherent $\mathcal{D}_X$ modules, $\mathrm{Mod}^X_{qc}(\mathcal{D}_Y)$: the category of quasi-coherent $\mathcal{D}_Y$ modules supported on $X$.

**Theorem 15.1.** *(Kashiwara's equivalence) The functor:*

$$i_+ : \mathrm{Mod}_{qc}(\mathcal{D}_X) \to \mathrm{Mod}^X_{qc}(\mathcal{D}_Y), \quad M \mapsto \mathcal{D}_{Y \leftarrow X} \otimes_{\mathcal{D}_X} M = M[\partial_{y_1}, \cdots, \partial_{y_l}],$$

*is an equivalence of categories. Its quasi-inverse is given by:*

$$i^\sharp : \mathrm{Mod}^X_{qc}(\mathcal{D}_Y) \to \mathrm{Mod}_{qc}(\mathcal{D}_X), \quad N \mapsto \ker(N \longrightarrow^{y} N).$$

*Proof.* There is a proof in [HTT08], I don't want to repeat it here. □

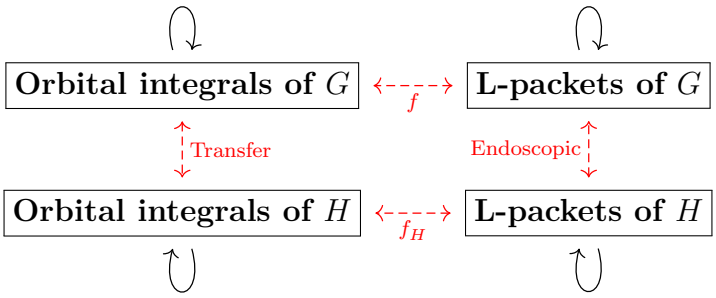
**Remark 15.2.** In the case of quasi-coherent sheaves, the equivalence does not hold. For example, Let $Y = \mathrm{Spec}(R)$, $X = \mathrm{Spec}(R/I) \subseteq Y$ a closed subscheme defined by ideal $I$. Then the functor:

$$i^* : \mathrm{Mod}^X_{qc}(\mathcal{O}_Y) \to \mathrm{Mod}_{qc}(\mathcal{O}_X), \quad N \mapsto R/I \otimes_R N.$$

is not an equivalence of categories. Since if $M$ is an $R$-module, then $M/I^2M$ is supported on $X$, but it does not defined over $R/I$.

**Corollary 15.3.** *Let $M$ be a quasi-coherent $\mathcal{D}_X$-module, if $Ch(M) = T^*_Y X$ for an irreducible closed smooth subvariety $Y \subset X$, then $M = i_+(N)$ for some integrable connection $N$ on $Y$.*

$$\mathrm{Mod}_{qc}(\mathcal{D}_X) \underset{i^\sharp}{\overset{i_+}{\rightleftarrows}} \mathrm{Mod}^X_{qc}(\mathcal{D}_Y)$$




## 16. Nilpotent orbits and degenerate Whittaker models

2025-10-19

**Degenerate Whittaker model**

$G$ be a finite extension of a reductive algebraic group defined over local field $F$, $\mathfrak{g}$ be its Lie algebra, fix a nontrivial character $\chi : F \to \mathbb{C}^1$.

A Whittaker pair is a pair $(S, \varphi)$ where $S \in \mathfrak{g}$ is a rational semi-simple element, $\varphi \in \mathfrak{g}^*$ such that $\mathrm{ad}^*(S)(\varphi) = -2\varphi$. Given such a Whittaker pair, we can define a degenerate Whittaker model $\mathcal{W}_{S,\varphi}$ as follows: Let $\mathfrak{g} = \bigoplus_{r\in\mathbb{Q}} \mathfrak{g}^S_r$ be the eigenspace decomposition with respect to $\mathrm{ad}(S)$, define

$$\mathfrak{u} = \mathfrak{g}_{\geq 1} = \bigoplus_{r\geq 1} \mathfrak{g}^S_r.$$

Define an anti-symmetric form $\omega_\varphi$ on $\mathfrak{u}$ by

$$\omega_\varphi(X, Y) = \varphi([X, Y]), \quad X, Y \in \mathfrak{u}.$$

Let $\mathfrak{n}$ be the kernel of $\omega_\varphi$, and $\mathfrak{n}' = \mathfrak{n} \cap \ker(\varphi)$. Then $\mathfrak{u}/\mathfrak{n}$ is a symplectic space with symplectic form induced by $\omega_\varphi$, and $\mathfrak{u}/\mathfrak{n}'$ is a Heisenberg Lie algebra with center $\mathfrak{n}/\mathfrak{n}'$. Corresponding groups are defined by exponentiation:

$$U = \exp(\mathfrak{u}), \quad N = \exp(\mathfrak{n}), \quad N' = \exp(\mathfrak{n}').$$

If $\varphi \neq 0$, there is a unique irreducible representation $(\rho_\varphi, \mathcal{S}_\varphi)$ of $U/N'$ with central character $\chi_\varphi : N/N' \to \mathbb{C}^\times$ defined by

$$\chi_\varphi(\exp(X)) = \chi(\varphi(X)), \quad X \in \mathfrak{n}.$$

Define the degenerate Whittaker model:

$$\mathcal{W}_{S,\varphi} = \mathrm{Ind}_U^G(\rho_\varphi).$$

If $\varphi = 0$, define $\mathcal{W}_{S,0} = \mathrm{Ind}_U^G(1_U)$.

**Remark 16.1.** In the special case when $(S, \varphi)$ comes from an $\mathfrak{sl}_2$-triple $(e, h, f)$ with $S = h$ and $\varphi$ corresponding to $f$ via the Killing form, we can see that $\mathfrak{n} = \mathfrak{g}_{\geq 2}$. Then $\mathcal{W}_f = \mathcal{W}_{S,\varphi}$ is called a generalized Whittaker model associated to the nilpotent orbit of $f$.

**Affine transformation group of real line**

$P_2(\mathbb{R}) = \mathbb{R}^\times \ltimes \mathbb{R}$, with multiplication:

$$(a, b) \cdot (a', b') = (aa', b + ab').$$

Its Lie algebra is $\mathfrak{p}_2(\mathbb{R}) = \mathbb{R} \times \mathbb{R}$ with bracket:

$$[(x_1, y_1), (x_2, y_2)] = (0, x_1y_2 - x_2y_1).$$

## 17. Spectral Sequences of Filtered Complexes

2025-10-21

**Spectral Sequences of Filtered Complexes**

$(F^\bullet, d)$ be a filtered complex, i.e. a complex $F^\bullet$ with an increasing filtration of subcomplexes:

$$\cdots \subseteq F_j^\bullet \subseteq F_{j+1}^\bullet \subseteq \cdots \subseteq F^\bullet.$$

Such that $d(F_j^i) \subseteq F_j^{i+1}$, and the filtration is exhaustive and separated:

$$\bigcup_{j\in\mathbb{Z}} F_j^i = F^i, \quad \bigcap_{j\in\mathbb{Z}} F_j^i = 0.$$

For any triple $(i, j, k)$, define:

- $Z_j^i(k) := \{u \in F_j^i \mid du \in F_{j-k}^{i+1}\}$;
- $B_j^i(k) := d(F_{j+k-1}^{i-1}) \cap F_j^i$;
- $E_j^i(k) := \frac{Z_j^i(k)+F_{j-1}^i}{B_j^i(k)+F_{j-1}^i}$.

So $d(Z_j^i(k)) \subseteq Z_{j-k}^{i+1}(k)$, which induces a differential $d_k : E_j^i(k) \to E_{j-k}^{i+1}(k)$.

We also define $E_k^{p,q} := E_{-p}^{p+q}(k)$.

**Example 17.1.** $E_j^i(0) = \frac{F_j^i}{F_{j-1}^i}$, so $E_0^{p,q} = \mathrm{Gr}_{-p}F^{p+q}$.

$E_j^i(1) = H^i(\mathrm{Gr}_j F^\bullet)$ , so $E_1^{p,q} = H^{p+q}(\mathrm{Gr}_{-p}F^\bullet)$. The first page of the spectral sequence is just the cohomology of the associated graded complex.

**Lemma 17.2.** $H^i(E_j^\bullet(k)) = E_j^i(k+1)$.

**Definition 17.3.** The spectral sequence associated to the filtered complex $(F^\bullet, d)$ is defined to be the collection of complexes $\{(E_\bullet^\bullet(k), d_k)\}_{k\geq 0}$.

They convergent to the cohomology of the original complex: $E_\infty^{p,q} \simeq \mathrm{Gr}_{-p}H^{p+q}(F^\bullet)$, if for any $i, j$, there exist $k \gg 0$ such that $E_j^i(k) = \mathrm{Gr}_j H^i(F^\bullet)$.

**Definition 17.4.** If there exist a $w \in \mathbb{Z}_{\geq 0}$ such that for any $i, j$, $F_j^i \cap d(F^{i-1}) \subseteq d(F_{j+w}^{i-1})$. Then the filtration is called $w$-regular.

**Theorem 17.5.** *If $F^\bullet$ is $w$-regular, then the spectral sequence associated to $(F^\bullet, d)$ converges to the cohomology of the original complex at page $w + 1$, i.e. for any $i, j$, $E_j^i(w + 1) \simeq \mathrm{Gr}_j H^i(F^\bullet)$.*

**Example 17.6.** $K^\bullet$ is a chain complex, $F$ a left exact functor, then there is a natural spectral sequence:

$$E_2^{p,q} = \mathbb{R}^{p+q}F(K^q) \Rightarrow \mathbb{R}^{p+q}F(K^\bullet).$$

(maybe:

$$E_1^{p,q} = \mathbb{R}^p F(K^q) \Rightarrow \mathbb{R}^{p+q} F(K^\bullet)?$$

)

Because we can use the Cartan-Eilenberg resolution to resolve $K^\bullet$, which is given by:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & K^0 & \longrightarrow & I^{0,0} & \longrightarrow & I^{0,1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & K^1 & \longrightarrow & I^{1,0} & \longrightarrow & I^{1,1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & K^2 & \longrightarrow & I^{2,0} & \longrightarrow & I^{2,1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \vdots & & \vdots & & \vdots & &
\end{array}$$

Then we can form a total complex $\mathrm{Tot}(I^{\bullet,\bullet})$ with a natural filtration, and apply the functor $F$ to get a filtered complex $(F(\mathrm{Tot}(I^{\bullet,\bullet})), d)$. The spectral sequence associated to this filtered complex is just the spectral sequence above.

A special case is the Hodge spectral sequence, when $X$ is a smooth algebraic variety over $\mathbb{C}$, $K^\bullet = \Omega_X^\bullet$ the de-Rham complex, and $F = \mathbb{R}\Gamma(X, -)$ the derived global section functor, the spectral sequence is:

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H_{\mathrm{dR}}^{p+q}(X).$$

## 18. Motivation for Quotient Stacks

2025-10-23

**Lemma 18.1.** *Let $H$ be a Lie group acting freely and properly on a smooth manifold $M$, then for any smooth manifold $T$, there is a natural isomorphism:*

$$\mathrm{Hom}_{sm}(T, M/H) \simeq \frac{\{(P \to^\pi T, f : P \to M) | \pi \text{ a principle } H\text{-bundle, } f \text{ is } H\text{-equivariant}\}}{\text{equivalence}}.$$

*Proof.* Given a smooth map $g : T \to M/H$, we can form the following Cartesian diagram:

$$\begin{array}{ccc}
P & \xrightarrow{f} & M \\
{\scriptstyle\pi}\downarrow & & \downarrow \\
T & \xrightarrow[g]{} & M/H
\end{array}$$

Since $H$ acts freely and properly on $M$, $\pi : P \to T$ is a principle $H$-bundle, and $f : P \to M$ is $H$-equivariant.

Conversely, given a principle $H$-bundle $\pi : P \to T$ and an $H$-equivariant map $f : P \to M$, we can define a smooth map $g : T \to M/H$ by sending $t \in T$ to the orbit of $f(p)$ where $p \in \pi^{-1}(t)$. □

**Remark 18.2.** Algebraic geometers use this to define the quotient stack $[M/H]$ even if the action is not free or proper, but $\mathrm{Hom}(T, [M/H])$ need to be a groupoid, not just a set of equivalence classes.

## 19. Unipotent Arthur Packets [ABV92]

2025-10-29

Let $G$ be a connected reductive algebraic group over $\mathbb{R}$, fix a weak $E$-group $\check{G}^\Gamma$. Which is an extension:

$$1 \to \check{G} \to \check{G}^\Gamma \to \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \to 1.$$

Here $\check{G}$ is the complex dual group of $G$. An $L$-parameter is a continuous homomorphism:

$$\phi : W_\mathbb{R} \to \check{G}^\Gamma,$$

such that the composition $W_\mathbb{R} \to {}^L\check{G} \to \mathrm{Gal}(\mathbb{C}/\mathbb{R})$ is the natural projection and $\phi(\mathbb{C}^*)$ consists of semisimple elements of $\check{G}$.

Given an $L$-parameter $\phi$, we can attach the following data:

- two semisimple elements $\lambda, \mu \in \check{G}$, defined by $\phi(e^t) = \exp(\lambda t + \mu \bar{t})$;
- $y = \exp(\pi i \lambda)\phi(j) \in \check{G}^\Gamma - \check{G}$.

The $L$-parameter $\phi$ is called *tempered* if the closure of $\phi(W_\mathbb{R})$ in the complex analytic topology is compact. This is equivalent to say that $\lambda + \mathrm{Ad}(y)\lambda$ is an elliptic element of $\check{\mathfrak{g}}$.

Fix a geometric parameter $(y, \Lambda)$, it lies in a variety $X_y(\mathcal{O}, \check{G}^\Gamma) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$. The $\check{G}$-equivariant microlocal geometry on $X_y(\mathcal{O}, \check{G}^\Gamma)$ is in some sense equivalent to the $K(y)$-equivariant microlocal geometry on $\check{G}(\lambda)_0/P(\Lambda)$.

Roughly speaking, there is a bijection between:

$$\boxed{\text{representation with minimal GK-dim}}$$
$$\updownarrow$$
$$\boxed{\text{parameters supported on regular orbits in } \check{G}(\Lambda)_0/P(\Lambda)}$$

**Definition 19.1.** An Arthur parameter is a homomorphism:

$$\psi : W_\mathbb{R} \times SL(2, \mathbb{C}) \to \check{G}^\Gamma,$$

such that the restriction to $W_\mathbb{R}$ $(\psi_0)$ is a tempered $L$-parameter, and the restriction to $SL(2, \mathbb{C})$ $(\psi_1)$ is algebraic.

For an arthur parameter $\psi$, we can define an $L$-parameter $\phi_\psi$ by:

$$\phi_\psi(w) = \psi(w, \begin{pmatrix} |w|^{1/2} & 0 \\ 0 & |w|^{-1/2} \end{pmatrix}).$$

Let $y_1 = \psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \check{G}$, and $\lambda_1 = d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \in \check{\mathfrak{g}}$, $(y_0, \lambda_0)$ is the parameter defined by the restriction of $\psi$ to $W_\mathbb{R}$.

Then we have $y = y_0 y_1, \lambda = \lambda_0 + \lambda_1$, since $\phi_\psi(e^t) = \exp(\lambda_0 t + \mu_0 \bar{t})\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix}$, and

$$\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix} = \exp\left( t \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} + \bar{t} \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right) = \exp(\lambda_1 t + \lambda_1 \bar{t}).$$

So $\phi_\psi(e^t) = \exp((\lambda_0 + \lambda_1)t + (\mu_0 + \lambda_1)\bar{t})$, hence $\lambda = \lambda_0 + \lambda_1$.

And

$$\begin{aligned} y &= \exp(\pi i \lambda)\phi_\psi(j) \\ &= \exp(\pi i \lambda_0) \exp\left( \pi i d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right) \psi_0(j) \\ &= \exp(\pi i \lambda_0)\psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \psi_0(j) \\ &= y_0 y_1. \end{aligned}$$

In addition of the two semisimple elements $\lambda, \mu$ and $y$ defined as above, we can also define a nilpotent element $E_\psi \in \check{\mathfrak{g}}$ by:

$$E_\psi = d\psi_1 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \check{\mathfrak{g}}.$$

$E_\psi$ enjoys the following properties:

- $\mathrm{Ad}(y)E_\psi = -E_\psi$, this means $E_\psi \in \mathfrak{s}(y)$;
- $[\lambda, E_\psi] = E_\psi$, in other words, $E_\psi \in \check{\mathfrak{g}}(\Lambda)_1 \subseteq \mathfrak{n}(\lambda)$ where $\check{\mathfrak{g}}(\Lambda) = \bigoplus_{r \in \mathbb{Z}} \check{\mathfrak{g}}_r$ is the eigenspace decomposition with respect to $\mathrm{ad}(\lambda)$.

$$\begin{array}{c} \text{Langlands parameter } (y, \lambda) \\ \downdownarrows \\ S = \check{G} \times^{K(y)} eP(\Lambda) \subseteq X_y(\mathcal{O}, \check{G}^\Gamma) = \check{G} \times^{K(y)} \check{G}(\Lambda)/P(\Lambda) \\ \downdownarrows \\ S_K = K(y) \cdot (eP(\Lambda)) \subseteq \check{G}(\Lambda)_0/P(\Lambda) \end{array}$$

Cotangent space at the base point:

$$T^*_{eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{n}(\lambda).$$

Conormal space at the base point of the $K(y)$-orbit:

$$T^*_{K(y),eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{s}(y) \cap \mathfrak{n}(\lambda).$$

So we can view $E_\psi$ as an element in the conormal space $T^*_{K(y),eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda))$ also in $T^*_{\check{G},\phi_\psi}(X(\check{G}^\Gamma))$.

**Definition 19.2.** An Arthur parameter $\psi$ is called *unipotent* if the restriction to $\mathbb{C}^* \subseteq W_{\mathbb{R}}$ is trivial.

**Remark 19.3.** In this case, $\psi : \mathrm{SL}_2(\mathbb{C}) \times \Gamma \to \check{G}^\Gamma$. So $\lambda = \lambda_1$

**Lemma 19.4.** *Suppose $\psi$ is unipotent. Consider the partial Springer resolution*

$$\mu : T^*(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \check{G}(\Lambda)_0 \times^{P(\Lambda)} \mathfrak{n}(\lambda) \to \mathcal{N}_{\mathcal{P}(\Lambda)},$$

*$E_\psi \in \mathcal{N}_{\mathcal{P}(\Lambda)} \cap \mathfrak{s}(y)$ is a regular element (belong to the Richardson orbit), and the fiber $\mu^{-1}(E_\psi)$ is a single point. This guarantees that $\mu$ has degree 1.*

*Proof.* We can see that

$$\mathfrak{p}(\Lambda) = \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)} + \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)^2} \cap \mathrm{Ad}(E_\psi)\check{\mathfrak{g}} + \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)^3} \cap \mathrm{Ad}(E_\psi)^2\check{\mathfrak{g}} + \cdots,$$

So the centralizer $\check{G}(\Lambda)^{E_\psi} \subseteq P(\Lambda)$, which means that the fiber $\mu^{-1}(E_\psi)$ is a single point. The operator $E_\psi$ carry the sum of non-negative eigenspaces of $\mathrm{ad}(\lambda)$ onto the sum of positive eigenspaces, i.e. $\mathrm{Ad}(E_\psi)\mathfrak{p}(\Lambda) = \mathfrak{n}(\Lambda)$. In other words, the differential of the map:

$$P(\Lambda) \to \mathfrak{n}(\lambda), \quad p \mapsto \mathrm{Ad}(p)E_\psi,$$

is surjective at the identity coset, so the orbit $\check{G}(\Lambda) \cdot E_\psi$ is dense in $\mathfrak{n}(\lambda)$, which means that $E_\psi$ is a regular element in $\mathcal{N}_{\mathcal{P}(\Lambda)}$. □

**Theorem 19.5.** *Fix an algebraic morphism $\psi : \mathrm{SL}_2(\mathbb{C}) \to \check{G}^\Gamma$, let $\lambda = d\psi \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$, $\mathcal{O} = \check{G} \cdot \lambda \subseteq \check{\mathfrak{g}}$ the corresponding semisimple orbits. Consider the open and closed subvariety of the geometric parameter space $X_y(\mathcal{O}, \check{G}^\Gamma) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$.*

*Then there are bijections between:*

- *equivalence classes of unipotent Arthur parameters supported on $X_y(\mathcal{O}, \check{G}^\Gamma)$;*
- *regular $K(y)$-orbits in $\check{G}(\Lambda)_0/P(\Lambda)$;*
- *regular $K(y)$-orbits in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.*

*Proof.* Given a unipotent Arthur parameter $\psi$, we can associate to it a nilpotent element $E_\psi \in \mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ as above, then $K(y) \cdot E_\psi$ is a regular orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.

Conversely, given a regular $K(y)$-orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$, let $e$ be a representative. We can construct a homomorphism $\psi_1' : \mathrm{SL}_2(\mathbb{C}) \to \check{G}$ such that $d\psi_1' \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = e$, and

$$\psi_1'(\theta_{\mathrm{SL}_2}(x)) = \theta_y(\psi_1'(x)).$$

Define $y_0 = y\psi_1' \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$

Then $(y_0, \psi_1')$ is the corresponding unipotent Arthur parameter. □

**Remark 19.6.** In other words, this theorem tell us the following diagram:

$$\begin{array}{ccc} \boxed{\check{G}\text{-orbits in } \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))} & \xleftrightarrow{1-1} & \boxed{K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \\ \uparrow & & \uparrow \\ \boxed{\text{unipotent A-parameters supported on } X_y(\mathcal{O}, \check{G}^\Gamma)} & \xleftrightarrow{1-1} & \boxed{\text{regular } K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \end{array}$$

**Theorem 19.7.** *Under the above setting, fix a unipotent Arthur parameter $\psi$ supported on $X_y(\mathcal{O}, \check{G}^\Gamma)$, let $S$ be the corresponding $\check{G}$-orbit in $X_y(\mathcal{O}, \check{G}^\Gamma)$, $S_K$ be the corresponding $K(y)$-orbit in $\check{G}(\Lambda)_0/P(\Lambda)$, and $\mathcal{O}_K$ be the corresponding $K(y)$-orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.*

*Then there are bijections between:*

- *irreducible $(\check{\mathfrak{g}}(\Lambda), K(y)^{alg})$-modules $M$ annihilated by $I_{\mathcal{P}(\Lambda)}$ such that*
$$\overline{\mathcal{O}_K} \subseteq \mathrm{AV}(M);$$
- *irreducible $K(y)^{alg}$-equivariant $\mathcal{D}$-modules $\mathcal{M}$ on $\check{G}(\Lambda)_0/P(\Lambda)$ such that*
$$T^*_{S_K}(\check{G}(\Lambda)_0/P(\Lambda)) \subseteq \mathrm{Ch}(\mathcal{M});$$
- *irreducible $\check{G}$-equivariant $\mathcal{D}$-modules $\mathcal{N}$ on $X_y(\mathcal{O}, \check{G}^\Gamma)$ such that*
$$T^*_S(X_y(\mathcal{O}, \check{G}^\Gamma)) \subseteq \mathrm{Ch}(\mathcal{N});$$
- *Arthur packets $\Pi^{Art}_\psi(G/\mathbb{R})$ associated to $\psi$.*

**Corollary 19.8.** *Fix a nilpotent orbit $\check{\mathcal{O}} \subseteq \check{\mathfrak{g}}$, attached to it is a semisimple orbit $\check{\mathcal{O}}_{ss} \subseteq \check{\mathfrak{g}}$ then*
$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \bigcup_{\psi \text{ unipotent}, \mathcal{O}_\psi = \check{\mathcal{O}}_{ss}} \Pi^{Art}_\psi(G/\mathbb{R}).$$

**Remark 19.9.**
$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \{\text{irreducible representations in } \mathrm{Irr}_{\check{\mathcal{O}}_{ss}}(G/\mathbb{R}) \text{ with minimal GK-dim}\}$$

**Example 19.10.** $G = G_1 \times G_1$ where $G_1$ is a complex connected reductive algebraic group viewed as a real group. Then $\check{G}^\Gamma = (\check{G}_1 \times \check{G}_1) \rtimes \{1, \delta\}$. A homomorphism $\psi : \mathrm{SL}_2(\mathbb{C}) \to \check{G}$ is determined by two homomorphisms $\psi_1, \psi_2 : \mathrm{SL}_2(\mathbb{C}) \to \check{G}_1$, we can check that $\psi$ comes from a unipotent Arthur parameter if and only if $\psi_1$ and $\psi_2$ are conjugate to each other. So unipotent Arthur parameters of $G$ are in bijection with homomorphisms $\psi_1 : \mathrm{SL}_2(\mathbb{C}) \to \check{G}_1$. Therefore, equivalence classes of unipotent Arthur parameters of $G_1$ are in bijection with nilpotent orbits in $\check{\mathfrak{g}}_1$.

## 20. Questions on the Unitary Dual

2025-11-04

**Some questions about unitary dual problem**

- How can we reduce the classification of irreducible unitary $(\mathfrak{g}, K)$-module to the classification of such module with real infinitesimal character?
- Why there always exist a $\mathfrak{u}_\mathbb{R}$-invariant Hermitian form on irreducible $(\mathfrak{g}, K)$-module with real infinitesimal character?
- what is Jantzen filtration of a standard $(\mathfrak{g}, K)$-module? Why it coincides with Hodge filtration?

Let $G$ be a connected reductive group over $\mathbb{C}$, $\mathcal{B}$ the variety of all Borel subalgebras of $\mathfrak{g}$, ${}^a\mathfrak{h}$ is the universal Cartan subalgebra. For any Borel subalgebra $\mathfrak{b} \in \mathcal{B}$, we have a canonical isomorphism:
$$\phi_\mathfrak{b} : {}^a\mathfrak{h} \simeq \mathfrak{h},$$
such that negative roots in ${}^a\mathfrak{h}^*$ correspond to roots in $\Delta(\mathfrak{h}, \mathfrak{n})$.

We can define a $G$-equivariant vector bundle $\mathcal{L}_\lambda$ over $\mathcal{B}$ for any $\lambda \in {}^a\mathfrak{h}^*$ by:

$$\mathcal{L}_\lambda|_{\mathfrak{b}} = \mathbb{C}_\lambda := \text{1-dimensional representation of } B \text{ defined by } \lambda \circ \phi_{\mathfrak{b}}^{-1}.$$

$$\boxed{\text{Weight lattice } \Lambda \subseteq {}^a h^*} \xleftrightarrow{1-1} \boxed{G\text{-equivariant line bundles on } \mathcal{B}}$$

Under this correspondence, $\mathcal{L}_\lambda$ is ample if and only if $\lambda$ is regular dominant. $\mathcal{L}_{-2\rho}$ is the canonical line bundle $\wedge^{top} T^*\mathcal{B}$ on $\mathcal{B}$.

## 21. Annihilator Varieties via Cells

2025-11-19

### 21.1. **compute annihilator variety.**

$$\boxed{\mathrm{Prim}_\nu(\mathrm{U}(\mathfrak{g}))} \xleftrightarrow{1-1} \boxed{\text{left cells in } \mathrm{Coh}_\nu(\mathfrak{g},\mathfrak{b})} \longrightarrow \boxed{\text{double cells in } \mathrm{Coh}_\nu(\mathfrak{g},\mathfrak{b})}$$

$$I \in \mathrm{Prim}(\mathrm{U}(\mathfrak{g})) \longmapsto \mathcal{C}^L_{\nu,I} \longmapsto \mathcal{C}^{LR}_{\nu,I}$$

$\mathrm{span}(\mathcal{C}^{LR}_{\nu,I})$ is a representation of the integral Weyl group $W(\nu)$, all irreducible constituents of $\mathrm{span}(\mathcal{C}^{LR}_{\nu,I})$ lies in a single double cell, and contains a unique special representation of $W(\nu)$

$$\sigma_{\nu,I}.$$

Then by results of Joseph, we have:

$$\mathrm{AV}(I) = \mathcal{O}(j^W_{W(\nu)}(\sigma_{\nu,I})).$$

Which means that to compute the annihilator variety of $\mathrm{L}(w\nu)$, we only need to compute the corresponding cell representation.

In the case of $(\mathfrak{g}, K)$-modules, the method is the same:

$$\begin{array}{c} X\text{: Irreducible } (\mathfrak{g},K)\text{-module} \\ \downarrow \\ C\text{: Harish-Chandra cell containing } X \\ \downarrow \\ \sigma_C\text{: special representation of } W(\nu) \text{ contained in the cell representation} \\ \downarrow \\ \mathcal{O}(j^W_{W(\nu)}(\sigma_C)) = \mathrm{AV}(X) \end{array}$$

**Remark 21.1.** The key point is the result of Joseph [Jos85] the Springer correspondence of cell representation gives the annihilator variety. However, I'm not fully understand the details.

21.2. **annihilator varieties are special.** $j$-induction of parabolic subgroups preserves special representations.

To prove that the annihilator variety of irreducible $(\mathfrak{g}, K)$-module is closure of a single special nilpotent orbit, only need to check the case when infinitesimal character has good parity and bad parity.

For example, in the case of $G = \mathrm{Sp}_{2n}(\mathbb{R})$, we have:

$$\mathrm{Coh}_{\Lambda_b}(\mathrm{SO}(n_b, n_b)) \otimes \mathrm{Coh}_{\Lambda_g}(\mathrm{Sp}_{2n_g}(\mathbb{R})) \simeq \mathrm{Coh}_{\Lambda}(\mathrm{Sp}_{2n}(\mathbb{R})).$$

We need to show that the $j$-induction of any special repn appears in $\mathrm{Coh}_{\Lambda_b}(\mathrm{SO}(n_b, n_b))$ and any special repn appears in $\mathrm{Coh}_{\Lambda_g}(\mathrm{Sp}_{2n_g}(\mathbb{R}))$ gives a special repn in $\mathrm{Coh}_{\Lambda}(\mathrm{Sp}_{2n}(\mathbb{R}))$.

- Problem: How to compute the $j$-induction explicitly?
- Problem: the special repn of each cell always appears?

## 22. Annihilator and Whittaker Support Problems

2025-11-21

**Some problems/plans**

- Use counting methods to prove that all annihilator variety of irreducible $(\mathfrak{g}, K)$-modules of orthogonal or special groups are special.
- Use counting methods to prove that all annihilator variety of irreducible genuine $(\mathfrak{g}, K)$-modules of metaplectic groups are metaplectic special.
- By computation, prove that in the case of metaplectic groups, metaplectic quasi-admissible equivalent to metaplectic special. solved! Yes
- Prove that for metaplectic groups, whittaker support of irreducible genuine $(\mathfrak{g}, K)$-modules are metaplectic quasi-admissible. Use arguments of Gomez-Gourevitch-Sahi
- Prove associated cycles coincides with whittaker cycles for irreducible $(\mathfrak{g}, K)$-modules of real classical groups and metaplectic groups.

## 23. Robinson-Schensted and Annihilator Varieties

2025-11-24

For a Weyl group $W$, elements of $W$ are participated into double cells, left cells, etc. Each double cell $\mathcal{C}^{LR}$ contains a unique special representation $\sigma_{\mathcal{C}^{LR}}$ of $W$. And each left cell $\mathcal{C}^{L}$ corresponds to a primitive ideal $I_{\mathcal{C}^{L}}$ of $\mathrm{U}(\mathfrak{g})$ with infinitesimal character $\rho$. For each element $w \in W$, there are two processes to get a nilpotent orbit $\mathcal{O}_w$ in $\mathfrak{g}^*$:

$$w \longmapsto \mathcal{C}_w^{LR} \longmapsto \sigma_{\mathcal{C}_w^{LR}} \longmapsto \mathcal{O}(\sigma_{\mathcal{C}_w^{LR}});$$

$$w \xmapsto{RS} P(w) \longmapsto \mathcal{O}_{P(w)} \longmapsto \mathcal{O}_{P(w),s}.$$

The question is: are these two nilpotent orbits the same? i.e. $\mathcal{O}(\sigma_{\mathcal{C}_w^{LR}}) = \mathcal{O}_{P(w),s}$?
answer: Yes! [BV82]

## 24. Linear Double Covers and Genuine Representations

2026-03-03

$G$ is a linear real reductive group, $\widetilde{G}$ is a double cover of $G$.

**Theorem 24.1.** *$\widetilde{G}_0$ is a linear group (i.e. there is an injective homomorphism $\widetilde{G}_0 \to \mathrm{GL}_n(\mathbb{R})$ for some $n$) equivalent to the existence of a finite-dimensional genuine representation of $\widetilde{G}$.*

*Proof.* If $\widetilde{G}_0$ is a linear group, there are two possibilities:

- $\varepsilon \notin \widetilde{G}_0$: then $\{1, \varepsilon\} \times \widetilde{G}_0$ is a subgroup of $\widetilde{G}$ with finite index, and it has a finite-dimensional representation such that $\varepsilon$ acts by $-1$, so we get a finite-dimensional genuine representation of $\widetilde{G}$ by induction.
- $\varepsilon \in \widetilde{G}_0$: denote $\widetilde{G}_0$ by $H$, then $H = A \times {}^0H$ as a Nash group, and ${}^0H$ is also a linear group. So there is a closed embedding of Nash groups ${}^0H \hookrightarrow \mathrm{GL}_n(\mathbb{R})$ for some $n$ (${}^0H$ has compact center), take Zariski closure of ${}^0H$ in $\mathrm{GL}_n(\mathbb{C})$ we get the complexification ${}^0H_{\mathbb{C}}$ of ${}^0H$, there is an algebraic matrix coefficient of ${}^0H_{\mathbb{C}}$ such that $\varepsilon$ acts on it by $-1$, and it generates a finite-dimensional genuine representation of $\widetilde{G}$ by induction.

Conversely, if there is a finite-dimensional genuine representation of $\widetilde{G}$, we also have two possibilities:

- $\varepsilon \notin \widetilde{G}_0$: then $\widetilde{G}_0 \to G$ is injective, since $G$ is a linear group, $\widetilde{G}_0$ is also a linear group.
- $\varepsilon \in \widetilde{G}_0$: then we can find a finite-dimensional representation $E_1$ of $\widetilde{G}_0$ such that $\varepsilon$ acts by $-1$. We also have a finite-dimensional faithful representation $E_2$ of $G$, we can regard it as a representation of $\widetilde{G}_0$ by pullback, and $\varepsilon$ acts trivially on $E_2$. So $E_1 \oplus E_2$ is a finite-dimensional faithful representation of $\widetilde{G}_0$.

□

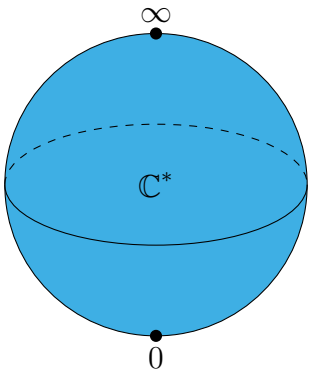
**Remark 24.2.** After I have a sleep, I find that $\widetilde{G}_0$ can be replaced by $\widetilde{G}$ in this theorem, and the proof is even easier.

## 25. Geometry of the flag variety of $\mathrm{SL}_2(\mathbb{C})$

2026-03-21

In this section, we illustrate the geometry of the flag variety of $\mathrm{SL}_2(\mathbb{C})$, which serve as a simple example for understanding more general phenomena.

$K = \mathbb{C}^*$ action has three orbits on $\mathbb{P}^1(\mathbb{C})$: two closed orbits $\{0\}$ and $\{\infty\}$, and one open orbit $\mathbb{C}^*$. The conormal bundle of the open orbit is the zero section, while the conormal bundle of the closed orbits are the cotangent fiber at 0 and $\infty$ respectively. So we have three conormal bundles, which are all Lagrangian subvarieties in $T^*\mathbb{P}^1(\mathbb{C})$.

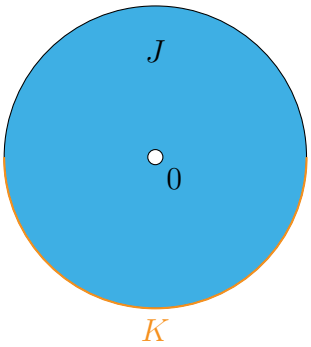



For a $K$-equivariant perverse sheaf $\mathcal{F}^\bullet$ on it, we calculate its characteristic cycle $\mathrm{CC}(\mathcal{F}^\bullet)$ use the formula

$$\chi_S^{mic} = \sum_{S'>S} c(S,S')\chi_{S'}^{loc},$$

only need to determine the coefficients $c(S,S')(=\sum_i(-1)^i H_c^i(J\cap S', K\cap S';\mathbb{C}))$ for $S'>S$, where $(J,K)$ is the normal Morse datum of S, which is well-defined up to homeomorphism.

- For the open orbit $\mathbb{C}^*$, we have $S'=\mathbb{C}^*$ and $c(\mathbb{C}^*,\mathbb{C}^*)=-1$.
- For the closed orbit $\{0\}$, we have $S'=\{0\},\mathbb{C}^*$, and $c(\{0\},\{0\})=1$, for $c(\{0\},\mathbb{C}^*)$, the pair $(J\cap S', K\cap S')$ is the following picture




therefore $c(\{0\},\mathbb{C}^*)=\sum_i(-1)^i H_c^i(J\cap\mathbb{C}^*, K\cap\mathbb{C}^*;\mathbb{C}) \stackrel{?}{=} -1$.

So we have

$$\begin{aligned}
\chi_0^{mic} &= \chi_0^{loc} - \chi_{\mathbb{C}^*}^{loc},\\
\chi_\infty^{mic} &= \chi_\infty^{loc} - \chi_{\mathbb{C}^*}^{loc},\\
\chi_{\mathbb{C}^*}^{mic} &= -\chi_{\mathbb{C}^*}^{loc}.
\end{aligned}$$

## 26. Perverse sheaves on $\mathbb{C}$

2026-03-22

Constructible sheaves on $\mathbb{C}=\mathbb{C}^*\sqcup\{0\}$. Some calculations comes from the lecture notes by Geordie Williamson [Wil15].

Denote $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$ the open inclusion, $i : \{0\} \hookrightarrow \mathbb{C}$ the closed inclusion. Then we have the following six functors:

$$D^b(\mathbb{C}^*) \underset{j_*}{\overset{j_!}{\underset{\longleftarrow j^*=j^! \longrightarrow}{\longrightarrow}}} D^b_c(\mathbb{C}) \underset{i^!}{\overset{i^*}{\underset{\longleftarrow i_*=i_! \longrightarrow}{\longrightarrow}}} D^b(\{0\})$$

$\mathcal{L}$ is a local system on $\mathbb{C}^*$.

$j_!\mathcal{L}$ has no quotient supported on $\{0\}$, $j_*$ has no subobject supported on $\{0\}$. $j_!\mathcal{L}[1]$ and $j_*\mathcal{L}[1]$ are perverse sheaves on $\mathbb{C}$.

**Remark 26.1.** Actually when $j : S \hookrightarrow X$ is affine locally closed embedding, $j_!$ and $j_*$ are $t$-exact for the perverse $t$-structure, so $j_!\mathcal{L}[\dim S]$ and $j_*\mathcal{L}[\dim S]$ are perverse sheaves on $X$.

By easy calculation of group cohomology, we have:

$$H^i(j_*\mathcal{L}[1]_0) = \begin{cases} 0 & i \neq 0, -1 \\ V_\mu & i = 0 \\ V^\mu & i = -1 \end{cases}$$

There is an exact sequence of perverse sheaves on $\mathbb{C}$:

$$0 \to i_*V^\mu \to j_!\mathcal{L}[1] \to j_*\mathcal{L}[1] \to i_*V_\mu \to 0.$$

## 27. Enhanced flag variety and line bundles on $\mathcal{B}$

2026-03-31

# I. The Enhanced Flag Variety and $G_0/N$ Isomorphism [BB93]

$\mathfrak{g}$ be a complex semisimple Lie algebra, ${}^a\mathfrak{h} = \mathfrak{b}/[\mathfrak{b},\mathfrak{b}]$ be the universal Cartan subalgebra, $G$ be the automorphism group of $\mathfrak{g}$, $G_0$ be the connected component of $G$, it is the adjoint group of $\mathfrak{g}$.

Attached to ${}^a\mathfrak{h}$ is a root system $\Delta$, and the Weyl group $W$. Although there isn't a well defined action of ${}^a\mathfrak{h} = \mathfrak{b}/[\mathfrak{b},\mathfrak{b}]$ on $\mathfrak{g}/\mathfrak{b} = \mathfrak{n}^*$, the action of ${}^a\mathfrak{h}$ on $(\mathfrak{n}/[\mathfrak{n},\mathfrak{n}])^*$ is well defined, and there is a canonical decomposition:

$$(\mathfrak{n}/[\mathfrak{n},\mathfrak{n}])^* = \bigoplus_{\alpha\in\Sigma} (\mathfrak{n}/[\mathfrak{n},\mathfrak{n}])^*_\alpha,$$

Where $\Sigma$ is the set of simple roots. Note that this choice of positive root system is opposite to the one usually used in representation theory, but it is more natural in geometry since dominant implies positive line bundle in this way.

The enhanced flag variety $\widetilde{\mathcal{B}}$ is defined as the variety of pairs $(\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma})$ where $\mathfrak{b}$ is a Borel subalgebra of $\mathfrak{g}$, and $\xi$ is a nonzero element in $(\mathfrak{n}/[\mathfrak{n},\mathfrak{n}])^*_\alpha$. There is a natural projection map $\pi : \widetilde{\mathcal{B}} \to \mathcal{B}$ which forgets the second component, and is an $H$ principal bundle.

There is a $G$ action on $\widetilde{\mathcal{B}}$ defined by $g \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma}) = (\mathrm{Ad}(g)\mathfrak{b}, \{\mathrm{Ad}^*(g)x_\alpha\}_{\alpha\in\Sigma})$. There is also an $H$ action on $\widetilde{\mathcal{B}}$ defined by $h \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma}) = (\mathfrak{b}, \{\alpha(h)x_\alpha\}_{\alpha\in\Sigma})$. These two actions is compatible in the following way:

$$g \cdot h \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma}) = \kappa_g(h) \cdot g \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma}),$$

where $\kappa : G \to \mathrm{Aut}(H)$ is the action as outer automorphisms of $H$.

The adjoint group $G_0$ acts transitively on $\widetilde{\mathcal{B}}$, and the stabilizer of a base point $(\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma})$ is $N$. So we have an isomorphism $\widetilde{\mathcal{B}} \simeq G_0/N$. Under this identification, the $G$ action on $\widetilde{\mathcal{B}}$ is given by $g \cdot (g_0N) = gg_0(N)$, and the $H$ action on $\widetilde{\mathcal{B}}$ is given by $h \cdot (g_0N) = g_0hN$.

# II. Line Bundles on $\mathcal{B}$ and Very Ampleness

Now $G$ is a connected reductive group.

**Lemma 27.1.** *Let $G \to \mathrm{GL}(V)$ be a finite-dimensional representation of $G$, $k \cdot v \subseteq V$ be a one-dimensional subspace, $P = \mathrm{Stab}_G(k \cdot v)$. Suppose that $k_\lambda$ be the one-dimensional representation of $P$ defined by the action of $P$ on $k \cdot v$, then $i^*(\mathcal{O}_{\mathbb{P}(V)}(1)) = \mathcal{O}_{G/P}(\lambda)$.*

*Proof.* Note that $\mathcal{O}_{\mathbb{P}(V)}(-1)$ is the tautological line bundle, $P$ action on the fiber of $\mathcal{O}_{\mathbb{P}(V)}(-1)$ at $k \cdot v$ is given by $k_\lambda$. So $k_{-\lambda}$ for $\mathcal{O}(1)$, also note that $\mathcal{O}_{G/P}(\lambda) = G \times^P k_{-\lambda}$. □

**Lemma 27.2.** *$\mathcal{O}_{\mathcal{B}}(\lambda)$ is very ample if and only if $\lambda$ is regular dominant ($\lambda \in X^+(H)$).*

*Proof.* Consider the Verma module $V^\lambda = \mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{h})} k_\lambda$. By the transitive group action of $G$ on $\mathcal{B}$, $\mathcal{O}(\lambda)$ is generated by global sections $\Gamma(\mathcal{B}, \mathcal{O}(\lambda)) = V^\lambda$. Hence there is a surjection $\Gamma(\mathcal{B}, \mathcal{O}(\lambda)) \twoheadrightarrow \mathcal{O}(\lambda) \otimes_k k_{eB} = k_\lambda$. Taking dual, we have $k_{-\lambda} \hookrightarrow (V^\lambda)^*$.

Since $\lambda$ is regular dominant, the stablizer of $k_{-\lambda}$ in $(V^\lambda)^*$ is the Borel $B$, by the previous lemma, we have a $G$-equivariant morphism $\mathcal{B} \hookrightarrow \mathbb{P}((V^\lambda)^*)$ such that $\mathcal{O}(\lambda)$ is the pullback of $\mathcal{O}(1)$, hence $\mathcal{O}(\lambda)$ is very ample. □

## 28. Tangent Maps for Associated Bundles

2026-04-05

$G$ is a complex algebraic group, $P$ is a closed subgroup of $G$ with a finite dimensional rational representation $V$. suppose that $\mathfrak{g} = \mathfrak{u} \oplus \mathfrak{p}$.

$$\begin{array}{ccc} Y = G \times^P V & \xrightarrow{\pi} & V \\ \downarrow & & \\ G/P & & \end{array}$$

There is an isomorphism:

$$\mathfrak{u} \oplus V \xrightarrow{\sim} T_{\overline{(1,\alpha)}}(G \times^P V),$$

which sends $(u, v)$ to $u^\sharp + v$, where $v \in V$ is regarded as a tangent vector of the fiber $V$ at $\alpha$, and $u^\sharp$ is the vector field on $Y$ generated by the action of $u \in \mathfrak{u} \subset \mathfrak{g}$.

The tangent map of $\pi$ at $\overline{(1, \alpha)}$ is:

$$\begin{aligned} d\pi_{\overline{(1,\alpha)}} : \mathfrak{u} \oplus V \to T_{\overline{(1,\alpha)}}(G \times^P V) \to V, \\ d\pi_{\overline{(1,\alpha)}}(u, v) = u \cdot \alpha + v, \end{aligned}$$

since $\pi(g, \alpha) = g \cdot \alpha$.

The above calculation is instrumental in extending the Kostant-Kirillov symplectic form of nilpotent orbits to the Jacobson-Morozov resolution of a nilpotent orbit:

$$\begin{array}{ccc} G \times^P \mathfrak{g}_{\geq 2} & \xrightarrow{\quad\pi\quad} & \overline{\mathbb{O}} \\ \uparrow & & \uparrow \\ G \times^P P \cdot e = G/Z_P(e) & \longrightarrow & \mathbb{O} \end{array}$$

$$\pi^* \omega_{KK,\alpha}((u,v),(x,y)) = \kappa(\alpha,[u,x]) + \kappa(u,y) - \kappa(v,x).$$

Where $(u,v),(x,y) \in \mathfrak{g}_{\leq -1} \oplus \mathfrak{g}_{\geq 2} = T_{\overline{(1,\alpha)}}(G \times^P \mathfrak{g}_{\geq 2})$.

This 2-form can be extended to a $G$-invariant 2-form $\omega_Y$ on $G \times^P \mathfrak{g}_{\geq 2}$.

It is easy to check that $\mathrm{Rad}(\omega_Y) = \mathfrak{g}_{-1} \oplus 0$, so this form is symplectic if and only if $\mathfrak{g}_{-1} = 0$, which is equivalent to the nilpotent orbit $\mathbb{O}$ is even.

## 29. Symplectic Geometry and Formal Quantization

2026-04-22

[Los22b]

### 29.1. **Examples of Symplectic Form and Moment map.**

**Proposition 29.1.** *The Kostant-Kirillov form $\omega_{KK}$ on a coadjoint orbit $\mathbb{O}$ is closed and hence symplectic.*

*Proof.* For $X \in \mathfrak{g}$ we have a vector field $X^\sharp$ on $\mathbb{O}$ defined by

$$\boxed{X^\sharp_\alpha = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathrm{Ad}^*(\exp(tX))\alpha = [X,\alpha]}$$

Define a smooth function $f_X = \kappa(-,X)$ on $\mathbb{O}$, then we have

$$\mathrm{d}f_X(Y^\sharp) = \kappa([Y,\alpha],X) = \kappa(\alpha,[X,Y]) = \omega_{KK}(X^\sharp,Y^\sharp).$$

So $\iota_{X^\sharp}\omega_{KK} = df_X$.

By Cartan's magic formula, we have

$$\mathcal{L}_{X^\sharp}\omega_{KK} = \mathrm{d}\iota_{X^\sharp}\omega_{KK} + \iota_{X^\sharp}\mathrm{d}\omega_{KK} = \mathrm{d}\mathrm{d}f_X + \iota_{X^\sharp}\mathrm{d}\omega_{KK} = \iota_{X^\sharp}\mathrm{d}\omega_{KK}.$$

Since $\omega_{KK}$ is $G$-invariant, $\mathcal{L}_{X^\sharp}\omega_{KK} = 0$, hence $\iota_{X^\sharp}\mathrm{d}\omega_{KK} = 0$ for all $X \in \mathfrak{g}$. Since the vector fields $X^\sharp$ span the tangent space of $\mathbb{O}$ at each point, we have $\mathrm{d}\omega_{KK} = 0$. □

Now, we consider the simplest example of Hamiltonian $G$-space.

Let $(V,\omega)$ be a complex symplectic space, and $G = \mathrm{Sp}(V,\omega)$

**Proposition 29.2.** *The moment map $\mu : V \to \mathfrak{g}^*$ is given by $(\mu(v),X) = \frac{1}{2}\omega(Xv,v)$ for $v \in V$ and $X \in \mathfrak{g}$.*

*Proof.* Set $H_X(v) = (\mu(v),X)$, we only need to check that $\iota_{X^\sharp}\omega = \mathrm{d}H_X$:

$$\begin{aligned} (\mathrm{d}H_X)_v(\delta v) &= \frac{1}{2}\omega(X\delta v, v) + \frac{1}{2}\omega(Xv,\delta v) \\ &= \omega(Xv,\delta v) \\ &= (\iota_{X^\sharp}\omega)_v(\delta v). \end{aligned}$$

Where $\delta v$ is a tangent vector of $V$ at $v$, which can be identified with an element in $V$ itself. □

**Remark 29.3.** Note that $X^\sharp(v) = X \cdot v$ for $v \in V$.

**Lemma 29.4.** *The symplectic group $G$ acts transitively on $V \setminus \{0\}$.*

*Proof.* Use induction on the dimension. □

**Proposition 29.5.** *The moment map $\mu : V \setminus \{0\} \to \mathfrak{g}^*$ is a double cover onto the minimal nilpotent orbit $\mathbb{O}_{min}$ in $\mathfrak{g}^*$.*

*Proof.* Use trace form on $\mathfrak{g}$ to identify $\mathfrak{g}$ with $\mathfrak{g}^*$, then we can check that $\mu(v) = (u \mapsto \omega(v,u)v) \in \mathfrak{g}$. $\mu(v)$ is nilpotent since $\mu(v)^2 = 0$, and the image of $\mu$ is a nonzero nilpotent orbit. It is the minimal nilpotent orbit since $\mu(v)$ has rank 1. $G$ acts transitively on $V \setminus \{0\}$, $\mu$ is a double cover onto $\mathbb{O}_{min}$. □

**Remark 29.6.** $\mathbb{O}_{min} = [2, 1, \cdots, 1]$ and $\pi_1^G(\mathbb{O}_{\min}) = \mathbb{Z}/2\mathbb{Z}$, the cover above is the universal cover of $\mathbb{O}_{min}$.

29.2. **Classical and Quantum Formal Darboux Theorems.** The classical Darboux theorem states that every point in a symplectic manifold has a neighborhood that is symplectomorphic to an open subset of the standard symplectic vector space. Here we investigate a formal algebraic analog of this theorem and its extension to quantization.

**Lemma 29.7.** *Let $A$ be a Poisson algebra and $\mathfrak{m}$ be its maximal ideal with $A/\mathfrak{m} = \mathbb{C}$. Then the Poisson bracket on $A$ induces a skew-symmetric form on $\mathfrak{m}/\mathfrak{m}^2$.*

*Proof.* Define the skew-symmetric form as:

$$\omega(a + \mathfrak{m}^2, b + \mathfrak{m}^2) = \{a, b\} \mod \mathfrak{m}.$$

□

Now let $A = \mathbb{C}[[x_1, \ldots, x_{2n}]]$ so that there is a unique maximal ideal $\mathfrak{m} = (x_1, \ldots, x_{2n})$ and $\mathfrak{m}/\mathfrak{m}^2$ is spanned by $x_i + \mathfrak{m}^2$. Suppose that the Poisson bracket on $A$ induces a non-degenerate skew-symmetric form on $\mathfrak{m}/\mathfrak{m}^2$. So $\{x_i, x_j\} = \omega_{ij} \mod \mathfrak{m}$ for some $\omega_{ij} \in \mathbb{C}$, then the matrix $(\omega_{ij})$ is non-degenerate skew symmetric.

**Theorem 29.8** (Formal Darboux Theorem)**.** *There exists a formal change of coordinates $y_i = y_i(x_1, \ldots, x_{2n}) \in \mathfrak{m}$ such that $\{y_i, y_j\} = \omega_{ij} \in \mathbb{C}$.*

*Proof.* Use induction, suppose that we have found $x_i^{(n)} \in \mathfrak{m}$ such that $\left\{x_i^{(n)}, x_j^{(n)}\right\} \equiv \omega_{ij} \mod \mathfrak{m}^n$ (i.e. $\left\{x_i^{(n)}, x_j^{(n)}\right\} = \omega_{ij} + R_{ij}^{(n)}$ for some $R_{ij}^{(n)} \in \mathfrak{m}^n$). We want to find $\delta_i^{(n)} \in \mathfrak{m}^{n+1}$ such that $\left\{x_i^{(n)} + \delta_i^{(n)}, x_j^{(n)} + \delta_j^{(n)}\right\} \equiv \omega_{ij} \mod \mathfrak{m}^{n+1}$.

Since

$$\begin{aligned}\left\{x_i^{(n)} + \delta_i^{(n)}, x_j^{(n)} + \delta_j^{(n)}\right\} &= \left\{x_i^{(n)}, x_j^{(n)}\right\} + \left\{\delta_i^{(n)}, x_j^{(n)}\right\} + \left\{x_i^{(n)}, \delta_j^{(n)}\right\} + \left\{\delta_i^{(n)}, \delta_j^{(n)}\right\} \\ &\equiv \left\{x_i^{(n)}, x_j^{(n)}\right\} + \left\{\delta_i^{(n)}, x_j^{(n)}\right\} + \left\{x_i^{(n)}, \delta_j^{(n)}\right\} \mod \mathfrak{m}^{n+1}\end{aligned}$$

$$\equiv \omega_{ij} + R_{ij}^{(n)} + \left\{x_i^{(n)}, \delta_j^{(n)}\right\} - \left\{x_j^{(n)}, \delta_i^{(n)}\right\} \mod \mathfrak{m}^{n+1}.$$

We need to solve the following equation for $\delta_i^{(n)}$:

$$\left\{x_i^{(n)}, \delta_j^{(n)}\right\} - \left\{x_j^{(n)}, \delta_i^{(n)}\right\} = -R_{ij}^{(n)} \mod \mathfrak{m}^{n+1}.$$

Note that $\left\{x_i^{(n)}, \delta_j^{(n)}\right\} - \left\{x_j^{(n)}, \delta_i^{(n)}\right\} = \omega_{ik} x_k^{(n)} \frac{\partial \delta_j^{(n)}}{\partial x_k} - \omega_{jk} x_k^{(n)} \frac{\partial \delta_i^{(n)}}{\partial x_k}$, we can solve the above equation by setting

$$\delta_i^{(n)} = -\frac{1}{n+1} \sum_{k,l} \omega^{kl} x_k^{(n)} R_{li}^{(n)}.$$

Set $x_i^{(n+1)} = x_i^{(n)} + \delta_i^{(n)}$, then $y_i = \lim_{n\to\infty} x_i^{(n)}$ satisfies $\{y_i, y_j\} \in \mathbb{C}$. □

**Theorem 29.9** (Quantum Formal Darboux Theorem)**.** *Let $\mathcal{A}_\hbar$ be a formal quantization of the Poisson algebra $A$. Then there are elements $\widehat{y}_i \in \mathcal{A}_\hbar$ such that*

- $\widehat{y}_i \equiv x_i \mod \hbar\mathcal{A}_\hbar$;
- $\frac{1}{\hbar}\left[\widehat{y}_i, \widehat{y}_j\right] \in \mathbb{C}$.

*Proof.* Choose lift of $y_i$ in $\mathcal{A}_\hbar$, denoted by $\widehat{y}_i^{(1)}$. Then $\frac{1}{\hbar}\left[\widehat{y}_i^{(1)}, \widehat{y}_j^{(1)}\right] \equiv \{y_i, y_j\} = \omega_{ij} \mod \hbar$, so we can write $\frac{1}{\hbar}\left[\widehat{y}_i^{(1)}, \widehat{y}_j^{(1)}\right] = \omega_{ij} + \hbar R_{ij}^{(1)}$ for some $R_{ij}^{(1)} \in \mathcal{A}_\hbar$ and $\omega_{ij} \in \mathbb{C}$.

Again, use induction. Suppose that we have found a lift $\widehat{y}_i^{(n)}$ of $y_i$ such that

$$\frac{1}{\hbar}\left[\widehat{y}_i^{(n)}, \widehat{y}_j^{(n)}\right] = \omega_{ij} + \hbar^n R_{ij}^{(n)}$$

for some $R_{ij}^{(n)} \in \mathcal{A}_\hbar$. We want to find $\delta_i^{(n)} \in \hbar^{n+1}\mathcal{A}_\hbar$ such that

$$\frac{1}{\hbar}\left[\widehat{y}_i^{(n)} + \delta_i^{(n)}, \widehat{y}_j^{(n)} + \delta_j^{(n)}\right] = \omega_{ij} + \hbar^{n+1} R_{ij}^{(n+1)}.$$

This is equivalent to solving the following equation for $\delta_i^{(n)}$:

$$\frac{1}{\hbar}\left[\widehat{y}_i^{(n)}, \delta_j^{(n)}\right] - \frac{1}{\hbar}\left[\widehat{y}_j^{(n)}, \delta_i^{(n)}\right] = -\hbar^n R_{ij}^{(n)} \mod \hbar^{n+1}.$$

Note that $\frac{1}{\hbar}\left[\widehat{y}_i^{(n)}, \delta_j^{(n)}\right] - \frac{1}{\hbar}\left[\widehat{y}_j^{(n)}, \delta_i^{(n)}\right] = \omega_{ik}\widehat{y}_k^{(n)} \frac{\partial \delta_j^{(n)}}{\partial \widehat{y}_k} - \omega_{jk}\widehat{y}_k^{(n)} \frac{\partial \delta_i^{(n)}}{\partial \widehat{y}_k}$, we can solve the above equation by setting

$$\delta_i^{(n)} = -\frac{1}{n+1} \sum_{k,l} \omega^{kl} \widehat{y}_k^{(n)} R_{li}^{(n)}.$$

Set $\widehat{y}_i^{(n+1)} = \widehat{y}_i^{(n)} + \delta_i^{(n)}$, then $\widehat{y}_i = \lim_{n\to\infty} \widehat{y}_i^{(n)}$ satisfies $\frac{1}{\hbar}\left[\widehat{y}_i, \widehat{y}_j\right] = \omega_{ij} \in \mathbb{C}$. □

## 29.3. **Localness of Differential Operators.**

Any differential operator on $A$ can extend to localizations and completions.

- Extension to the Localization $S^{-1}A$. Let $A$ be a commutative algebra over a field $k$. A $k$-linear map $L : A \to A$ is a differential operator of order $\leq k$ if for any $a_0, \ldots, a_k \in A$, the iterated commutator vanishes:

$$[\ldots[[L, a_0], a_1], \ldots, a_k] = 0$$

where $[L, a](b) = L(ab) - aL(b)$. We denote the space of such operators as $\mathrm{Diff}^k(A)$.The General Extension FormulaFor any $L \in \mathrm{Diff}^k(A)$, the unique extension $\widetilde{L} \in \mathrm{Diff}^k(S^{-1}A)$ is determined by its action on the reciprocal of an element $s \in S$. The formula uses the iterated adjoint action $\mathrm{ad}_s(L) = [L, s]$:

$$\widetilde{L}\left(\frac{1}{s}\right) = \sum_{j=0}^{k} (-1)^j s^{-(j+1)} \mathrm{ad}_s^j(L)(1)$$

Applying this to a general fraction $\frac{a}{s}$ via the Leibniz-like property for higher-order operators, we get:

$$\widetilde{L}\left(\frac{a}{s}\right) = \sum_{j=0}^{k} (-1)^j s^{-(j+1)} \mathrm{ad}_s^j(L)(a).$$

- Specific Cases for Low OrderOrder 0 (Multiplication by $f \in A$):

$$\widetilde{L}\left(\frac{a}{s}\right) = s^{-1}(f \cdot a) = \frac{fa}{s}$$

Order 1 (Derivations, where $L(1) = 0$):Using the general formula for $j = 0, 1$:

$$\widetilde{L}\left(\frac{a}{s}\right) = s^{-1}L(a) - s^{-2}[L, s](a) = \frac{L(a)}{s} - \frac{L(sa) - sL(a)}{s^2} = \frac{sL(a) - aL(s)}{s^2}$$

Order 2 (e.g., a Laplacian $\Delta$):

$$\widetilde{L}\left(\frac{a}{s}\right) = \frac{\Delta(a)}{s} - \frac{\Delta(sa) - s\Delta(a)}{s^2} + \frac{\Delta(s^2a) - 2s\Delta(sa) + s^2\Delta(a)}{s^3}.$$

- Extension to the Completion $\widehat{A}$. Let $I \subset A$ be an ideal. The completion is the inverse limit $\widehat{A} = \varprojlim A/I^n$.The Continuity PrecisionA differential operator $L$ of order $k$ is strictly continuous with respect to the $I$-adic topology because it satisfies:

$$L(I^n) \subseteq I^{n-k} \quad \text{for all } n \geq k$$

This filtration ensures that for any Cauchy sequence $(a_n)$ in $A$, the sequence $(L(a_n))$ is also Cauchy in the $I$-adic sense. The extension $\widehat{L} : \widehat{A} \to \widehat{A}$ is defined by taking the limit of the action on the truncated sequences:

$$\widehat{L}\left(\lim_{n\to\infty} a_n\right) = \lim_{n\to\infty} L(a_n)$$

Because $L$ is a polynomial-type operator (it "sees" only a finite neighborhood of any point algebraically), this limit is always well-defined and unique.

- The Poisson Bracket CaseA Poisson bracket $\{\cdot, \cdot\}$ is a bilinear map that is a derivation (order 1 operator) in each slot.On $S^{-1}A$: Since $L_f = \{f, \cdot\}$ is a derivation, the extension is forced by the first-order formula derived above. For $a/s, b/t \in S^{-1}A$:

$$\left\{\frac{a}{s}, \frac{b}{t}\right\} = \frac{1}{s^2t^2}\left(st\{a, b\} - at\{s, b\} - bs\{a, t\} + ab\{s, t\}\right)$$

On $\widehat{A}$: Since the bracket is a derivation in each variable, it is continuous in each variable. By the universal property of completions of bilinear maps, it extends uniquely:

$$\{\widehat{a}, \widehat{b}\}_{\widehat{A}} = \lim_{n\to\infty} \{a_n, b_n\}_A$$

This ensures that $\widehat{A}$ inherits the Poisson algebra structure from $A$ without any ambiguity.

## 30. Normality of Nilpotent Orbit Closures

2026-04-24

### Normality of Closure of Nilpotent Orbits

**Theorem 30.1** (Main theorem)**.** *For a nilpotent orbit $\mathbb{O}$ in a complex semisimple Lie algebra $\mathfrak{g}$, its closure $\overline{\mathbb{O}}$ is normal if and only if $\mathbb{C}[\mathbb{O}] = \mathbb{C}[\overline{\mathbb{O}}]$.*

**Lemma 30.2** (Serre's criterion for normality)**.** *$A$ is a commutative Noetherian ring. Then $A$ is normal if and only if it satisfies the following two conditions:*

($R_1$) *For any prime ideal $\mathfrak{p}$ of $A$ with $\mathrm{ht}(\mathfrak{p}) \leq 1$, the localization $A_\mathfrak{p}$ is a regular local ring.*
($S_2$) *For any prime ideal $\mathfrak{p}$ of $A$, we have $\mathrm{depth}(A_\mathfrak{p}) \geq \min\{2, \mathrm{ht}(\mathfrak{p})\}$.*

*Proof of Main Theorem.* ($\Rightarrow$) If $\overline{\mathbb{O}}$ is normal, then by Serre's criterion, we get information about the depth inequality, and then we can apply the algebraic Hartogs theorem [Har77, Chapter III, Exercise 3.5] to conclude that $\mathbb{C}[\mathbb{O}] = \mathbb{C}[\overline{\mathbb{O}}]$.

($\Leftarrow$) If $\mathbb{C}[\mathbb{O}] = \mathbb{C}[\overline{\mathbb{O}}]$, then $\mathbb{C}[\overline{\mathbb{O}}]$ is a domain and integrally closed in its field of fractions, Since $\mathbb{C}[\mathbb{O}]$ is smooth. Hence $\mathbb{C}[\overline{\mathbb{O}}]$ is normal. □

## 31. Filtered Quantization of the Nilpotent Cone

2026-04-25

[Los22b]

**Lemma 31.1.** *Let $A$ be a filtered algebra, $I \subseteq A$ an ideal, we have a short exact sequence:*

$$0 \to \mathrm{gr}\, I \to \mathrm{gr}\, A \to \mathrm{gr}(A/I) \to 0.$$

*Proof.* $I$ has an induced filtration $I_{\leq n} = I \cap A_{\leq n}$, and $\mathrm{gr}\, I = \bigoplus_n I_{\leq n}/I_{\leq n-1}$ is a graded ideal of $\mathrm{gr}\, A$. It is easy to see that the map $\mathrm{gr}\, A \to \mathrm{gr}(A/I)$ is surjective, and $\mathrm{gr}\, I \to \mathrm{gr}\, A$ is injective. For an element $a + A_{\leq i-1} \in \mathrm{gr}_i A$, then

$$\begin{aligned}
& a + A_{\leq i-1} \mapsto 0 \in \mathrm{gr}_i(A/I) \\
\Leftrightarrow\ & \overline{a} \in (A/I)_{\leq i-1} \\
\Leftrightarrow\ & \exists\ a' \in A_{\leq i-1} \text{ such that } a - a' \in I \\
\Leftrightarrow\ & a + A_{\leq i-1} \in I_{\leq i}/I_{\leq i-1}.
\end{aligned}$$

□

**Remark 31.2.** If $I = \langle f_1, \cdots, f_n \rangle$, then $\langle \sigma(f_1), \cdots, \sigma(f_n) \rangle \subseteq \mathrm{gr}\, I$.

**Proposition 31.3.** *$A$ is a $\mathbb{Z}_{\geq 0}$ graded Noetherian Poisson algebra with $\deg\{,\} = -d < 0$, $\mathcal{A}$ is its filtered quantization, and $\mathcal{I} = \langle \widehat{f}_1, \cdots, \widehat{f}_n \rangle$ ($\widehat{f}_i \in \mathcal{A}_{d_i}$), $f_i = \widehat{F}_i + \mathcal{A}_{\leq d_i - 1}$, with the following conditions:*

- *$f_1, \cdots, f_n$ is a regular sequence in $A$;*
- *$\left[\widehat{f}_i, \widehat{f}_j\right] = \sum_{i=1}^r \widehat{c}_{ij}^k \widehat{f}_k$, with $\widehat{c}_{ij}^k \in \mathcal{A}_{\leq d_i + d_j - d - d_k}$.*

*Then $\mathrm{gr}\, \mathcal{I} = \langle f_1, \cdots, f_n \rangle$.*

*Proof.* We only need to check that $\operatorname{gr}\mathcal{I} \subseteq \langle f_1, \cdots, f_n\rangle$. Assume on the contrary that there exists a positive integer $e$ and $\widehat{b_i} \in \mathcal{A}_{\leq e-d_i}$ such that $\sigma(\sum_{i=1}^n \widehat{b_i}\widehat{f_i}) \notin \langle f_1, \cdots, f_n\rangle$. Also assume $e$ is the smallest integer with this property.

Note that $\sum_{i=1}^n b_i f_i \in A_e$ must be zero, otherwise $\sigma(\sum_{i=1}^n \widehat{b_i}\widehat{f_i}) = \sum_{i=1}^n b_i f_i \in \langle f_1, \cdots, f_n\rangle$. So by the vanishing of the first Koszul homology, we can find $b_{ij} \in A$ such that $b_i = \sum_{j=1}^n b_{ij} f_j$ and $b_{ij} = -b_{ji}$ for $i \neq j$. By restriction to the degree $d_i$ we can assume that $b_{ij} \in A_{\leq e-d_i-d_j}$, take lifts $\widehat{b}_{ij} \in \mathcal{A}_{\leq e-d_i-d_j}$, such that $\widehat{b}_{ij} = -\widehat{b}_{ji}$. Then $\widehat{b}_i' := \widehat{b}_i - \sum_{j=1}^n \widehat{b}_{ij}\widehat{f}_j \in \mathcal{A}_{<e-d_i}$.

$$\begin{aligned}
\sum_{i=1}^n \widehat{b}_i\widehat{f}_i &= \sum_{i=1}^n \widehat{b}_i'\widehat{f}_i + \sum_{i,j=1}^n \widehat{b}_{ij}\widehat{f}_j\widehat{f}_i \\
&= \sum_{i=1}^n \widehat{b}_i'\widehat{f}_i + \sum_{j<k} \widehat{b}_{jk}[\widehat{f}_j, \widehat{f}_k] \\
&= \sum_{i=1}^n \widehat{b}_i'\widehat{f}_i + \sum_{j<k} \widehat{b}_{jk} \sum_{i=1}^n \widehat{c}_{jk}^i \widehat{f}_i \\
&= \sum_{i=1}^n \left(\widehat{b}_i' + \sum_{j<k} \widehat{b}_{jk}\widehat{c}_{jk}^i\right) \widehat{f}_i.
\end{aligned}$$

But $\widehat{b}_i' + \sum_{j<k} \widehat{b}_{jk}\widehat{c}_{jk}^i \in \mathcal{A}_{<e-d_i}$, which is a contradiction to the minimality of $e$. □

A discovery: Harish-Chandra's isomorphism can be viewed as a quantized version of the Chevalley restriction theorem.

**Theorem 31.4** (Chevalley restriction theorem)**.** *Let $\mathfrak{g}$ be a complex reductive Lie algebra, $\mathfrak{h}$ be a Cartan subalgebra, $W$ be the Weyl group. Then the restriction map $\mathbb{C}[\mathfrak{g}]^G \to \mathbb{C}[\mathfrak{h}]^W$ is an isomorphism.*

**Theorem 31.5** (Harish-Chandra isomorphism)**.** *Let $\mathfrak{g}$ be a complex reductive Lie algebra, $\mathfrak{h}$ be a Cartan subalgebra, $W$ be the Weyl group. Then there is an isomorphism $Z(\mathfrak{g}) = \mathcal{U}(\mathfrak{g})^G \to \mathbb{C}[\mathfrak{h}]^W$.*

**Corollary 31.6.** *$\mathfrak{h}/W$ parametrizes both the filtered Poisson deformations and the filtered quantizations of $\mathcal{N}$.*

Some phenomena of separation of variables:

- $\mathbb{C}[\mathfrak{g}] = \mathbb{C}[\mathfrak{g}]^G \otimes_{\mathbb{C}} \mathcal{H}$;
- $\mathbb{C}[\mathfrak{h}] = \mathbb{C}[\mathfrak{h}]^W \otimes_{\mathbb{C}} \mathcal{H}$;
- $\mathbb{C}[V] = \mathbb{C}[V]^{\mathrm{SO}(V)} \otimes_{\mathbb{C}} \mathcal{H}(V) = \mathbb{C}[r^2] \otimes_{\mathbb{C}} \mathcal{H}(V)$. (Spherical harmonics)

*Proof of the first separation of variables.* There is an isomorphism $\mathbb{C}[\mathfrak{g}] = \mathbb{C}[\mathfrak{g}]_+^G\mathbb{C}[\mathfrak{g}] \oplus \mathcal{H}$.

Choose a homogeneous vector space basis $\overline{b}_i$ of $\mathbb{C}[\mathfrak{g}]/(\mathbb{C}[\mathfrak{g}]_+^G\mathbb{C}[\mathfrak{g}])$, and let $b_i$ be a homogeneous lift of $\overline{b}_i$ in $\mathcal{H}$. Then $b_i$ is a vector space basis of $\mathcal{H}$. By graded Nakayama's lemma, $b_i$ generates $\mathbb{C}[\mathfrak{g}]$ as a $\mathbb{C}[\mathfrak{g}]^G$-module, we claim that $b_i$ is a free basis of $\mathbb{C}[\mathfrak{g}]$ as a $\mathbb{C}[\mathfrak{g}]^G$-module, so that the following is an isomorphism

$$\mathbb{C}[\mathfrak{g}]^G \otimes_{\mathbb{C}} \mathcal{H} \to \mathbb{C}[\mathfrak{g}], \quad f \otimes b \mapsto fb.$$

If not, then there exists $r_i \in \mathbb{C}[\mathfrak{g}]^G$ such that $\sum_i r_i b_i = 0$, and not all $r_i$ are zero. Projection to $\mathcal{H}$ we have $\sum_i \overline{r}_i \overline{b}_i = 0$, since $\overline{b}_i$ is a vector space basis, so $\overline{r}_i = 0$ for all $i$ which means $r_i \in \mathbb{C}[\mathfrak{g}]^G_+ \mathbb{C}[\mathfrak{g}] = \langle f_1, \cdots, f_n \rangle$, assume that $r_i = \sum_j r_{ij} f_j$ then

$$0 = \sum_i r_i b_i = \sum_{i,j} r_{ij} f_j b_i = \sum_j f_j \underbrace{\left( \sum_i r_{ij} b_i \right)}_{a_j}.$$

Since $f_i$ form a regular sequence, the first Koszul homology vanishes, which means that there exists $s_{jk}$ such that $a_j = \sum_k s_{jk} f_k$ and $s_{jk} = -s_{kj} \in \mathbb{C}[\mathfrak{g}]$. So we have

$$\sum_k s_{jk} f_k = a_j = \sum_i r_{ij} b_i.$$

This implies that $r_{ij} \in \langle f_1, \cdots, f_n \rangle$, suppose $r_{ij} = \sum_k r_{ijk} f_k$, then

$$0 = \sum_{i,j} r_{ij} f_j b_i = \sum_{i,j,k} r_{ijk} f_k f_j b_i = \sum_{j,k} f_j f_k \underbrace{\left( \sum_i r_{ijk} b_i \right)}_{a_{jk}}.$$

Again by the vanishing of the second Koszul homology, there exists $s_{jkl}$ such that $a_{jk} = \sum_l s_{jkl} f_l$ and $s_{jkl} = -s_{jlk}$, so we have

$$\sum_l s_{jkl} f_l = a_{jk} = \sum_i r_{ijk} b_i.$$

This implies that $r_{ijk} \in \langle f_1, \cdots, f_n \rangle$, and we can continue this process to conclude that $r_i \in \langle f_1, \cdots, f_n \rangle^m$ for all $m$, which implies that $r_i = 0$. □

## 32. Divisors and Picard Groups

2026-04-26

Some relations between Picard groups, Cartier divisors, and Weil divisors.

Pic is the group of isomorphism classes of line bundles. Cl is the group of Weil divisors modulo linear equivalence. CaCl is the group of Cartier divisors modulo linear equivalence.

- If $X$ is integral, then $\mathrm{Pic}(X) \cong \mathrm{CaCl}(X)$.
- If $X$ is integral and normal, then $\mathrm{Pic}(X) \hookrightarrow \mathrm{Cl}(X)$.
- If $X$ is integral and locally factorial/Smooth, then $\mathrm{Pic}(X) \cong \mathrm{Cl}(X)$.

If $X$ is integral and normal, by Serre's criterion for normality, the singular locus of $X$ has codimension at least 2, hence $\mathrm{Cl}(X) \cong \mathrm{Cl}(X^{reg}) \cong \mathrm{Pic}(X^{reg})$.

## 33. Symplectic Leaves of Nilpotent Orbits

2026-04-27

[Los22b]

We prove that for a nilpotent orbit $\mathbb{O}$ in a complex semisimple Lie algebra $\mathfrak{g}$,

$$\overline{\mathbb{O}}^{reg} = \mathbb{O}.$$

**Lemma 33.1.** *$\overline{\mathbb{O}}^{reg}$ is symplectic.*

*Proof.* Since $\overline{\mathbb{O}}$ is Poisson, it has a Poisson bivector field $\pi \in \Gamma(\overline{\mathbb{O}}, \wedge^2 T_{\overline{\mathbb{O}}})$, and $\pi$ is non-degenerate on $\mathbb{O}$, and we know that the zero locus of a section of a line bundle on the smooth variety $\overline{\mathbb{O}}^{reg}$ has codimension at most 1, but $\operatorname{codim}_{\overline{\mathbb{O}}^{reg}}(\{\pi = 0\}) \geq 2$, so $\pi$ is non-vanishing on $\overline{\mathbb{O}}^{reg}$. □

Let $X$ be a Poisson variety, and $X_1 = X \backslash X^{reg}$, $X_{k+1} = X_k \backslash X_k^{reg}$. We say that $X$ has finitely many symplectic leaves if each $X_k^{reg}$ is symplectic. For a symplectic leaf, we mean an irreducible (connected) component of $X_k^{reg}$ for some $k$.

**Lemma 33.2.** *Symplectic leaves are in one-to-one correspondence with prime Poisson ideals of* $\mathbb{C}[X]$.

**Lemma 33.3.** $\overline{\mathbb{O}}$ *has finitely many symplectic leaves, and the symplectic leaves are exactly the nilpotent orbits contained in* $\overline{\mathbb{O}}$.

*Proof.* The second statement is due to the fact that Poisson bracket is generated by Lie bracket. □

**Lemma 33.4.** *A smooth symplectic variety has no proper Poisson subvarieties, hence it is a symplectic leaf.*

*Proof.* Suppose $X$ is a smooth symplectic variety, and let $x \in X$ be a point.

$$T_x X \xrightarrow{\pi_x} T_x^* X$$

is an isomorphism.

Under this isomorphism $H_f$ corresponds to $\mathrm{d}f$, so we can see that $\operatorname{span}\{H_f \mid f \in \mathbb{C}[X]\} = T_x X$, but Poisson subvariety must tangent to all Hamiltonian vector fields, so it must be the whole $X$. □

*Proof of* $\overline{\mathbb{O}}^{reg} = \mathbb{O}$. Since $\overline{\mathbb{O}}$ has finitely many symplectic leaves, $\overline{\mathbb{O}}^{reg}$ is a symplectic leaf, hence it is a nilpotent orbit. Since $\mathbb{O} \subseteq \overline{\mathbb{O}}^{reg}$, we have $\overline{\mathbb{O}}^{reg} = \mathbb{O}$. □

**Remark 33.5.** The argument above works for any coadjoint orbits in a complex Lie algebra, not necessarily semisimple.

**Remark 33.6.** For a nilpotent orbit cover $\widetilde{\mathbb{O}}$, we have a finite surjective morphism

$$\pi : \operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}]) \to \overline{\mathbb{O}},$$

$\operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$ has finitely many symplectic leaves, and we also know that $\operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])^{reg}$ is symplectic, it has no proper Poisson subvarieties, hence it is a symplectic leaf. By in this situation, symplectic leaves are not in one-to-one correspondence with $G$-orbits any more, so $\widetilde{\mathbb{O}}$ may properly contained in $\operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])^{reg}$.

Actually in the case of exceptional Lie algebras, there exist some nilpotent orbits $\mathbb{O}$ such that $\operatorname{codim}_{\overline{\mathbb{O}}}(\overline{\mathbb{O}} \backslash \mathbb{O}) = 2$ but $\operatorname{codim}_X(X \backslash X^{reg}) \geq 4$, where $X = \operatorname{Spec}(\mathbb{C}[\mathbb{O}])$.

## 34. Richardson Orbits and Symplectic Resolutions

2026-04-27

34.1. **A fundamental fact I often forget.** $G$ be a complex connected reductive algebraic, $\mathcal{B}$ be the flag variety of $G$, then

$$T\mathcal{B} = \{(b, \xi) \,|\, b \in \mathcal{B}, \quad \xi \in \mathfrak{g}/b\}.$$

For any $X \in \mathfrak{g}$, the vector field $X^\sharp$ on $\mathcal{B}$ generated by $X$-action is given by

$$X^\sharp(b) = (b, X + b).$$

We also have the cotangent bundle

$$T^*\mathcal{B} = \{(b, \xi) \,|\, b \in \mathcal{B}, \quad \xi \in b^\perp \subseteq \mathfrak{g}^*\}.$$

For a symmetric subgroup $K$ of $G$, we have:

$$T^*_K\mathcal{B} = \{(b, \xi) \,|\, b \in \mathcal{B}, \quad \xi \in b^\perp \cap \mathfrak{k}^\perp\}.$$

We have the following Cartesian diagram:

$$\begin{array}{ccc} T^*_K\mathcal{B} & \hookrightarrow & T^*\mathcal{B} \\ \downarrow & & \downarrow \\ \mathfrak{k}^\perp & \hookrightarrow & \mathfrak{g}^* \end{array}$$

34.2. **Richardson orbits.** For a parabolic $P \subseteq G$, $T^*(G/P) = T^*\mathcal{P}$ has a unique open $G$-orbit.

Since there is a bijection between $P$-orbits in $\mathfrak{u}$ and $G$-orbits in $T^*\mathcal{P}$, the unique open $G$-orbit in $T^*\mathcal{P}$ corresponds to the unique open $P$-orbit in $\mathfrak{u}$.

We also have the moment map $\mu : T^*\mathcal{P} \to \mathfrak{g}^*$, the image of $\mu$ is the closure of a nilpotent orbit $\mathbb{O}_\mathcal{P}$, which is called the Richardson orbit associated to $\mathcal{P}$.

We have the following Cartesian diagram:

$$\begin{array}{ccc} \widetilde{\mathbb{O}}_\mathcal{P} & \hookrightarrow & T^*\mathcal{P} \\ \downarrow{\scriptstyle \mu} & & \downarrow{\scriptstyle \mu} \\ \mathbb{O}_\mathcal{P} & \hookrightarrow & \overline{\mathbb{O}}_\mathcal{P} \end{array}$$

where $\widetilde{\mathbb{O}}_\mathcal{P}$ is the unique open $G$-orbit in $T^*\mathcal{P}$, it is also a finite cover of $\mathbb{O}_\mathcal{P}$.

When the orbit $\mathbb{O}_\mathcal{P}$ is even, the moment map $\mu$ is just the Jacobson-Morozov resolution of $\overline{\mathbb{O}}_\mathcal{P}$, and $\widetilde{\mathbb{O}}_\mathcal{P} = \mathbb{O}_\mathcal{P}$. So all the even orbits are Richardson orbits, and they admit symplectic resolutions.

## 35. Equivariant Structures on the Structure Sheaf

2026-04-29

Let $\Gamma$ be a discrete group acting on a complex algebraic variety $X$, we want to classify the $\Gamma$-equivariant structures on the structure sheaf $\mathcal{O}_X$. (Note that in the case of discrete group actions, the $\Gamma$-equivariant structure on a sheaf $\mathcal{F}$ is just a collection of isomorphisms $\phi_\gamma : \mathcal{F} \to \gamma^*\mathcal{F}$ for each $\gamma \in \Gamma$, such that the cocycle condition holds.[Ach21])

**Lemma 35.1.** *There is a natural $\Gamma$-equivariant structure on $\mathcal{O}_X$ given by the translation of functions, i.e. for $g \in \Gamma$, the isomorphism $T_\gamma : \mathcal{O}_X \to \gamma^*\mathcal{O}_X$ is given by $T_\gamma(f) = f(\gamma^{-1}(\cdot))$.*

**Theorem 35.2.** *The isomorphism classes of $\Gamma$-equivariant structures on $\mathcal{O}_X$ are in one-to-one correspondence with $H^1(\Gamma, \mathcal{O}_X^*(X))$.*

*Proof.* Let $\phi_\gamma : \mathcal{O}_X \to \gamma^*\mathcal{O}_X$ be a $\Gamma$-equivariant structure on $\mathcal{O}_X$, compose with the inverse translation isomorphism $T_\gamma^{-1} : \gamma^*\mathcal{O}_X \to \mathcal{O}_X$, we get an automorphism $\psi_\gamma = T_\gamma^{-1} \circ \phi_\gamma$ of $\mathcal{O}_X$, which is given by multiplication by some invertible function $f_\gamma \in \mathcal{O}_X^*(X)$. At this time, $\phi$ is given by the explicit formula

$$\begin{aligned} \phi_\gamma : \quad \mathcal{O}_X(U) &\longrightarrow \mathcal{O}_X(\gamma \cdot U) \\ f &\longmapsto f_\gamma(\gamma^{-1}(\cdot)) f(\gamma^{-1}(\cdot)) \end{aligned}$$

The compatibility condition explicitly says that the following two morphisms are equal:

$$\mathcal{O}_X(U) \xrightarrow{\phi_\gamma} \mathcal{O}_X(\gamma \cdot U) \xrightarrow{\phi_\delta} \mathcal{O}_X(\delta\gamma \cdot U)$$

$$f \longmapsto f_\gamma(\gamma^{-1}(\cdot)) f(\gamma^{-1}(\cdot)) \longmapsto f_\delta(\delta^{-1}(\cdot)) f_\gamma(\gamma^{-1}\delta^{-1}(\cdot)) f(\gamma^{-1}\delta^{-1}(\cdot))$$

$$\mathcal{O}_X(U) \xrightarrow{\phi_{\delta\gamma}} \mathcal{O}_X(\delta\gamma \cdot U)$$

$$f \longmapsto f_{\delta\gamma}((\delta\gamma)^{-1}(\cdot)) f((\delta\gamma)^{-1}(\cdot))$$

This means that $f_{\delta\gamma} = f_\delta \cdot (\delta \cdot f_\gamma)$, so $f_\gamma$ is a 1-cocycle of $\Gamma$ with coefficients in $\mathcal{O}_X^*(X)$.

Conversely, given a 1-cocycle $f_\gamma$, we can define a $\Gamma$-equivariant structure on $\mathcal{O}_X$ by the explicit formula above, and the compatibility condition is automatically satisfied.

Finally, two $\Gamma$-equivariant structures $\phi$ and $\phi'$ are isomorphic if and only if there exists an automorphism $\psi : \mathcal{O}_X \to \mathcal{O}_X$ such that the following diagram commutes for all $\gamma \in \Gamma$:

$$\begin{array}{ccc} \mathcal{O}_X & \xrightarrow{\phi_\gamma} & \gamma^*\mathcal{O}_X \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \gamma^*\psi} \\ \mathcal{O}_X & \xrightarrow{\phi'_\gamma} & \gamma^*\mathcal{O}_X \end{array}$$

Since the automorphism $\psi$ is given by multiplication by some invertible function $h \in \mathcal{O}_X^*(X)$, the above diagram commutes if and only if $\frac{f'_\gamma}{f_\gamma} = \frac{h(\gamma^{-1}(\cdot))}{h(\cdot)}$, which means that $f_\gamma$ and $f'_\gamma$ are cohomologous. □

**Corollary 35.3.** *If $\mathcal{O}_X^*(X) = \mathbb{C}^*$, then all the $\Gamma$-equivariant structures on $\mathcal{O}_X$ is given by characters in $\mathfrak{X}(\Gamma)$: $\mathbb{C}_\chi \mapsto \mathbb{C}_\chi \otimes \mathcal{O}_X$*

*Proof.*

$$\mathfrak{X}(\Gamma) = \mathrm{Hom}(\Gamma, \mathbb{C}^*) \cong H^1(\Gamma, \mathcal{O}_X^*(X)).$$

□

## 36. Principal Nilpotent Elements

2026-04-30

Let $\mathfrak{g}$ be a complex reductive Lie algebra, fix a Borel $\mathfrak{b}$ and a Cartan $\mathfrak{h} \subseteq \mathfrak{b}$, we have the root space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$, where $\Phi$ is the set of roots. $\Pi \subseteq \Phi^+$ be the set of simple roots, for each $\alpha \in \Pi$, we can choose a nonzero root vector $X_\alpha \in \mathfrak{g}_\alpha$, then

$$X = \sum_{\alpha \in \Pi} X_\alpha$$

lies in the principal nilpotent orbits $\mathbb{O}_{prin}$.

Let $h = \sum_{\alpha\in\Phi^+} \check{\alpha}$

**Lemma 36.1.** $[h, X] = 2X$

*Proof.* For each $\alpha \in \Pi$, we have $\alpha(h) = 2$, since $s_\alpha(h) = h - 2\check{\alpha}$. □

Suppose that $h = \sum_{\alpha\in\Pi} r_\alpha\check{\alpha}$. For each $\alpha \in \Pi$ we can find $Y_\alpha \in \mathfrak{g}_{-\alpha}$ such that $[X_\alpha, Y_\alpha] = r_\alpha\check{\alpha}$. Set

$$Y = \sum_{\alpha\in\Pi} Y_\alpha,$$

then we have $[h, Y] = -2Y$ and $[X, Y] = h$, so $\{X, h, Y\}$ is an $\mathfrak{sl}_2$-triple.

We have a weight decomposition $\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_i$, where $\mathfrak{g}_i = \{x \in \mathfrak{g} \mid [h, x] = ix\}$.

**Lemma 36.2.** $\mathfrak{g}_0 = \mathfrak{h}$, $\mathfrak{g}_{\geq 2} = \mathfrak{n}$.

Suppose $\mathbb{O} = G \cdot X$. Then the Jacobson-Morozov resolution of $\overline{\mathbb{O}}$ is just the Springer resolution $T^*\mathcal{B} \to \overline{\mathbb{O}}_{prin} = \mathcal{N}$, so $\mathbb{O}$ is the principal nilpotent orbit.

## 37. Conical Symplectic Singularities

2026-05-15

We summarize some facts about conical symplectic singularities and their partial Poisson resolutions, with emphasis on the case of nilpotent covers of complex semisimple Lie algebras. The reference is [Los22b].

All the things are on the classical side, we will discuss the quantum side later.

### 37.1. Conical symplectic singularity.

**Definition 37.1.** For normal irreducible Poisson variety $X$, we say that $X$ has symplectic singularity if the following conditions are satisfied:

- The Poisson structure on $X^{reg}$ is nondegenerate denoted by $\omega^{reg}$;
- There is a resolution of singularity $\pi : Y \to X$ such that $\pi^*(\omega^{reg}) \in \Gamma(\pi^{-1}(X^{reg}); \Omega^2_Y)$ extends to a 2-form $\widetilde{\omega}$ in $\Gamma(Y; \Omega^2_Y)$. ($\widetilde{\omega}$ is a closed 2-form, but possibly degenerate)

**Definition 37.2.** A conical symplectic singularity is a normal irreducible Poisson variety with symplectic singularity which is also a conic variety.

**Proposition 37.3.** *For a nilpotent orbit $\mathbb{O}$, $\mathrm{Spec}(\mathbb{C}[\mathbb{O}])$ is a conic symplectic singularity.*

*Proof.* $X^{reg}$ is symplectic since the boundary has codimension at least 2. Use the Jacobson-Morozov resolution, we can construct the extension of symplectic form explicitly. □

**Proposition 37.4.** *For a nilpotent cover $\widetilde{\mathbb{O}}$, $\mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$ is a conic symplectic singularity.*

*Proof.* This follows from the previous proposition and Namikawa's results on symplectic singularities. □

37.2. **Partial Poisson resolution, $\mathbb{Q}$-factorial terminalization.**

**Definition 37.5.** Let $X$ be a normal Poisson variety. A morphism $\pi : Y \to X$ is called partial Poisson resolution if:

- It's proper and birational;
- It's Poisson.

**Proposition 37.6.** *If $X$ has symplectic singularity then so does $Y$.*

**Proposition 37.7.** *Let $X$ be a conical symplectic singularity, $\pi : Y \to X$ is a partial Poisson resolution with $\operatorname{codim}_Y(Y^{sing}) \geq 4$. Then $\mathbb{C}[Y^{reg}] \simeq \mathbb{C}[Y] \simeq \mathbb{C}[X]$ and $H^1(Y^{reg}; \mathcal{O}_Y) = H^2(Y^{reg}; \mathcal{O}_Y) = 0$.*

*Proof.* The first assertion is due to symplectic singularities have rational singularities, and Hartogs theorem.

The second assertion due to the short exact sequence:

$$R\Gamma_{Y^{sing}} \to \mathrm{id} \to R\iota_*\iota^* \xrightarrow{+1},$$

where $\iota : Y^{reg} \to Y$. And the fact that symplectic singularity is Cohen-Macaulay. □

**Theorem 37.8.** *Let $X$ be a conical symplectic singularity. For a partial Poisson resolution $\pi : Y \to X$ the following are equivalent:*

- *$\pi$ is maximal Poisson resolution;*
- *$Y$ is $\mathbb{Q}$-factorial and terminal.*

In this case, $Y$ is called a $\mathbb{Q}$-factorial terminalization of $X$.

For a nilpotent cover $\widetilde{\mathbb{O}}$, suppose that there exists a nilpotent cover $\widetilde{\mathbb{O}}_L$ of a Levi subalgebra such that $\mathrm{Bind}_L^G(\widetilde{\mathbb{O}}_L) = \widetilde{\mathbb{O}}$. Set $X = \mathrm{Spec}(\mathbb{C}[\mathbb{O}])$, $X_L = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}_L])$, and $Y = G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \,|\, \alpha|_{\mathfrak{l}} = \mu(x)\}$.

**Proposition 37.9.** *The moment map $\mu : Y \to \mathfrak{g}^*$ factor through $X \to \mathfrak{g}^*$, and $\mu : Y \to X$ is a partial Poisson resolution.*

**Proposition 37.10.** *$Y$ is $\mathbb{Q}$-factorial and terminal, if and only if $X_L$ is.*

**Theorem 37.11.** *$X_L$ is $\mathbb{Q}$-factorial and terminal if and only if $\widetilde{\mathbb{O}}_L$ is birationally rigid.*

37.3. **Poisson deformation.**

**Definition 37.12.** Let $X$ be a conical symplectic singularity and $B$ is a finitely generated positively graded algebra. A graded Poisson deformation of $X$ over $\mathrm{Spec}(B)$ is a Poisson scheme $X_{\mathrm{Spec}(B)}$ over $\mathrm{Spec}(B)$ and a Poisson isomorphism $X_0 \xrightarrow{\sim} X$ such that:

- $X_{\mathrm{Spec}(B)}$ is flat over $\mathrm{Spec}(B)$;
- $X_{\mathrm{Spec}(B)}$ is graded ($\exists \quad \mathbb{C}^\times \circlearrowleft X_{\mathrm{Spec}(B)}$), such that $\{,\}$ has degree $-d$ (which is the same as $X$).

**Remark 37.13.** $B$ lies in the Poisson center of $X_{\mathrm{Spec}(B)}$, but I don't know if it is contained in the definition.

In the case of nilpotent cover, we can construct a graded Poisson deformation of $Y$:

$$Y_{\mathfrak{z}} := G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \,|\, \alpha|_{\mathfrak{l}} - \mu(x) \in \mathfrak{z}\}.$$

Here $\mathfrak{z} = (\mathfrak{l}/[\mathfrak{l},\mathfrak{l}])^*$

**Proposition 37.14.** *For any $\lambda \in \mathfrak{z}$, the image of $\mu : Y_\lambda \to \mathfrak{g}^*$ is the closure of a coadjoint orbit $\mathbb{O}_\lambda$, and*

$$\mu : Y_\lambda \to \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}_\lambda}])$$

*is a partial Poisson resolution.*

**Theorem 37.15** (Namikawa: Universal Poisson Deformation)**.** *Set $\mathfrak{h}_X := H^2(Y^{reg};\mathbb{C})$ for a $\mathbb{Q}$-factorial terminalization of $X$, then there exist a reflection group $W_X \subseteq \mathrm{GL}(\mathfrak{h}_X)$ and a graded Poisson deformation $X_{\mathfrak{h}_X/W_X}$ with the following universal property:*

*For any graded Poisson deformation $X_{\mathrm{Spec}(B)}$ of $X$ there exist a unique graded algebra homomorphism $\mathbb{C}[\mathfrak{h}_X]^{W_X} \to B$ such that there is an isomorphism of graded Poisson deformations*

$$\mathrm{Spec}(B) \times_{\mathfrak{h}_X/W_X} X_{\mathfrak{h}_X/W_X} \xrightarrow{\sim} X_{\mathrm{Spec}(B)}.$$

$$\begin{array}{ccc} \mathrm{Spec}(B) \times_{\mathfrak{h}_X/W_X} X_{\mathfrak{h}_X/W_X} & \longrightarrow & X_{\mathfrak{h}_X/W_X} \\ \downarrow & & \downarrow \\ \mathrm{Spec}(B) & \xrightarrow{\exists!} & \mathfrak{h}_X/W_X \end{array}$$

In the case of nilpotent cover $X = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$, take birationally rigid nilpotent cover $\widetilde{\mathbb{O}}_L$ such that $\mathrm{Bind}_L^G(\widetilde{\mathbb{O}}_L) = \widetilde{\mathbb{O}}$, we have a Poisson deformation $Y_{\mathfrak{z}}$ of $Y = G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \,|\, \alpha|_{\mathfrak{l}} = \mu(x)\}$ then $X_{\mathfrak{z}} = \mathrm{Spec}(\mathbb{C}[Y_{\mathfrak{z}}])$ is a graded deformation of $X$, by Namikawa's theorem, there is a unique morphism $\mathfrak{z} \to \mathfrak{h}_X/W_X$

the universal Poisson deformation of $X$ can be constructed as $\mathrm{Spec}(\mathbb{C}[Y_{\mathfrak{z}}])/W_X = X_{\mathfrak{z}}/W_X$.

$$\begin{array}{ccc} Y & \hookrightarrow & Y_{\mathfrak{z}} \\ \downarrow & & \downarrow \\ & & X_{\mathfrak{z}} \\ & & \downarrow \\ X & \hookrightarrow & X_{\mathfrak{z}}/W_X \end{array}$$

37.4. **Summarize.**

TABLE 1. Conical Symplectic Singularities: General Theory vs. Nilpotent Covers

| | **General Theory** | **Nilpotent Cover Example** |
|---|---|---|
| **Singularity** | Normal irreducible Poisson variety $X$ with symplectic singularities | $X = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$ for a nilpotent cover $\widetilde{\mathbb{O}}$. |
| **Partial Resolution** | Proper birational Poisson morphism $\pi : Y \to X$. | Generalized Springer map $\mu : Y \to X$ with $Y = G \times^P (X_L \times \mathfrak{p}^\perp)$. |
| **Terminalization** | $\mathbb{Q}$-factorial terminalization $\iff$ maximal partial Poisson resolution. | $Y$ is $\mathbb{Q}$-factorial and terminal $\iff$ $\widetilde{\mathbb{O}}_L$ is birationally rigid. |
| **Graded Deformation** | Poisson scheme $X_{\mathrm{Spec}(B)}$ flat over a graded algebra $B$. | Deformation $Y_{\mathfrak{z}}$ constructed via the center $\mathfrak{z} = (\mathfrak{l}/[\mathfrak{l},\mathfrak{l}])^*$. |
| **Universal Deformation** | Base space $\mathfrak{h}_X/W_X$ where $\mathfrak{h}_X := H^2(Y^{reg};\mathbb{C})$. | Universal deformation constructed as $X_{\mathfrak{z}}/W_X = \mathrm{Spec}(\mathbb{C}[Y_{\mathfrak{z}}])/W_X$. |

## 38. Zariski's Main Theorem

2026-05-16

**Theorem 38.1** (Zariski's Main Theorem)**.** *Let $f : X \to Y$ be a quasi-finite, separated morphism of schemes, where $Y$ is quasi-compact and quasi-separated. Then there exists a factorization:*

$$X \xrightarrow{j} X' \xrightarrow{g} Y$$

*where $j$ is an open immersion and $g$ is a finite morphism.*

*Proof.* $f_*\mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras. Let $\mathcal{A}$ be the integral closure of $\mathcal{O}_X$ in $f_*\mathcal{O}_Y$, then take $X' = \underline{\mathrm{Spec}}_Y(\mathcal{A})$. □

The intuition is that quasi-finite morphisms have finite fibers, but the fibers can drop abruptly, the theorem tells us that we can appropriately extend the scheme $Y$ to avoid this bad phenomenon.

**Example 38.2.** $\mathbb{C}^\times \to \mathbb{C}$ can extend to $\mathbb{C}^\times \hookrightarrow \mathbb{C} \xrightarrow{\sim} \mathbb{C}$.

**Example 38.3.** For nilpotent cover, the morphism $\widetilde{\mathbb{O}} \to \overline{\mathbb{O}}$ can extend to

$$\widetilde{\mathbb{O}} \hookrightarrow \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}]) \to \overline{\mathbb{O}}.$$

## 39. Twisted Differential Operators

2026-05-17

$X$ be a smooth variety, $\mathcal{L}$ be a line bundle on $X$. Choose a local trivialization for $\mathcal{L}$ by affine open subsets:

$$X = \bigcup_i X_i, \quad \phi_i : \mathcal{L}|_{X_i} \xrightarrow{\sim} \mathcal{O}_{X_i}.$$

Suppose that $\phi_j \circ \phi_i^{-1} : \mathcal{O}_{X_{ij}} \xrightarrow{\sim} \mathcal{O}_{X_{ij}}$ is given by $f_{ji} \in \mathcal{O}^\times(X_{ij})$.

We can define a sheaf of twisted differential operators $\mathcal{D}_\mathcal{L}$ via the local trivialization:

$$\psi_i : \mathcal{D}_\mathcal{L} \xrightarrow{\sim} \mathcal{D}_X|_{X_i},$$

such that $\psi_j \circ \psi_i^{-1} : \mathcal{D}_{X_{ij}} \xrightarrow{\sim} \mathcal{D}_{X_{ij}}$ is given by $f \mapsto f$ and $\xi \mapsto \xi - \frac{\xi(f_{ji})}{f_{ji}}$.

**Theorem 39.1.** *$\mathcal{D}_\mathcal{L}$ acts on $\mathcal{L}$.*

*Proof.* For each local trivialization $X_i$ and $X \in \mathcal{D}_\mathcal{L}(X_i)$, $s \in \mathcal{L}(X_i)$, define

$$X \cdot s = \phi_i^{-1}(\psi_i(X) \cdot \phi_i(s)).$$

We check it is well defined on the overlap of $X_i$ and $X_j$, i.e.

$$\phi_j \circ \phi_i^{-1}(\psi_i(X) \cdot \phi_i(s)) = \psi_j(X) \cdot \phi_j(s).$$

If $\psi_i(X) = f \in \mathcal{O}(X_{ij})$, the verification is easy.

If $\psi(X) = \xi \in \mathrm{Vect}(X_{ij})$ is a vector field, then left hand side is

$$f_{ji}\xi \cdot \phi_i(s).$$

Right hand side is

$$(\xi - \frac{\xi(f_{ji})}{f_{ji}}) \cdot (f_{ji} \cdot \phi_i(s)) = \xi(f_{ji})\phi_i(s) + f_{ji}\xi(\phi_i(s)) - \xi(f_{ji})\phi_i(s) = f_{ji}\xi \cdot \phi_i(s).$$

□

## 40. Quantization of Nonaffine Varieties

2026-05-18

40.1. **Deformation and Quantization of nonaffine schemes.** Let $X$ be a graded normal quasi-projective Poisson variety, the Poisson bracket has degree $-d$. According to Sumihiro's Theorem, any normal variety $X$ with a $\mathbb{G}_m$-action can be covered by $\mathbb{G}_m$-invariant affine open subvarieties $U_i$.

**Definition 40.1.** A filtered quantization of $\mathcal{O}_X$ is a pair $(\mathcal{D}, \theta)$ where

- $\mathcal{D}$ is a filtered sheaf of algebra $\mathcal{D}$ in the conical topology of $X$ such that $[\mathcal{D}_{\leq i}, \mathcal{D}_{\leq j}] \subseteq \mathcal{D}_{\leq i+j-d}$;
- $\theta : \mathrm{gr}(\mathcal{D}) \xrightarrow{\sim} \mathcal{O}_X$ is a graded Poisson isomorphism.

**Remark 40.2** (A very important situation)**.** When $X = \underline{\mathrm{Spec}}_Y(\mathcal{A})$ for a variety $Y$ and a quasi-coherent graded ring of Poisson algebra $\mathcal{A}$ on $Y$, we can focus on the variety $Y$: the filtered quantization of $\mathcal{O}_X$ on $X$ is the same as filtered quantization of $\mathcal{A}$ on $Y$.

Similarly, we can define filtered Poisson deformation of $X$.

40.2. **A very important example of Hamiltonian reduction.** $Y$ be an algebraic variety, $H$ is an algebraic group. $\pi : \widetilde{Y} \to Y$ is a principal $H$-bundle. Then we have the following reduction:

**Theorem 40.3.** $T^*(\widetilde{Y}) /\!/\!/_0 H \xrightarrow{\sim} T^*Y$.

*Proof.* The moment map $\mu : T^*\widetilde{Y} \to \mathfrak{h}^*$ is given by $\langle \mu(\alpha), X\rangle = X^\sharp_{\widetilde{Y}}(\alpha)$, so $\mu^{-1}(0)$ consists of the cotangent vector fields which vanish on the vertical tangent spaces:

$$\mu^{-1}(0) = \widetilde{Y} \times_Y T^*Y.$$

□

**Theorem 40.4.** $\mathcal{D}_{\widetilde{Y}} /\!\!/\!\!/_\lambda H \xrightarrow{\sim} \mathcal{D}^\lambda_Y$.

*Here $\mathcal{D}^\lambda_Y$ is the twisted differential operators on $\mathcal{L}^\lambda = \widetilde{Y} \times^H \mathbb{C}_\lambda$.*

These theorems verify the quantization commutes with reduction principle.

**Example 40.5.** $\pi : \tilde{Y} = G/U \to Y = G/P$ is a principle $L$-bundle. We can define

$$\mathcal{D}^\lambda_{G/P} := \mathcal{D}_{G/U} /\!\!/\!\!/_{\lambda-\rho_{G/P}} L.$$

Then

$$\mathcal{D}^\lambda_{G/P} = \mathrm{Diff}(\mathcal{L}^{\lambda-\rho_{G/P}}, \mathcal{L}^{\lambda-\rho_{G/P}}).$$

40.3. **Quantization of nilpotent covers.**

Suppose $\mathrm{Bind}^G_L(\widetilde{\mathbb{O}}_L) = \widetilde{\mathbb{O}}$, where $\widetilde{\mathbb{O}}_L$ is birationally rigid.

$Y = G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \,|\, \alpha|_{\mathfrak{l}} = \mu(x)\}$ is a $\mathbb{Q}$-factorial terminalization of $X = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$.

$Y$ is affine over $G/P$ satisfies the situation of the previous remark.

$G/U \to G/P$ is an $L$-principal bundle, $T^*(G/U) \times X_L \to G/U$ is affine.

We can consider all of these quantizations, Hamiltonian reductions as sheaves over the smooth variety $G/P$:

We have a quantization $\mathcal{D}_{G/U} \otimes \mathcal{A}_L$ of $T^*(G/U) \times X_L$ as a sheaf of algebras over $G/U$. For each $\lambda \in \mathfrak{z}$, we can consider the Hamiltonian reduction

$$\mathcal{D}^\lambda_{G/P} := \mathcal{D}_{G/U} \otimes \mathcal{A}_L /\!\!/\!\!/_{\lambda-\rho_{G/P}} L.$$

As sheaves of algebras over $G/P$. They are all the quantizations of $Y$.

There is also sheaf of $\mathfrak{z}$-algebra $\mathcal{D}_\mathfrak{z}$ over $G/P$ as the universal quantization of $Y$, such that $\mathbb{C}_{\lambda-\rho_{G/P}} \otimes_\mathfrak{z} \mathcal{D}_\mathfrak{z} \cong \mathcal{D}^\lambda_{G/P}$. This is the quantization of $Y_\mathfrak{z} \to Y$.

## 41. Harish-Chandra Bimodules

2026-06-08

$\mathfrak{g}$ be a Lie algebra, $\mathcal{B}$ be a $U(\mathfrak{g})$-bimodule. A good filtration on $\mathcal{B}$ is a filtration:

$$0 = \mathcal{B}_{-1} \subseteq \mathcal{B}_0 \subseteq \mathcal{B}_1 \subseteq \cdots \subseteq \mathcal{B}_n \subseteq \cdots,$$

such that

- $\bigcup_{i=0}^\infty \mathcal{B}_i = \mathcal{B}$;
- $[U_i(\mathfrak{g}), \mathcal{B}_j] \subseteq \mathcal{B}_{i+j-1}$, $U_i(\mathfrak{g}) \cdot \mathcal{B}_j \subseteq \mathcal{B}_{i+j}$;
- $\mathrm{gr}(\mathcal{B})$ is finitely generated as $S(\mathfrak{g})$-module.

**Theorem 41.1.** *$\mathcal{B}$ has a good filtration if and only if $\mathcal{B}$ is finitely-generated (left or right $U(\mathfrak{g})$-module) and adjoint action of $\mathfrak{g}$ on $\mathcal{B}$ is locally finite.*

*Proof.* □

**Proposition 41.2.** *If $\mathcal{B}$ is locally finite under the adjoint action of $\mathfrak{g}$, then $\mathcal{B}$ is left f.g. $\Leftrightarrow$ bi-f.g. $\Leftrightarrow$ right f.g.*

*Proof.* □

## 42. Principal Connections and Picard Groups

2026-06-29

$X$ be a smooth manifold, $\mathcal{V}$ a vector bundle on $X$, it's associated frame bundle is the principal bundle $\mathcal{M}$. Note that $\mathcal{V} = \mathcal{M} \times^{\mathrm{GL}_n} \mathbb{C}^n$.

A connection on $\mathcal{V}$ is a map $\nabla : \Gamma(X, TX) \to \Gamma(X, \mathfrak{gl}(\mathcal{V}))$. While a connection on $\mathcal{M}$ is a GL-equivariant bundle map $\theta : \mathcal{E}_M \to \mathfrak{gl}(V)_M$, split the Atiyah sequence:

$$0 \to \mathfrak{gl}(V)_M \to \mathcal{E}_M \to TX \to 0.$$

Restriction of $\theta$ to $TX$ gives a connection on $\mathcal{V}$, this is a bijection.

---

Here we compute some Picard groups of symplectic singularities, and their Poisson resolutions.

$\widetilde{\mathbb{O}}$ be a nilpotent cover of a complex semisimple Lie group $G$, $\widetilde{\mathbb{O}_L}$ be a nilpotent cover of a Levi subgroup $L$. And there is a Poisson resolution

$$\pi : Y = G \times^P (X_L \times \mathfrak{p}^\perp) \to X.$$

**Proposition 42.1.**
- $\mathrm{Pic}(X) = 0$;
- $\mathrm{Pic}(X^{reg}) = X^*(\widetilde{H})$, *where* $\widetilde{H}$ *is the stabilizer in* $\widetilde{G}$ *of a point in* $\widetilde{\mathbb{O}}$.

*Proof.* First assertion is due to $X$ is conical.

For the second assertion, since $\widetilde{\mathbb{O}} \subseteq X^{reg} \subseteq X$ and $\mathrm{codim}_X(X \backslash \widetilde{\mathbb{O}}) \geq 2$, we have

$$\mathrm{Pic}(X^{reg}) = \mathrm{Pic}(\widetilde{\mathbb{O}}).$$

Since $\widetilde{\mathbb{O}}$ is a homogeneous space of $\widetilde{G}$, $\widetilde{\mathbb{O}} \simeq \widetilde{G}/\widetilde{H}$ ($\widetilde{G}$ is the simply-connected cover of $G$),

$$\mathrm{Pic}(\widetilde{\mathbb{O}}) = \mathrm{Pic}(\widetilde{G}/\widetilde{H}) = \mathrm{Pic}^{\widetilde{G}}(\widetilde{G}/\widetilde{H}) = X^*(\widetilde{H}).$$

The second equality is due to the fact that $\widetilde{G}$ is simply-connected, so any line bundle on $\widetilde{G}/\widetilde{H}$ admits a $\widetilde{G}$-equivariant structure. And $\mathcal{O}_X$ has only one $\widetilde{G}$-equivariant structure, since $H^1(G; \mathcal{O}_X^*(X)) = 0$. □

**Proposition 42.2.** *There is a short exact sequence*

$$0 \to \mathrm{Pic}(Y) \to \mathrm{Pic}(Y^{reg}) \to \mathrm{Pic}(X_L^{reg}) \to 0.$$

*Proof.* Take the open Bruhat cell $U^- \subseteq G/P$.

Then

$$\pi^{-1}(U^-) \simeq U^- \times (X_L \times \mathfrak{p}^\perp).$$

$U^-$ and $\mathfrak{p}^\perp$ are affine spaces, so $\mathrm{Cl}(\pi^{-1}(U^-)) = \mathrm{Cl}(X_L) = \mathrm{Pic}(X_L^{reg})$.

The map $\mathrm{Pic}(Y^{reg}) \to \mathrm{Pic}(X_L^{reg})$ is just the restriction map $\mathrm{Cl}(Y) \to \mathrm{Cl}(\pi^{-1}(U^-))$, and the kernel is $\mathrm{Cl}(Y \backslash \pi^{-1}(U^-))$.

$$\mathrm{Pic}(Y) \xrightarrow{div} \mathrm{Cl}(Y) \xrightarrow{res} \mathrm{Cl}(\pi^{-1}(U^-)),$$

the composition can also realized as

$$\mathrm{Pic}(Y) \xrightarrow{res} \mathrm{Pic}(\pi^{-1}(U^-)) \xrightarrow{div} \mathrm{Cl}(\pi^{-1}(U^-)),$$

while $\mathrm{Pic}(\pi^{-1}(U^-)) = 0$ the composition is zero.

On the other hand, if a divisor $D$ on $Y$ is mapped to zero in $\mathrm{Cl}(\pi^{-1}(U^-))$, then $D$ is linearly equivalent to a divisor supported on $Y\backslash\pi^{-1}(U^-)$, it is linearly equivalent to $\sum_i n_i\pi^{-1}(D_i)$ (suppose that $(G/P)\backslash U^- = \cup_i D_i$, where $D_i$ are divisors.), and $\pi^{-1}(D_i) = \mathrm{div}(\pi^*(\mathcal{O}(D_i)))$ they are Cartier divisors, so $D$ is Cartier. This shows that the kernel of $\mathrm{Cl}(Y) \to \mathrm{Cl}(\pi^{-1}(U^-))$ is $\mathrm{Pic}(Y)$. □

**Proposition 42.3.** $\mathrm{Pic}(Y) = \mathrm{Pic}(G/P) = X^*(\widetilde{L})$.

*Proof.* Then $\pi^* : \mathrm{Pic}(G/P) \to \mathrm{Pic}(Y)$ is injective, since $\pi$ has a section $G/P \hookrightarrow Y$. For any line bundle $\mathcal{L}$ on $Y$ the proof of previous proposition shows that $\mathcal{L}$ is isomorphic to some $\mathcal{O}(\sum_i \pi^{-1}(D_i)) = \pi^*(\mathcal{O}(\sum_i D_i))$. □

**Corollary 42.4.** *If $\widetilde{\mathbb{O}_L}$ is birationally rigid, then we have a canonical isomorphism:*

$$c_1 : (\mathfrak{l}/[\mathfrak{l},\mathfrak{l}])^* = \mathrm{Pic}(Y) \otimes_{\mathbb{Z}} \mathbb{C} \xrightarrow{res} \mathrm{Pic}(Y^{reg}) \otimes_{\mathbb{Z}} \mathbb{C} \to H^2(Y^{reg}, \mathbb{C}).$$

The next theorem is the computation of the period map for the quantization of $Y$.

**Theorem 42.5.** *Under the period map*

$$\mathrm{Per} : \mathrm{Quant}(Y) = \mathrm{Quant}(Y^{reg}) \to H^2(Y^{reg}, \mathbb{C}).$$

*$\mathcal{D}_\lambda$ maps to $c_1(\lambda)$.*

## 43. Formal quantization and Star product

2026-07-25

For a Poisson variety $X$ a formal quantization of $X$ is a pair $(\mathcal{D}_\hbar, \theta)$ where $\mathcal{D}_\hbar$ is sheaf of $\mathbb{C}[[h]]$-algebra flat over $\mathbb{C}[[\hbar]]$ and complete in $\hbar$-adic topology, and $\theta : \mathcal{D}_\hbar/(\hbar) \xrightarrow{\sim} \mathcal{O}_X$ is a Poisson isomorphism. Isomorphism classes of formal quantizations of $X$ is denoted by $Quant(X)$

When $X = \mathrm{Spec}(A)$ is affine, use micro-localization, we can identify the formal quantization of $X$ with a formal quantization of $A$.

A star product on $A$ is a $\mathbb{C}[[\hbar]]$ bi-linear map

$$\star : A[[\hbar]] \times A[[\hbar]] \to A[[\hbar]], \quad f \star g = \sum_{i=0}^{\infty} D_i(f,g)\hbar^i,$$

where $D_i : A \times A \to A$ are bi-differential operators, such that

- $D_0(f,g) = fg$;
- $D_1(f,g) - D_1(g,f) = \{f,g\}$;
- $\star$ is associative, 1 is the unit.

Two star products $\star$ and $\star'$ are called equivalent if there is a $\mathbb{C}[[\hbar]]$-linear automorphism $T : A[[\hbar]] \to A[[\hbar]]$,

$$T = \mathrm{id} + \sum_{i=0}^{\infty} T_i\hbar^i, \quad T_i : A \to A,$$

such that $T(f \star g) = T(f) \star' T(g)$.

**Theorem 43.1.** *There is a bijection:*

$$Quant(X) \xrightarrow{\sim} \{Star\ products\ on\ A\}/equivalence.$$

*Proof.* Step 1: For any formal quantization $A_\hbar$ of $A$, there exist a $\mathbb{C}[[\hbar]]$-linear map $\Phi : A[[\hbar]] \xrightarrow{\sim} A_\hbar$.

By definition of formal quantization, we have an isomorphism $\theta : A_\hbar/(\hbar) \simeq A$. We can choose a $\mathbb{C}$-linear section $s : A \to A_\hbar$ of $\theta$, then extend it to a $\mathbb{C}[[\hbar]]$-linear map $\Phi : A[[\hbar]] \to A_\hbar$, $\sum a_i\hbar^i \mapsto \sum s(a_i)\hbar^i$. This map is surjective, for any $a \in A_\hbar$, let $a_0 = \theta(a) \in A$, then $a - s(a_0) \in \hbar A_\hbar$, continue this process, we can find $a_i$ for each $i \in \mathbb{N}$ such that $a = \sum s(a_i)\hbar^i$. This map is injective, if $\sum s(a_i)\hbar^i = 0$, then $s(a_0) \in \hbar A_\hbar$, so $a_0 = \theta \circ s(a_0) = 0$, continue this process, we can find $a_i = 0$ for each $i \in \mathbb{N}$.

So we can transport the product on $A_\hbar$ to $A[[\hbar]]$, which satisfies the conditions of star product without the bi-differential condition. The choice of $\Phi$ is not unique, if we choose another $\Phi'$, then $\Phi' \circ \Phi^{-1}$ is a $\mathbb{C}[[\hbar]]$-linear automorphism of $A[[\hbar]]$, which is of the form $T = \mathrm{id} + \sum_{i=0}^{\infty} T_i\hbar^i$, and $T$ gives an equivalence between the two transported products.

Step 2: There exist a choice of $\Phi$ such that the transported product on $A[[\hbar]]$ satisfies the bi-differential condition.

We need the fact that $HH^i(A) \simeq HH^i_{\mathrm{diff}}(A)$.

□

## 44. Filtered Quantization and Graded Quantization

2026-08-31

$A$ be a positively graded commutative algebra with a Poisson bracket $\{,\}$ of degree $-d$.

### 44.1. Notes version. [Los22b]

- A filtered quantization of $A$ is a pair $(\mathcal{A}, \theta)$ where $\mathcal{A}$ is a filtered algebra with $[\mathcal{A}_{\le i}, \mathcal{A}_{\le j}] \subseteq \mathcal{A}_{\le i+j-d}$, and $\theta : \mathrm{gr}(\mathcal{A}) \xrightarrow{\sim} A$ is a graded Poisson isomorphism.
- A graded quantization of $A$ is a pair $(\mathcal{A}_\hbar, \theta)$ where $\mathcal{A}_\hbar$ is a graded $\mathbb{C}[\hbar]$-algebra flat over $\mathbb{C}[\hbar]$ such that $[\mathcal{A}_\hbar, \mathcal{A}_\hbar] \subseteq \hbar^d\mathcal{A}_\hbar$, and $\theta : \mathcal{A}_\hbar/(\hbar) \xrightarrow{\sim} A$ is a Poisson isomorphism. ($\deg \hbar = 1$)

The above two quantizations are equivalent. The equivalence is given by the Rees construction:

$$\mathcal{A}_\hbar = \bigoplus_{i=0}^{\infty} \mathcal{A}_{\le i}\hbar^i.$$

**Example 44.1.** For a complex Lie algebra $\mathfrak{g}$, $S(\mathfrak{g})$ is a graded Poisson algebra with $\deg \mathfrak{g} = 2$. (We need the degree 2 to deal with the case of nilpotent covers.)

We have a natural filtered quantization of $S(\mathfrak{g})$, which is the pair $(U(\mathfrak{g}), \iota)$, where $U(\mathfrak{g})$ is equipped with the doubled PBW filtration.

By Rees construction, we can get a graded quantization:

$$U_\hbar(\mathfrak{g}) = \bigoplus_{i=0}^{\infty} U_{\le \lfloor \frac{i}{2} \rfloor}(\mathfrak{g})\hbar^i = T(\mathfrak{g})[\hbar]/(X \otimes Y - Y \otimes X - \hbar^2[X, Y]).$$

### 44.2. LY23 version. [LY23, Section 2]

- A filtered quantization of $A$ is a triple $(\mathcal{A}, \zeta, \iota)$, where $\mathcal{A}$ is a filtered algebra with $[\mathcal{A}_{\le i}, \mathcal{A}_{\le j}] \subseteq \mathcal{A}_{\le i+j-d}$, $\zeta : \mu_d \to \mathrm{Aut}_{fil}(\mathcal{A})$ such that $\zeta(t) \cdot \mathrm{gr}(\mathcal{A})_i = t^i \,\mathrm{gr}(\mathcal{A})_i$, and $\iota : \mathrm{gr}(\mathcal{A}) \xrightarrow{\sim} A$ is a graded Poisson isomorphism.

- A graded quantization of $A$ is a pair $(\mathcal{A}_\hbar, \theta)$, where $\mathcal{A}_\hbar$ is a graded $\mathbb{C}[\hbar]$-algebra flat over $\mathbb{C}[\hbar]$ such that $[\mathcal{A}_\hbar, \mathcal{A}_\hbar] \subseteq \hbar\mathcal{A}_\hbar$, and $\theta : \mathcal{A}_\hbar/(\hbar) \xrightarrow{\sim} A$ is a Poisson isomorphism. ($\deg \hbar = d$)

The above two quantizations are equivalent too. The equivalence is given by the Rees construction:

$$\mathcal{A}_\hbar = \bigoplus_{i=0}^{\infty} (\mathcal{A}_{\leq i} \cap \mathcal{A}_{i+d\mathbb{Z}}) u^i.$$

Here $u^d = \hbar$, $\mathcal{A}_{i+d\mathbb{Z}} = \{a \in \mathcal{A} \mid \zeta(t) \cdot a = t^i a\}$.

This definition compatible well with the definition of formal quantization in many other papers, for example, [BK04] and [LY21].

**Example 44.2.** For a complex Lie algebra $\mathfrak{g}$, $S(\mathfrak{g})$ is a graded Poisson algebra with $\deg \mathfrak{g} = 2$.

There is a natural filtered quantization of $S(\mathfrak{g})$, which is the triple $(U(\mathfrak{g}), \zeta = 1, \iota)$, where $U(\mathfrak{g})$ is equipped with the doubled PBW filtration, and $\iota : \mathrm{gr}(U(\mathfrak{g})) \xrightarrow{\sim} S(\mathfrak{g})$ is the natural isomorphism.

By Rees construction, we can get a graded quantization:

$$U_\hbar(\mathfrak{g}) = \bigoplus_{i=0}^{\infty} (U_{\leq \lfloor \frac{i}{2} \rfloor}(\mathfrak{g}) \cap U_{i+2\mathbb{Z}}(\mathfrak{g})) u^i = \bigoplus_{i=0}^{\infty} U_i(\mathfrak{g}) \hbar^i = T(\mathfrak{g})[\hbar]/(X \otimes Y - Y \otimes X - \hbar[X, Y]).$$

## 45. Quantum Slices Theorem

2026-09-02

$A$ be a graded Poisson algebra with a Poisson bracket of degree $-d$ over $\mathbb{C}$. Let $(\mathcal{A}, \zeta, \iota)$ be a filtered quantization of $A$, and $(\mathcal{A}_\hbar, \theta)$ be the corresponding graded quantization.

For any smooth closed point $e \in \mathrm{Spec}(A)$, denote by $m$ the corresponding maximal ideal, $A^\wedge = \varprojlim A/m^n$ is the completion of $A$ with respect to $m$. $m_\hbar$ the corresponding maximal ideal in $\mathcal{A}_\hbar$, and $\mathcal{A}_\hbar^\wedge = \varprojlim \mathcal{A}_\hbar/m_\hbar^n$ the completion of $\mathcal{A}_\hbar$ at $m_\hbar$.

There is a skew symmetric bilinear form $\omega$ on $T_e^* \mathrm{Spec}(A) = m/m^2$ defined by:

$$\omega(\xi + m^2, \eta + m^2) = \{\xi, \eta\} + m \in A/m = \mathbb{C}.$$

Choose a completion $V$ of the radical $\ker(\omega)$ in $T_e^* \mathrm{Spec}(A)$, then $V$ is a symplectic vector space. Denote by $\hat{\mathcal{W}}_\hbar(V)$ the completed Weyl algebra of $V$ over $\mathbb{C}[[\hbar]]$.

Consider the composition:

$$\pi : \hat{m}_\hbar \to \hat{m} \to m/m^2 \to V.$$

**Theorem 45.1** (Quantum Slice)**.** *There exist a lift $\iota : V \to m_\hbar$ such that:*

- $\pi \circ \iota = \mathrm{id}_V$;
- $\frac{1}{\hbar}[\iota(u), \iota(v)] = \omega(u, v)$;
- *Let $\mathcal{A}'_\hbar$ denote the centralizer of $\iota(V)$ in $\mathcal{A}_\hbar^\wedge$, then the multiplication map gives an isomorphism:*

$$\hat{\mathcal{W}}_\hbar(V) \widehat{\otimes}_{\mathbb{C}[[\hbar]]} \mathcal{A}'_\hbar \xrightarrow{\sim} \mathcal{A}_\hbar^\wedge.$$

*Proof.* For any $x, y \in V$ such that $\omega(x, y) = 1$, for any lift $x_1, y_1$ we have $\frac{1}{\hbar}[x_0, y_0] - 1 \in \hat{m}_\hbar$.

We try to used induction to construct lifts $x_k, y_k \in \hat{m}_\hbar$ such that

$$\frac{1}{\hbar}[x_k, y_k] - 1 \in \hat{m}_\hbar^k, \quad x_k - x_{k-1}, y_k - y_{k-1} \in \hat{m}_\hbar^k.$$

Note that we have a well defined linear morphism:

$$T_k = \overline{\frac{1}{\hbar}[x_k, y_k \cdot (-)]} : \hat{m}_\hbar^k / \hat{m}_\hbar^{k+1} \to \hat{m}_\hbar^k / \hat{m}_\hbar^{k+1}.$$

Suppose that $\frac{1}{\hbar}[x_k, y_k] = e_k \in \hat{m}_\hbar^k$, then $T_k = 1 + \bar{y_k} \cdot D$, where $D$ is a derivation of $\hat{m}_\hbar^k / \hat{m}_\hbar^{k+1}$. □

## 46. Symplectic Lie algebra and Weyl algebra

2026-09-04

Let $(V, \omega)$ be a symplectic vector space over $\mathbb{C}$.
As representations of $\mathrm{Sp}(V)$, we have the isomorphism:

$$\begin{array}{rccc} \Phi\colon & S^2(V) & \longrightarrow & \mathfrak{sp}(V) \\ & u \odot v & \longmapsto & (z \mapsto \omega(u,z)v + \omega(v,z)u). \end{array}$$

Let $\mathcal{W}(V) = T(V)/(u \otimes v - v \otimes u - \omega(u,v))$ be the Weyl algebra.
There is an embedding:

$$\begin{array}{rccc} \iota\colon & S^2(V) & \longrightarrow & \mathcal{W}(V) \\ & u \odot v & \longmapsto & \frac{1}{2}(u \otimes v + v \otimes u). \end{array}$$

**Proposition 46.1.** *The composition $\eta = \iota \circ \Phi^{-1} : \mathfrak{sp}(V) \to \mathcal{W}(V)$ is a Lie algebra homomorphism.*

*Proof.* For any $v \in V$ and $a \odot b \in S^2(V)$, we have:

$$[\iota(a \odot b), v] = [\frac{1}{2}(a \otimes b + b \otimes a), v] = \omega(a,v)b + \omega(b,v)a = \Phi(a \odot b)(v).$$

Since $\Phi$ is $\mathrm{Sp}(V)$ equivariant, we have

$$[A, \Phi(u \odot v)] = \Phi(A \cdot (u \odot v)) = \Phi(Au \odot v + u \odot Av), \quad A \in \mathfrak{sp}(V), u, v \in V.$$

The rest is a direct computation. □

**Remark 46.2.** The first formula in the proof shows that the action of $\mathrm{Sp}(V)$ on $\mathcal{W}(V)$ is Hamiltonian, the quantum comoment map is given by $\eta : \mathfrak{sp}(V) \to \mathcal{W}(V)$.

## 47. Koszul Duality Pattern in Category $\mathcal{O}$

2026-09-17

We summarize some general pictures of Koszul duality pattern in category $\mathcal{O}$.

47.1. **Soergel Functor.** $\mathfrak{g}$ be a complex semisimple Lie algebra, $\check{\mathfrak{g}}$ be the Langlands dual Lie algebra of $\mathfrak{g}$. Fix Cartan subalgebras $\mathfrak{h}$ and $\mathfrak{h}^*$, and Borel subalgebras $\mathfrak{b}$ and $\check{\mathfrak{b}}$. Their Weyl groups are the same $W$. $\mathcal{O}^{pr}$ and $\check{\mathcal{O}}^{pr}$ be the principal blocks of the category $\mathcal{O}$.

**Theorem 47.1.** *There are canonical algebra isomorphisms:*

$$\mathrm{End}_{\mathcal{O}^{pr}}(P(w_0\cdot 0)) \simeq \mathrm{Ext}^*_{D^b(\check{\mathcal{O}}^{pr})}(\check{L}(0),\check{L}(0)) \simeq C.$$

*Here* $C = \mathbb{C}[\mathfrak{h}^*]/\mathbb{C}[\mathfrak{h}^*]^W_+$.

We can define functors:

$$\mathbb{V} : \mathcal{O}^{pr} \to C - mod_{ugr}, \quad M \mapsto \mathrm{Hom}_{\mathcal{O}^{pr}}(P(w_0\cdot 0), M).$$

**Theorem 47.2.**
- $\mathbb{V}$ *is a fully faithful functor on projective objects, and it induces an equivalence of categories:*

$$\mathcal{O}^{pr} - \mathrm{Proj} \xrightarrow{\sim} \mathrm{SMod}_{ugr}.$$

*Here* SMod *is the additive subcategory of Soergel modules in* $C - mod_{ugr}$*;*
- $\mathbb{V}(P(w\cdot 0)) = \bar{B}_w$.

On the Koszul dual side, we have the functor:

$$\check{\mathbb{V}}_K : D^b(\check{\mathcal{O}}^{pr}) \to C - mod_{gr}, \quad M \mapsto \bigoplus_{i\in\mathbb{Z}} \mathrm{Hom}_{D^b}(\check{L}(0), M[i]).$$

**Theorem 47.3.**
- $\check{\mathbb{V}}_K$ *is fully faithful on irreducible objects, and it induces an equivalence of categories:*

$$D^b(\check{\mathcal{O}}^{pr})_{ss} \xrightarrow{\sim} \mathrm{SMod}_{gr}.$$

- $\check{\mathbb{V}}_K(\check{L}(w\cdot 0)) = \bar{B}_{w^{-1}w_0}[-l(w_0)]$

Combining the above two equivalences of categories, we get an isomorphism of algebras:

**Theorem 47.4.**

$$\mathrm{End}_{\mathcal{O}^{pr}}(\bigoplus_{w\in W} P(w\cdot 0)) \simeq \mathrm{Ext}^*_{\check{\mathcal{O}}^{pr}}(\bigoplus_{w\in W} \check{L}(w\cdot 0), \bigoplus_{w\in W} \check{L}(w\cdot 0)).$$

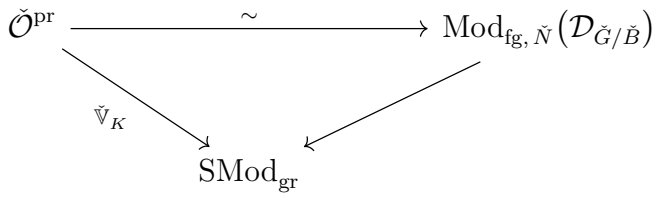
47.2. **Geometry of Dual Side.** On the dual side, the functor $\check{\mathbb{V}}_K$ has geometric interpretation. Let $\check{G}$ be the Langlands dual group of $G$, $\check{B}$ be the Borel subgroup of $\check{G}$, and $\check{\mathcal{B}} = \check{G}/\check{B}$ be the flag variety. Then we have an equivalence of categories:

$$\check{\mathcal{O}}^{pr} \xrightarrow{\sim} \mathrm{Mod}_{fg,\check{N}}(\mathcal{D}_{\check{G}/\check{B}}), \quad \check{L}(w^{-1}w_0\cdot 0) \mapsto \mathrm{IC}(\check{B}w\check{B}/\check{B}, trivial).$$

$$D^b(\check{\mathcal{O}}^{pr}) \xrightarrow{\sim} D^b(\mathrm{Mod}_{fg,\check{N}}(\mathcal{D}_{\check{G}/\check{B}})) \xrightarrow{\sim} D^b_{\check{N}}(\check{G}/\check{B}).$$

Under these equivalences, the functor $\check{\mathbb{V}}_K$ can be realized as:

$$\check{\mathbb{V}}_K : D^b_{\check{N}}(\check{G}/\check{B}) \to C - mod_{gr}, \quad \mathcal{F} \mapsto \bigoplus_{i\in\mathbb{Z}} \mathrm{Hom}_{D^b_{\check{N}}(\check{G}/\check{B})}(\mathcal{O}_{\check{G}/\check{B}}, \mathcal{F}[i]).$$




Then we can see that $\check{\mathbb{V}}_K(\check{L}(w^{-1}w_0\cdot 0)) = \mathrm{IH}^*_N(BwB/B, trivial)[?]$, so $\mathrm{IH}^*_N(BwB/B, trivial)$ is indecomposible in $\mathrm{SMod}_{gr}$, and use BBDG decomposition theorem it is isomorphic to $\bar{B}_w$ up to a shift.

## 48. Examples of resolution of the nilpotent orbit closure

2026-09-22

Let $G = \mathrm{Sp}_4(\mathbb{C})$, and let $\mathbb{O} \subset \mathfrak{sp}_4(\mathbb{C})$ be the nilpotent orbit corresponding to the partition $[2,2]$. Let $P_1$ be the stabilizer of a line in the standard representation, and let $P_2$ be the stabilizer of a Lagrangian subspace. We will show that the moment maps

$$\mu_1 : T^*(G/P_1) \longrightarrow \overline{\mathbb{O}}, \qquad \mu_2 : T^*(G/P_2) \longrightarrow \overline{\mathbb{O}}$$

have generic degrees 2 and 1, respectively.

Write

$$V = \mathrm{span}_{\mathbb{C}}\{p_1, p_2, q_1, q_2\},$$

with symplectic form satisfying

$$\omega(p_i, p_j) = \omega(q_i, q_j) = 0, \qquad \omega(p_i, q_j) = \delta_{ij}.$$

Choose $e \in \mathbb{O}$ defined by

$$e(p_i) = 0, \qquad e(q_i) = p_i, \qquad i = 1, 2.$$

In particular,

$$\mathrm{im}(e) = \ker(e) = W, \qquad W := \mathrm{span}\{p_1, p_2\}.$$

Using an invariant nondegenerate bilinear form to identify $\mathfrak{g}$ with $\mathfrak{g}^*$, we have

$$T^*(G/P_i) \cong G \times^{P_i} \mathfrak{u}_i, \qquad \mu_i([g, x]) = \mathrm{Ad}(g)x,$$

where $\mathfrak{u}_i$ is the nilpotent radical of $\mathfrak{p}_i$. Thus the fiber over $e$ parametrizes the corresponding subspaces whose stabilizers have nilpotent radicals containing $e$.

- The Lagrangian case. The variety $G/P_2$ parametrizes Lagrangian subspaces $M \subset V$. For such an $M$,

  $$e \in \mathfrak{u}_M \quad \Longleftrightarrow \quad \mathrm{im}(e) \subseteq M \subseteq \ker(e).$$

  Since both the image and kernel of $e$ equal $W$, necessarily $M = W$. Conversely, $W$ is Lagrangian and satisfies these conditions. Therefore

  $$\mu_2^{-1}(e) = \{W\}, \qquad \#\mu_2^{-1}(e) = 1.$$

- The line case. Every line in $V$ is isotropic, so $G/P_1 \cong \mathbb{P}(V)$. For a line $\ell \subset V$, the nilpotent radical of its stabilizer consists of the endomorphisms in $\mathfrak{sp}(V)$ that strictly lower the flag

$$0 \subset \ell \subset \ell^\perp \subset V.$$

Consequently,

$$e \in \mathfrak{u}_\ell \quad \Longleftrightarrow \quad e(\ell) = 0, \qquad e(\ell^\perp) \subseteq \ell, \qquad e(V) \subseteq \ell^\perp.$$

The first condition implies $\ell \subset W$, so write

$$\ell = \mathbb{C}(ap_1 + bp_2), \qquad [a : b] \in \mathbb{P}^1.$$

Then

$$\ell^\perp = \operatorname{span}\{p_1, p_2, bq_1 - aq_2\}.$$

Since $e(V) = W \subseteq \ell^\perp$, the third condition is automatic. The second condition is equivalent to

$$bp_1 - ap_2 \in \mathbb{C}(ap_1 + bp_2),$$

or, equivalently,

$$\det \begin{pmatrix} a & b \\ b & -a \end{pmatrix} = 0.$$

Thus

$$a^2 + b^2 = 0.$$

Over $\mathbb{C}$, this equation has exactly two projective solutions,

$$[a : b] = [1 : i] \qquad \text{and} \qquad [a : b] = [1 : -i].$$

Hence

$$\mu_1^{-1}(e) = \{\mathbb{C}(p_1 + ip_2),\, \mathbb{C}(p_1 - ip_2)\}, \qquad \#\mu_1^{-1}(e) = 2.$$

Finally, these computations also justify the asserted images of the moment maps. Both maps are proper and $G$-equivariant, and their images contain $e$, hence contain $\overline{\mathbb{O}}$. Since

$$\dim T^*(G/P_1) = \dim T^*(G/P_2) = 6 = \dim \mathbb{O},$$

their images are exactly $\overline{\mathbb{O}}$. They are therefore generically finite. In characteristic zero, a generically finite dominant morphism is generically étale; equivariance then implies that the fibers over the open orbit $\mathbb{O}$ are reduced. The above cardinalities consequently give

$$\deg \mu_1 = 2, \qquad \deg \mu_2 = 1.$$

## 49. Oscillator Representations as Quantization

2026-09-23

Let $G_{\mathbb{R}} = \mathrm{Sp}_{2n}(\mathbb{R})$, and let $V$ be its complexified standard representation. Throughout, assume $n \geq 2$.

The minimal nilpotent orbit in $\mathfrak{sp}(V)$ has Jordan type $(2, 1^{2n-2})$ and admits the description

$$\mathcal{O}_{\min} \simeq (V \setminus \{0\})/\{\pm 1\}.$$

Its double cover is

$$\widetilde{\mathcal{O}}_{\min} = V \setminus \{0\}.$$

The canonical quantizations of their affinizations are, respectively,

$$\mathcal{A}_0^V = \mathcal{W}(V), \qquad \mathcal{A}_0^{V/\{\pm 1\}} = \mathcal{W}(V)^{\{\pm 1\}} = \mathcal{W}_{\mathrm{even}}(V),$$

where

$$\mathcal{W}(V) = \mathcal{W}_{\mathrm{even}}(V) \oplus \mathcal{W}_{\mathrm{odd}}(V)$$

is the decomposition into eigenspaces for the involution induced by $v \mapsto -v$.

Since the only boundary leaf of $\overline{\mathcal{O}_{\min}}$ is the origin, of codimension $2n \geq 4$, the map

$$V \longrightarrow V/\{\pm 1\}$$

is almost étale. Consequently,

$$I_0(\widetilde{\mathcal{O}}_{\min}) = I_0(\mathcal{O}_{\min}),$$

and this common ideal is the Joseph ideal.

Choose a Lagrangian decomposition

$$V = L_1 \oplus L_2.$$

The subgroup preserving each summand is

$$K_{\mathbb{C}} = \mathrm{Stab}(L_1) \cap \mathrm{Stab}(L_2) \simeq \mathrm{GL}(L_1),$$

with $L_2 \simeq L_1^*$. Writing $\mathfrak{sp}(V) = \mathfrak{k} \oplus \mathfrak{p}$, we have

$$\mathcal{O}_{\min} \cap \mathfrak{p} = \mathcal{O}_K^+ \sqcup \mathcal{O}_K^-, \qquad \mathcal{O}_K^+ = \frac{L_1 \setminus \{0\}}{\{\pm 1\}}, \qquad \mathcal{O}_K^- = \frac{L_2 \setminus \{0\}}{\{\pm 1\}}.$$

We describe the admissible orbit data on $\mathcal{O}_K^+$. Choose a nonzero vector $e_1 \in L_1$, and let $e = [e_1] \in \mathcal{O}_K^+$. Its stabilizer is

$$H = (K_{\mathbb{C}})_e = \{g \in \mathrm{GL}(L_1) : ge_1 = \varepsilon(g)e_1,\ \varepsilon(g) \in \{\pm 1\}\}.$$

Relative to a decomposition $L_1 = \mathbb{C}e_1 \oplus L_1'$, its elements have the form

$$g = \begin{pmatrix} \varepsilon & b \\ 0 & A \end{pmatrix}, \qquad A \in \mathrm{GL}_{n-1}(\mathbb{C}).$$

Pass to the complexified maximal compact subgroup of the metaplectic group:

$$\widetilde{K} = \left\{(g, z) \in \mathrm{GL}(L_1) \times \mathbb{C}^\times : z^2 = \det g\right\},$$

and let $\widetilde{H}$ be the preimage of $H$.

Let $\delta_+$ denote the character of $H$ on $\bigwedge^{\mathrm{top}} T_e^* \mathcal{O}_K^+$. Then

$$\delta_+(g) = \varepsilon(g)^n (\det g)^{-1}.$$

Using the half-canonical convention, admissibility requires

$$d\chi = \frac{1}{2}\, d\delta_+ = -\frac{1}{2}\,\mathrm{tr}$$

on the stabilizer Lie algebra. There are exactly two irreducible admissible characters:

$$\chi_{+,0}(g,z) = z^{-1}, \qquad \chi_{+,1}(g,z) = \varepsilon(g) z^{-1}.$$

They define the admissible line bundles

$$\mathcal{E}_{+,a} = \widetilde{K} \times_{\widetilde{H}} \mathbb{C}_{\chi_{+,a}}, \qquad a = 0, 1.$$

Thus the two admissible orbit data on $\mathcal{O}_K^+$ are

$$(\mathcal{O}_K^+, \mathcal{E}_{+,0}), \qquad (\mathcal{O}_K^+, \mathcal{E}_{+,1}).$$

Quantizing these data as in [LY23] produces the even and odd irreducible constituents of the oscillator representation for the corresponding polarization. Both have annihilator

$$\mathrm{Ann}_{U(\mathfrak{sp}(V))}(M) = I_0(\widetilde{\mathcal{O}}_{\min}) = I_0(\mathcal{O}_{\min}).$$

The analogous construction on $\mathcal{O}_K^-$ gives the contragredient oscillator constituents.

**Remark 49.1.** In the case of $SL_2(\mathbb{C})$, $\mathcal{N} = \{(x,y,z) \in \mathbb{C}^3 \mid x^2 + yz = 0\}$.

There is an isomorphism:

$$\mathbb{C}[a,b]^{\{\pm 1\}} = \mathbb{C}[a^2, ab, b^2] \cong \mathbb{C}[x,y,z]/(x^2+yz), \qquad x \mapsto ab,\ y \mapsto a^2,\ z \mapsto -b^2$$

$\mathcal{N}_\theta = \mathrm{Spec}((\mathbb{C}[a,b]/(a^2+b^2=0))^{\{\pm 1\}}) = (\mathbb{C})/\{\pm 1\} \cup (\mathbb{C})/\{\pm 1\}$.

Its Springer resolution is given by $T^*\mathbb{P}^1$, with exceptional fiber $\mathbb{P}^1$.

## 50. Structure of Cell Representations

2026-09-24

The argument below uses Perron–Frobenius theory to prove multiplicity one of the special representation in a Harish–Chandra cell. It is generated by ChatGPT Astra; the references have been checked against the sources, but the mathematical argument itself still needs review.

50.1. **Set-up.** Let $G$ be a real reductive group in Harish–Chandra's class and let $\mathfrak{g}$ be the complexified semisimple Lie algebra. Fix a regular infinitesimal character $\lambda \in \mathfrak{h}^*$, without assuming that $\lambda$ is integral. Let $Q$ be the root lattice and put

$$\Lambda = \lambda + Q, \qquad W_\lambda = W(\Lambda) = \{w \in W(\mathfrak{g},\mathfrak{h}) \mid w\lambda - \lambda \in Q\}.$$

Thus $W_\lambda$ is the integral Weyl group of the block. Let

$$\mathrm{Coh}_\Lambda(K_0(\mathcal{HC}))$$

be the coherent-continuation representation. We write $V = \mathrm{Coh}_\Lambda(K_0(\mathcal{HC}))$, and denote its basal (coherent-family) basis by $\{e_Y\}$. If $\Psi_Y$ is the coherent family corresponding to $e_Y$, then

$$(w\Psi)(\nu) = \Psi(w^{-1}\nu), \qquad w \in W_\lambda,\ \nu \in \Lambda.$$

For a regular dominant $\nu \in \Lambda$, the value $\Psi_Y(\nu)$ is an irreducible Harish–Chandra module whenever $e_Y$ is basal. We abbreviate

$$\mathrm{GKdim}(Y) := \mathrm{GKdim}\, \Psi_Y(\nu).$$

The value is independent of the chosen regular dominant parameter in the block, by the translation principle.

For a basal element $e_Y$, let $\langle Y\rangle$ denote the smallest basal $W_\lambda$-subrepresentation of $V$ containing $e_Y$. The left–right preorder is

$$Y \leq_{LR} X \quad \Longleftrightarrow \quad \langle Y\rangle \subseteq \langle X\rangle.$$

The associated equivalence relation is the Harish–Chandra cell relation. Fix a cell $C$ and $X \in C$. Put

$$V_{\leq X} = \operatorname{span}\{e_Y : Y \leq_{LR} X\}, \qquad V_{<X} = \operatorname{span}\{e_Y : Y <_{LR} X\},$$

and define the cell representation

$$V_C = V_{\leq X}/V_{<X}.$$

The cosets of $e_Y$ for $Y \in C$ form a basis of $V_C$.

Tensoring with finite-dimensional representations preserves associated varieties and gives the usual monotonicity of Gelfand–Kirillov dimension along the preorder. Consequently, if

$$d = \operatorname{GKdim}(X),$$

then $\operatorname{GKdim}(Y) \leq d$ for every $Y \leq_{LR} X$, and all elements of the cell $C$ have GK dimension $d$.

**Assumption 50.1** (Strict GK-dimension drop)**.** *For every $Y <_{LR} X$ one has*

$$\operatorname{GKdim}(Y) < \operatorname{GKdim}(X) = d.$$

*Equivalently, every proper lower cone contributes only modules of GK dimension strictly smaller than $d$.*

**Remark 50.2.** Assumption 50.1 is exactly the descent input needed below. It is often used implicitly in the older formulation of the cell argument, but it should be stated separately. Barbasch–Ma–Sun–Zhu point out that the corresponding strict inequality in Vogan's Corollary 14.11 has not been established in the literature in this generality; see their Remark 4.16 [BMSZ22]. The proof in this section is organized conditionally on Assumption 50.1. Section 58 supplies a proof of that input using positive projective functors and the Kildetoft–Mazorchuk positive-idempotent theorem.

50.2. **The fixed-degree Bernstein map.** Let $M$ be a finite-length Harish–Chandra module with a good filtration. If $\operatorname{GKdim}(M) \leq d$, write

$$\dim F_n M = \frac{b_d(M)}{d!} n^d + O(n^{d-1}),$$

where $b_d(M) = 0$ when $\operatorname{GKdim}(M) < d$. Equivalently, $b_d(M) = \operatorname{Deg}(M)$ when $\operatorname{GKdim}(M) = d$ and $b_d(M) = 0$ otherwise. The leading Hilbert coefficient is independent of the good filtration and is additive in short exact sequences; see Vogan [Vog78]. Hence $b_d$ defines an additive map on the Grothendieck group of modules whose GK dimension is at most $d$.

If $F$ is finite-dimensional, then

$$\operatorname{GKdim}(F \otimes M) = \operatorname{GKdim}(M), \qquad b_d(F \otimes M) = (\dim F) b_d(M).$$

Indeed, tensoring a good filtration with the finite-dimensional space $F$ multiplies the leading Hilbert coefficient by $\dim F$.

For a coherent family $\Psi$ in the lower cone of $X$, define on the lattice $\Lambda$

$$f_\Psi(\nu) = b_d(\Psi(\nu)).$$

The defining identity for coherent families is

$$F \otimes \Psi(\nu) = \sum_{\mu \in \mathrm{wt}(F)} \Psi(\nu + \mu),$$

where weights are counted with multiplicity. Applying $b_d$ gives

$$(\dim F) f_\Psi(\nu) = \sum_{\mu \in \mathrm{wt}(F)} f_\Psi(\nu + \mu).$$

These finite-difference relations are precisely the relations used in the Bernstein–King polynomiality theorem: $f_\Psi$ extends uniquely to a $W_\lambda$-harmonic polynomial, which we denote by

$$\beta_d(\Psi) \in \mathcal{H}_d \subseteq S(\mathfrak{h}).$$

Here $\mathcal{H}_d$ is the corresponding harmonic-polynomial space. For a basal family $\Psi_Y$ of GK dimension $d$, $\beta_d(\Psi_Y)$ is, up to a nonzero scalar depending only on the normalization, the Goldie-rank (or King character) polynomial of $\mathrm{Ann}(\Psi_Y(\nu))$; if $\mathrm{GKdim}(Y) < d$, it is zero. This is the Bernstein-degree form of Vogan's character-polynomial theorem [Vog82, Thm. 14.10], together with King's and Joseph's results [Kin81; Jos80].

**Lemma 50.3.** *The map $\beta_d$ is $W_\lambda$-equivariant:*

$$\beta_d(w\Psi) = w\beta_d(\Psi), \qquad w \in W_\lambda.$$

*Proof.* For $\nu \in \Lambda$ we compute directly from the coherent-continuation action:

$$\begin{aligned} f_{w\Psi}(\nu) &= b_d\big((w\Psi)(\nu)\big) \\ &= b_d\big(\Psi(w^{-1}\nu)\big) \\ &= f_\Psi(w^{-1}\nu). \end{aligned}$$

The right-hand side is the restriction to $\Lambda$ of the polynomial $(w\beta_d(\Psi))(\nu) = \beta_d(\Psi)(w^{-1}\nu)$. Since the root-lattice coset is Zariski dense in its affine span, equality on $\Lambda$ implies equality of the polynomial extensions. Thus

$$\beta_d(w\Psi) = w\beta_d(\Psi).$$

□

**Proposition 50.4.** *Under Assumption 50.1, $\beta_d$ vanishes on $V_{<X}$ and therefore induces a $W_\lambda$-equivariant map*

$$\overline{\beta}_C : V_C \longrightarrow \mathcal{H}_d.$$

*Proof.* If $Y <_{LR} X$, Assumption 50.1 gives $\mathrm{GKdim}(Y) < d$, so by definition $\beta_d(\Psi_Y) = 0$. Therefore $V_{<X} \subseteq \ker \beta_d$, and $\beta_d$ factors through the quotient $V_C$. Equivariance follows from Lemma 50.3. □

50.3. **The special representation in the image.** Let $D$ be the double cell attached to the Harish–Chandra cell $C$ in the highest-weight coherent-continuation representation. The attachment means that the annihilator left cell of every $\Psi_Y$, $Y \in C$, lies in the same double cell $D$; this is the content of the standard comparison theorem for Harish–Chandra and highest-weight coherent families [BMSZ22, Lem. 4.12].

Let $\sigma_C = \sigma_D$ be the special irreducible representation of $W_\lambda$ associated with $D$. Joseph's Goldie-rank theorem identifies the span of the Goldie-rank polynomials of the left cells in $D$ with an irreducible representation isomorphic to $\sigma_D$ [BMSZ22, Prop. 3.24]; see also the Springer-theoretic formulation in [BV83].

For $Y \in C$, the polynomial $p_Y = \beta_d(\Psi_Y)$ is a nonzero scalar multiple of the Goldie-rank polynomial attached to the annihilator left cell of $\Psi_Y$. Hence each $p_Y$ belongs to the $\sigma_C$-isotypic component of $\mathcal{H}_d$. We use only this isotypic assertion; at this stage we do not identify all the $p_Y$ with one fixed concrete copy of $\sigma_C$ inside $\mathcal{H}_d$. At this stage one must not conclude that the image is one copy of $\sigma_C$. Set

$$E = \operatorname{im} \overline{\beta}_C.$$

Then $E$ is a nonzero $W_\lambda$-subrepresentation contained in the $\sigma_C$-isotypic component. Since representations of the finite group $W_\lambda$ are semisimple, there is an integer $k \geq 1$ such that

$$E \simeq \sigma_C^{\oplus k}.$$

The purpose of the next subsection is to prove $k = 1$ rather than assuming it.

50.4. **Perron–Frobenius argument.** Let $S_\lambda$ be the simple reflections of $W_\lambda$. On the cell basis of $V_C$, the wall-crossing operators

$$\Theta_s = 1 + s, \qquad s \in S_\lambda,$$

have nonnegative integral matrices. Their directed graph generates the left–right cell, so the graph obtained from all $\Theta_s$ is strongly connected. This is the based-module form of the wall-crossing description of the Harish–Chandra cell preorder; see Vogan's coherent-continuation construction [Vog81]. The based-module and Perron–Frobenius formalism used below is the one of [KM16, §5.1].

Define

$$A = I + \sum_{s \in S_\lambda} \Theta_s = I + \sum_{s \in S_\lambda} (1 + s) \in \operatorname{End}(V_C).$$

The matrix of $A$ in the cell basis is nonnegative, has positive diagonal, and is irreducible. Positive diagonal entries make it primitive. Therefore the Perron–Frobenius theorem gives:

(1) a real eigenvalue $r > 0$;
(2) a vector

$$v = \sum_{Y \in C} a_Y e_Y, \qquad a_Y > 0,$$

with $Av = rv$;
(3) the eigenvalue $r$ has one-dimensional eigenspace on $V_C$.

Choose a regular dominant $\nu \in \Lambda$. Every $\Psi_Y(\nu)$ with $Y \in C$ is irreducible of GK dimension $d$, hence

$$p_Y(\nu) = b_d(\Psi_Y(\nu)) > 0.$$

It follows that

$$\overline{\beta}_C(v)(\nu) = \sum_{Y \in C} a_Y p_Y(\nu) > 0.$$

Thus $\overline{\beta}_C(v) \neq 0$. Since $\overline{\beta}_C$ is $W_\lambda$-equivariant,

$$A\overline{\beta}_C(v) = r\overline{\beta}_C(v).$$

The vector lies in $E \simeq \sigma_C^{\oplus k}$, so $r$ is an eigenvalue of $A$ on the irreducible representation $\sigma_C$.

Write

$$m = [\sigma_C : V_C]$$

for the multiplicity of $\sigma_C$ in the cell representation. Complete reducibility gives

$$V_C \simeq \sigma_C^{\oplus m} \oplus U,$$

where $U$ has no constituent isomorphic to $\sigma_C$. Because $r$ is an eigenvalue on $\sigma_C$, each of the $m$ copies contributes an independent $r$-eigenvector for $A$ on $V_C$. The Perron–Frobenius eigenspace is one-dimensional, so

$$m \leq 1.$$

On the other hand, $E \simeq \sigma_C^{\oplus k}$ is a quotient of $V_C$, so $k \leq m$. Since $k \geq 1$, we obtain

$$1 \leq k \leq m \leq 1, \qquad \text{hence} \qquad k = m = 1.$$

This is the mechanism of [KM16, Cor. 5.1]: a Perron–Frobenius element for a based module determines a special subquotient occurring with multiplicity one.

**Theorem 50.5** (Conditional multiplicity-one theorem)**.** *Assume Assumption 50.1. Then the special representation $\sigma_C$ occurs in the Harish–Chandra cell representation $V_C$ with multiplicity exactly one. Moreover, the Bernstein-degree map induces a surjection*

$$\overline{\beta}_C : V_C \twoheadrightarrow \sigma_C,$$

*and its image is exactly one copy of $\sigma_C$.*

*Proof.* Proposition 50.4 gives the map. The Goldie-rank polynomial description shows that its image is a nonzero sum of copies of $\sigma_C$. The Perron–Frobenius argument proves that the number of copies in the image and the multiplicity in the cell representation are both equal to one. □

50.5. **Logical status of the argument.** The proof has three independent inputs:

(1) the Bernstein–King theorem, which turns the fixed-degree function $\nu \mapsto b_d(\Psi(\nu))$ into a harmonic polynomial and identifies it with a Goldie-rank polynomial up to scalar;
(2) Joseph's theorem that the Goldie-rank polynomials in one double cell generate one irreducible special $W_\lambda$-representation;
(3) Assumption 50.1, which makes the fixed-degree map descend from the lower cone to the cell quotient.

The last item is the only step in this organization that is not supplied by the formal Bernstein-degree calculation; it is addressed separately in Section 58. Once descent is available, the argument does not assume that the image contains only one copy of $\sigma_C$; multiplicity one is forced by Perron–Frobenius.

The references and the quoted numbering (Vogan's Theorem 14.10 and Corollary 14.11, and Lemma 4.12, Proposition 3.24, Remark 4.16 of [BMSZ22]) have been checked against

the sources. The attribution of the Springer-correspondence formulation to [BV83] is the standard one, but the precise statement in that paper has not been read in full.

## 51. Atlas Parameter

2026-09-28

An atlas parameter is a triple $p = (x, \lambda, \nu)$.

Here

- $x \in K\backslash G/B$ is a $KGB$-element, it essentially encodes a Cartan $H = TA$ and a chosen positive system;
- $\lambda \in (X^* + \rho)/(1 - \theta_x)X^*$ encodes a genuine character of the $\rho$-double cover of $T$;
- $\nu \in (X^* \otimes \mathbb{Q})^{-\theta_x}$ encodes a character of $A$.

The infinitesimal character of the atlas parameter is given by

$$\gamma = \frac{1 + \theta_x}{2}\lambda + \nu \in X^* \otimes \mathbb{Q}.$$

For $G(\mathbb{R}) = SL_2(\mathbb{R})$, identify $X^* = \mathbb{Z}$ using the fundamental weight. Then the positive root is 2, $\rho = 1$, and the Weyl group acts by $s \mapsto -s$.

There are three KGB elements:

| $x$ | $H$ | $T$ | $\theta_x$ |
|---|---|---|---|
| $0, 1$ | $SO(2)$ | $SO(2)$ | $+1$ |
| $2$ | $\mathbb{R}^\times$ | $\{\pm 1\}$ | $-1$ |

*Compact Cartan.* For $x = 0, 1$, we have $\theta_x = 1$, so $\nu = 0$ and

$$\gamma = \frac{1 + \theta_x}{2}\lambda + \nu = \lambda.$$

For $n \geq 2$, the two discrete series representations are

$$D_n^+ : \quad p = (0, n-1, 0), \quad K\text{-weights} : \quad n, n+2, n+4, \ldots,$$
$$D_n^- : \quad p = (1, n-1, 0), \quad K\text{-weights} : \quad -n, -n-2, -n-4, \ldots.$$

Each $K$-type occurs with multiplicity one, and both representations have infinitesimal character $[n-1] \in \mathbb{C}/\{\pm 1\}$. For $n = 1$, these parameters give the two limits of discrete series.

*Split Cartan.* For $x = 2$, we have $\theta_x = -1$ and

$$H = \mathbb{R}^\times = \{\pm 1\}\mathbb{R}_{>0}.$$

Consequently,

$$\lambda \in (\mathbb{Z} + 1)/2\mathbb{Z}, \qquad \nu = s \in \mathbb{Q}, \qquad \gamma = s.$$

Write

$$I(\varepsilon, s) = \mathrm{Ind}_{B(\mathbb{R})}^{SL_2(\mathbb{R})}\big(\mathrm{sgn}(a)^\varepsilon |a|^s\big), \qquad \varepsilon \in \{0, 1\},$$

where induction is normalized and $a$ denotes the diagonal coordinate in $\mathrm{diag}(a, a^{-1})$. The relation with Atlas parameters is

$$\varepsilon \equiv \lambda - 1 \pmod 2, \qquad \nu = s.$$

Thus convenient representatives are

$$I(0, s) : \quad (2, 1, s), \qquad I(1, s) : \quad (2, 2, s).$$

Their $K$-weights are respectively $2\mathbb{Z}$ and $2\mathbb{Z} + 1$, each with multiplicity one.

Some examples are:

| Atlas parameter | Standard module | Structure | Infinitesimal character |
|---|---|---|---|
| $(0,1,0)$ | $D_2^+$ | irreducible | [1] |
| $(1,1,0)$ | $D_2^-$ | irreducible | [1] |
| $(2,1,0)$ | $I(0,0)$ | irreducible | [0] |
| $(2,2,0)$ | $I(1,0)$ | $D_1^+ \oplus D_1^-$ | [0] |
| $(2,1,1)$ | $I(0,1)$ | reducible, quotient $\mathbf{1}$ | [1] |
| $(2,2,1)$ | $I(1,1)$ | irreducible | [1] |

In particular, there is an exact sequence

$$0 \longrightarrow D_2^+ \oplus D_2^- \longrightarrow I(0,1) \longrightarrow \mathbf{1} \longrightarrow 0.$$

Thus the final Atlas parameter $(2,1,1)$ labels the trivial representation as its irreducible Langlands quotient, although its standard module is reducible. The parameter $(2,2,0)$ is non-final: its standard module is the direct sum of the two limits of discrete series.

## 52. Calculation of Lusztig-Vogan Modules

2026-09-29

Let $G$ be a connected complex reductive algebraic group, and fix a Borel subgroup

$$B \supseteq T$$

with maximal torus $T$. Let $\alpha$ be a simple root, and let $P_\alpha$ be the corresponding minimal parabolic subgroup containing $B$.

Write

$$U_\alpha := R_u(P_\alpha)$$

for the unipotent radical of $P_\alpha$, and define

$$H_\alpha := (\ker \alpha)^0 \subset T.$$

Then

$$\widetilde{G}_\alpha := P_\alpha/(U_\alpha H_\alpha)$$

is a connected semisimple group of rank one. Hence it is isogenous to $SL_2(\mathbb{C})$. Its adjoint quotient is

$$G_\alpha^{\mathrm{ad}} := \widetilde{G}_\alpha / Z(\widetilde{G}_\alpha) \simeq PGL_2(\mathbb{C}).$$

It is this adjoint rank-one group that acts effectively on the rank-one fiber below.

**The rank-one fiber.** Consider the natural projection

$$\pi_\alpha : \mathcal{B} = G/B \longrightarrow G/P_\alpha.$$

This is a $\mathbb{P}^1$-fibration, since

$$P_\alpha/B \simeq \mathbb{P}^1.$$

Take

$$x = B \in G/B, \qquad z = \pi_\alpha(x) = P_\alpha.$$

Then

$$\pi_\alpha^{-1}(z) = P_\alpha/B \simeq \mathbb{P}^1.$$

The action of $P_\alpha$ on this fiber factors through the rank-one quotient, and its effective action is

$$G_\alpha^{\mathrm{ad}} \simeq PGL_2(\mathbb{C}) \curvearrowright \mathbb{P}^1.$$

Thus every simple root $\alpha$ gives a canonical rank-one $PGL_2$-geometry inside the flag variety.

**Reduction of the $K$-orbit geometry.** Let $K \subset G$ be a symmetric subgroup, and let

$$\mathcal{O} = K \cdot x$$

be the $K$-orbit through $x$. Put

$$K_z := \mathrm{Stab}_K(z).$$

Since $\pi_\alpha$ is $K$-equivariant,

$$\mathcal{O} \cap \pi_\alpha^{-1}(z) = K_z \cdot x.$$

The action of $K_z$ on the fiber factors through its image

$$\overline{K}_z \subset G_\alpha^{\mathrm{ad}} \simeq PGL_2(\mathbb{C}).$$

Hence

$$\boxed{\mathcal{O} \cap \pi_\alpha^{-1}(z) = \overline{K}_z \cdot x.}$$

Therefore the local behavior of a $K$-orbit in the $\alpha$-direction is completely determined by the rank-one action

$$\overline{K}_z \curvearrowright \mathbb{P}^1.$$

The action of the simple Hecke generator $T_{s_\alpha}$ in the Lusztig–Vogan module is determined by this rank-one geometry, together with the behavior of the equivariant local system on the relevant $K$-orbits.

The singularities of the closures of these $K$-orbits, which control the local-system bookkeeping, are computed in [LV83]; the duality between the $K$-orbit module and the character module on which the $T_{s_\alpha}$ act is the content of [Vog82].

**Computing the rank-one stabilizer $\bar{K}_z$.** Assume that $K = G^\theta$ for an involution $\theta$, and choose the representative $x = B$ so that the Cartan $T \subset B$ is $\theta$-stable. Then

$$z = \pi_\alpha(x) = P_\alpha, \qquad K_z = K \cap P_\alpha.$$

Let

$$\mathfrak{q}_\alpha := \mathfrak{p}_\alpha / (\mathfrak{u}_\alpha + \mathfrak{h}_\alpha) \simeq \mathfrak{sl}_2.$$

Choose the usual basis

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The image of $B$ in the rank-one quotient is the standard Borel

$$\mathfrak{b}_\alpha = \mathbb{C}H \oplus \mathbb{C}E.$$

Thus the point $x = B$ corresponds to the standard point

$$x = [1 : 0] \in \mathbb{P}^1.$$

The Lie algebra of the effective stabilizer is the image

$$\bar{\mathfrak{k}}_z = \mathrm{Im}\left(\mathfrak{k} \cap \mathfrak{p}_\alpha \longrightarrow \mathfrak{q}_\alpha\right).$$

The different root types can now be read off from this rank-one calculation.

*Compact imaginary root.* Suppose

$$\theta(\alpha) = \alpha$$

and $\alpha$ is compact imaginary. Then, after normalizing the root vectors,

$$\theta(E) = E, \qquad \theta(F) = F, \qquad \theta(H) = H.$$

Hence

$$\bar{\mathfrak{k}}_z = \mathfrak{sl}_2.$$

Therefore

$$\bar{K}_z = PGL_2(\mathbb{C}),$$

and its action on $\mathbb{P}^1$ is transitive:

$$\boxed{\bar{K}_z \cdot x = \mathbb{P}^1.}$$

Thus in the compact imaginary case,

$$\mathcal{O} \cap \pi_\alpha^{-1}(z) = \mathbb{P}^1.$$

*Noncompact imaginary root.* Suppose

$$\theta(\alpha) = \alpha$$

and $\alpha$ is noncompact imaginary. Then

$$\theta(E) = -E, \qquad \theta(F) = -F, \qquad \theta(H) = H.$$

Consequently

$$\boxed{\bar{\mathfrak{k}}_z = \mathbb{C}H.}$$

Thus

$$\bar{K}_z^0 = T_\alpha$$

is the standard maximal torus of $PGL_2$.

In homogeneous coordinates,

$$T_\alpha = \left\{ \begin{bmatrix} t & 0 \\ 0 & t^{-1} \end{bmatrix} \right\}$$

acts by

$$[u : v] \longmapsto [tu : t^{-1}v].$$

Its orbit decomposition is

$$\boxed{\mathbb{P}^1 = \{[1:0]\} \sqcup \{uv \neq 0\} \sqcup \{[0:1]\}.}$$

The middle orbit is isomorphic to $\mathbb{C}^\times$.

Since

$$x = [1 : 0],$$

the point $x$ is a torus-fixed point.

Hence for the connected stabilizer,

$$T_\alpha \cdot x = \{x\}.$$

The full group $\bar{K}_z$ may contain the nontrivial Weyl element of

$$N(T_\alpha)/T_\alpha \simeq \mathbb{Z}/2\mathbb{Z}.$$

It exchanges the two fixed points

$$[1 : 0] \longleftrightarrow [0 : 1].$$

Thus schematically,

$$\begin{array}{c|l} \bar{K}_z = T_\alpha & \{[1:0]\},\ \{[0:1]\} \text{ are separate orbits} \\ \bar{K}_z = N(T_\alpha) & \{[1:0],[0:1]\} \text{ is one orbit.} \end{array}$$

This is the geometric origin of the type I/type II distinction for noncompact imaginary roots.

*Real root.* Now suppose

$$\theta(\alpha) = -\alpha.$$

After normalizing root vectors, we may take

$$\theta(E) = F, \qquad \theta(F) = E, \qquad \theta(H) = -H.$$

Therefore

$$\boxed{\bar{\mathfrak{k}}_z = \mathbb{C}(E+F).}$$

Since

$$E + F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is semisimple, $\bar{K}_z^0$ is again a maximal torus:

$$\bar{K}_z^0 = S_\alpha \simeq \mathbb{C}^\times.$$

However this torus is different from the standard torus above. Indeed,

$$E + F \notin \mathfrak{b}_\alpha,$$

so $S_\alpha$ does not fix

$$x = [1:0].$$

The eigenlines of $E+F$ are

$$p_+ = [1:1], \qquad p_- = [1:-1].$$

These are exactly the two fixed points of $S_\alpha$ on $\mathbb{P}^1$. Hence

$$\boxed{\mathbb{P}^1 = \{p_+\} \sqcup S_\alpha \cdot x \sqcup \{p_-\},}$$

with

$$S_\alpha \cdot x = \mathbb{P}^1 - \{p_+, p_-\} \simeq \mathbb{C}^\times.$$

Therefore

$$\boxed{\mathcal{O} \cap \pi_\alpha^{-1}(z) = \mathbb{P}^1 - \{p_+, p_-\}.}$$

Again the full group may be either

$$S_\alpha \qquad \text{or} \qquad N(S_\alpha).$$

If it is $N(S_\alpha)$, the nontrivial Weyl element exchanges the two boundary points $p_+$ and $p_-$.

Thus the type I/type II distinction records whether these two boundary points remain two different $K$-orbits or become one $K$-orbit.

Notice that for real roots there is also an additional *local-system condition*, namely Vogan's parity condition [Vog81]. The orbit geometry itself is always the open-torus geometry above; the parity condition determines whether the relevant equivariant local system extends across the boundary and hence whether the inverse Cayley transform occurs.

*Complex root.* Finally suppose that $\alpha$ is complex:

$$\theta(\alpha) \neq \pm\alpha.$$

First assume

$$\theta(\alpha) > 0.$$

For

$$X_\alpha \in \mathfrak{g}_\alpha,$$

the vector

$$X_\alpha + \theta(X_\alpha)$$

belongs to $\mathfrak{k} \cap \mathfrak{p}_\alpha$. Modulo

$$\mathfrak{u}_\alpha + \mathfrak{h}_\alpha,$$

the $\theta(X_\alpha)$-term disappears, since

$$\theta(\alpha) > 0, \qquad \theta(\alpha) \neq \alpha.$$

Thus its image is $E$.

The Cartan direction $H$ also survives, whereas the $F$-direction does not. Consequently

$$\boxed{\bar{\mathfrak{k}}_z = \mathbb{C}H \oplus \mathbb{C}E = \mathfrak{b}_\alpha.}$$

Hence

$$\bar{K}_z = B_\alpha.$$

The Borel $B_\alpha$ acts on $\mathbb{P}^1$ with two orbits:

$$\boxed{\mathbb{P}^1 = \{[1:0]\} \sqcup \big(\mathbb{P}^1 - \{[1:0]\}\big).}$$

The second orbit is $\mathbb{A}^1$.

Since $x = [1:0]$,

$$\boxed{\bar{K}_z \cdot x = \{x\}.}$$

If instead

$$\theta(\alpha) < 0,$$

the same calculation gives the opposite Borel:

$$\bar{\mathfrak{k}}_z = \mathbb{C}H \oplus \mathbb{C}F, \qquad \bar{K}_z = B_\alpha^-.$$

Now $B_\alpha^-$ fixes the opposite point $[0:1]$, while

$$x = [1:0]$$

lies in its open orbit. Hence

$$\boxed{\bar{K}_z \cdot x = \mathbb{P}^1 - \{[0:1]\} \simeq \mathbb{A}^1.}$$

**Rank-one summary.** The essential calculations are therefore

| root type | $\bar{\mathfrak{k}}_z$ | $\bar{K}_z^0 \cdot x$ |
|---|---|---|
| compact imaginary | $\mathfrak{sl}_2$ | $\mathbb{P}^1$ |
| noncompact imaginary | $\mathbb{C}H$ | $\{x\}$ |
| real | $\mathbb{C}(E+F)$ | $\mathbb{P}^1 - \{p_+, p_-\} \simeq \mathbb{C}^\times$ |
| complex, $\theta\alpha > 0$ | $\mathbb{C}H \oplus \mathbb{C}E$ | $\{x\}$ |
| complex, $\theta\alpha < 0$ | $\mathbb{C}H \oplus \mathbb{C}F$ | $\mathbb{A}^1$. |

Thus the case-by-case Lusztig–Vogan formulas ultimately come from the very small list of subgroup actions

$$PGL_2, \qquad T, \qquad N(T), \qquad B, \qquad B^-$$

on

$$\mathbb{P}^1 = PGL_2/B.$$

The Hecke-module (Kazhdan–Lusztig) description of characters in which the operators $T_{s_\alpha}$ act is proved in the integral case in [Vog83].

## 53. Hamiltonian Reduction of Differential Operators

2026-10-13

$H$ be an algebraic group over $\mathbb{C}$, and $\pi : \tilde{Y} \to Y$ is a principal $H$-bundle ($H$ acts on the right).

For any character $\chi \in X^*(H)$, we have the associated line bundle

$$\mathcal{L}_\chi := \tilde{Y} \times^H \mathbb{C}_\chi \longrightarrow Y,$$

where $H$ acts on $\tilde{Y} \times \mathbb{C}_\chi$ by $h \cdot (y, z) = (yh^{-1}, \chi(h)z)$.x

There is an identification of $\Gamma(Y, \mathcal{L}_\chi)$ with $\{F \in \Gamma(\tilde{Y}, \mathcal{O}_{\tilde{Y}}) \,|\, F(yh) = \chi^{-1}(h)F(y)\}$ by:

$$F \longmapsto (y \mapsto [y; F(y)]) \in \Gamma(Y, \mathcal{L}_\chi).$$

**Example 53.1.** $G = \mathrm{SL}_2(\mathbb{C})$, $H = \{\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \,|\, t \in \mathbb{C}^*\}$, $B = \{\begin{pmatrix} t & n \\ 0 & t^{-1} \end{pmatrix} \,|\, t \in \mathbb{C}^*, n \in \mathbb{C}\}$. Let $\chi_n : B \to \mathbb{C}^*$, $\begin{pmatrix} t & n \\ 0 & t^{-1} \end{pmatrix} \mapsto t^n$. Then we have the associated line bundle $\mathcal{L}_{\chi_n} = G \times^B \mathbb{C}_{\chi_n} \longrightarrow G/B$.

If we identify $G/B \xrightarrow{\sim} \mathbb{P}^1$, $gB \mapsto [g \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}]$. Then we have identification of $\mathcal{L}_n$ with $\mathcal{O}_{\mathbb{P}^1}(-n)$. Actually, we only need to check the case when $n = 1$:

$$\begin{array}{rccc} \phi\colon & \mathcal{L}_1 & \longrightarrow & \mathcal{O}_{\mathbb{P}^1}(-1) \\ & [g, z] & \longmapsto & ([g \begin{pmatrix} 1 \\ 0 \end{pmatrix}], zg \begin{pmatrix} 1 \\ 0 \end{pmatrix}) \end{array}$$

(Check that $\phi$ is well-defined and bijective.)

Let $\mathcal{D}_{\tilde{Y}}$ be the sheaf of differential operators on $\tilde{Y}$.

We have a quantum comoment map $\Phi : \mathfrak{h} \to \mathcal{D}_{\tilde{Y}}$ defined by

$$\Phi(\xi)(f) := \frac{d}{dt}\Big|_{t=0} f(y \cdot \exp(t\xi)), \qquad f \in \mathcal{O}_{\tilde{Y}}, \quad y \in \tilde{Y}, \quad \xi \in \mathfrak{h}.$$

The Hamiltonian reduction of $\mathcal{D}_{\tilde{Y}}$ with respect to the $H$-action and homomorphism $\lambda \in (\mathfrak{h}/[\mathfrak{h},\mathfrak{h}])^*$ is defined as follows:

$$\mathcal{D}_{\tilde{Y}} /\!\!/\!\!/_\lambda H := \pi_* \left(\mathcal{D}_{\tilde{Y}}/\mathcal{D}_{\tilde{Y}} \cdot (\Phi - \lambda)(\mathfrak{h})\right)^H,$$

For any $F \in \Gamma(Y, \mathcal{L}_\chi)$, we have:

$$\Phi(\xi)(F) = \frac{d}{dt}\Big|_{t=0} F(y \cdot \exp(t\xi)) = \frac{d}{dt}\Big|_{t=0} \chi^{-1}(\exp(t\xi))F(y) = -d\chi(\xi)F(y).$$

So the action of $\mathcal{D}_{\tilde{Y}}$ on $\Gamma(Y, \mathcal{L}_\chi)$ factors through the Hamiltonian reduction $\mathcal{D}_{\tilde{Y}} /\!\!/\!\!/_{-d\chi} H$.

**Remark 53.2.** Actually, we have an isomorphism of algebras:

$$\mathcal{D}_{\tilde{Y}} /\!\!/\!\!/_{-d\chi} H \simeq \mathcal{D}_{\mathcal{L}_\chi},$$

where $\mathcal{D}_{\mathcal{L}_\chi}$ is the sheaf of differential operators on the line bundle $\mathcal{L}_\chi$.

## 54. Orbits on Partial Flag Variety

2026-10-17

We summarize some basic facts about the orbits of a symmetric subgroup $K$ on a partial flag variety $\mathcal{P} \simeq G/P$, where $P$ is a parabolic subgroup of $G$.

$G$ is connected complex reductive algebraic group. $\sigma$ is a real form on $G$, $\theta$ is the corresponding Cartan involution, and $K = G^\theta$ is the symmetric subgroup.

**Proposition 54.1.** *If $P_1, P_2 \subset G$ are real parabolic subgroups, then:*

$$\boxed{P_1 \sim_{G(\mathbb{R})} P_2 \iff P_1 \sim_{K_\mathbb{R}} P_2 \iff P_1 \sim_K P_2.}$$

*Proof.* If $P_1 \sim_{G(\mathbb{R})} P_2$, since $G(\mathbb{R}) = K_\mathbb{R} \cdot P_1$, we can see that $P_1 \sim_{K_\mathbb{R}} P_2$.

If $P_1 \sim_K P_2$, then

$$P_2 = kP_1k^{-1}, \quad k \in K.$$

Because $P_i$ are real,

$$P_2 = \sigma(P_2) = \sigma(k)\sigma(P_1)\sigma(k^{-1}) = \sigma(k)P_1\sigma(k^{-1}).$$

Hence

$$\sigma(k)^{-1}k \in N_G(P_1) = P_1.$$

Suppose $k = k_0 \exp(iX)$ for $k_0 \in K_\mathbb{R}$ and $X \in \mathfrak{k}_\mathbb{R}$. Then

$$\sigma(k)^{-1}k = \exp(2iX) \in P_1.$$

Since $P_1$ is the stabilizer of some flag in some faithful finite dimensional representation of $G$, $\exp(2iX)$ is positive semisimple, hence

$$\exp(iX) \in P_1.$$

Therefore

$$P_2 = kP_1k^{-1} = k_0 \exp(iX)P_1 \exp(-iX)k_0^{-1} = k_0P_1k_0^{-1},$$

which shows that $P_1 \sim_{K_\mathbb{R}} P_2$. □

**Proposition 54.2.** *If $P_1, P_2 \subset G$ are $\theta$-stable parabolic subgroups, then:*

$$\boxed{P_1 \sim_{G(\mathbb{R})} P_2 \iff P_1 \sim_{K_{\mathbb{R}}} P_2 \iff P_1 \sim_K P_2.}$$

*Proof.* Very similar to the proof of the previous proposition. □

**Proposition 54.3.** *If $P$ is a real parabolic subgroup of $G$, then the $K$-orbit $K \cdot P$ is the unique open orbit $\mathcal{P}$.*

*Proof.* If $P$ is a real parabolic subgroup in $\mathcal{P}$, then

$$T_P(K \cdot P) = (\mathfrak{k} + \mathfrak{p})/\mathfrak{p} = \mathfrak{g}/\mathfrak{p} = T_P\mathcal{P}.$$

□

**Proposition 54.4.** *If $P$ is a $\theta$-stable parabolic subgroup of $G$, then the $K$-orbit $K \cdot P$ is closed in $\mathcal{P}$. Conversely, if there exist a $\theta$-stable parabolic subgroup in $\mathcal{P}$, then any closed $K$-orbit in $\mathcal{P}$ consists of $\theta$-stable parabolic subgroups.*

*Proof.* If $P$ is a $\theta$-stable parabolic subgroup in $\mathcal{P}$, then $K \cap P$ is a parabolic subgroup of $K$. Hence $K/(K \cap P)$ is projective, which shows that the $K$-orbit $K \cdot P$ is closed in $\mathcal{P}$.

Conversely, write $\mathcal{P} = G \cdot P$, where $\theta(P) = P$, and suppose that $Q = K \cdot P_1$ is closed in $\mathcal{P}$. Let $\mathcal{B}$ denote the full flag variety of $G$, and let

$$\pi : \mathcal{B} \longrightarrow \mathcal{P}$$

be the natural projection, sending a Borel subgroup to the unique parabolic subgroup in $\mathcal{P}$ containing it.

Since $Q$ is closed, $\pi^{-1}(Q)$ is a nonempty closed $K$-stable subvariety of $\mathcal{B}$. It therefore contains a closed $K$-orbit $\widetilde{Q}$. By $K$-equivariance, $\pi(\widetilde{Q}) = Q$. Thus we may choose a Borel subgroup $B \in \widetilde{Q}$ with $B \subset P_1$. By the closed-orbit criterion for the full flag variety, $\theta(B) = B$.

Write $P_1 = gPg^{-1}$. Since $\theta(P) = P$, we have

$$\theta(P_1) = \theta(g)P\theta(g)^{-1} \in \mathcal{P}.$$

Moreover,

$$B = \theta(B) \subset \theta(P_1).$$

Hence $P_1$ and $\theta(P_1)$ are parabolic subgroups of the same type containing the same Borel subgroup $B$. By uniqueness of the parabolic subgroup of a fixed type containing $B$, they coincide. Thus $\theta(P_1) = P_1$.

Finally, for every $k \in K$,

$$\theta(kP_1k^{-1}) = k\theta(P_1)k^{-1} = kP_1k^{-1}.$$

Therefore every parabolic subgroup in $Q$ is $\theta$-stable. □

## 55. Geometric Explanation of $\tau$-Invariants

2026-10-19

55.1. **Conventions and full-flag localization.** Let $G$ be a connected complex reductive algebraic group, equipped with a real form $\sigma$ and a corresponding algebraic Cartan involution $\theta$. Put $K = G^\theta$ and $\mathfrak{g} = \mathrm{Lie}(G)$. We allow $K$ to be disconnected. Equivariance of a $\mathcal{D}$-module will always mean *strong algebraic equivariance*: the differentiated group action agrees with the action coming from differential operators.

Fix a Borel subgroup $B \subseteq G$ and a Cartan subgroup $H \subseteq B$. Let $\Phi^+$ be the positive roots occurring in $\mathfrak{b}$, let $\Pi$ be the simple roots, and put

$$\rho = \frac{1}{2} \sum_{\beta \in \Phi^+} \beta.$$

For $J \subseteq \Pi$, let $P_J \supseteq B$ be the standard parabolic subgroup with Levi simple roots $J$, and write

$$\mathcal{B} = G/B, \qquad \mathcal{P}_J = G/P_J, \qquad \pi_J : \mathcal{B} \longrightarrow \mathcal{P}_J.$$

Thus $\mathcal{P}_\varnothing = \mathcal{B}$ and $\mathcal{P}_\Pi$ is a point. For a singleton, write $\mathcal{P}_\alpha = \mathcal{P}_{\{\alpha\}}$ and $\pi_\alpha = \pi_{\{\alpha\}}$. The fibers of $\pi_\alpha$ are $\mathbb{P}^1$. More generally, the fiber of $\pi_J$ is the full flag variety of the Levi subgroup $L_J$; in particular, it is connected and simply connected.

All sheaves $\mathcal{D}_Y$ below are the *ordinary untwisted* sheaves of algebraic differential operators, acting on the left. Define

$$A_J = \Gamma(\mathcal{P}_J, \mathcal{D}_{\mathcal{P}_J}), \qquad I_J = \ker\bigl(U(\mathfrak{g}) \longrightarrow A_J\bigr).$$

Before imposing a localization theorem, distinguish the image algebra

$$R_J = \operatorname{im}\bigl(U(\mathfrak{g}) \longrightarrow A_J\bigr) \simeq U(\mathfrak{g})/I_J$$

from the full algebra $A_J$ of global differential operators. Their equality in the ordinary untwisted setting is proved in Proposition 55.2; it is not an assumption about arbitrary twisted differential operators. For $\mathcal{P} = \mathcal{P}_J$, the notation of the original setup is $J = S_{\mathfrak{l}}$, and we also write $I_{\mathcal{P}} = I_J$. Set $I_{\mathcal{B}} = I_\varnothing$, $I_\alpha = I_{\{\alpha\}}$, and $R = U(\mathfrak{g})/I_{\mathcal{B}}$. The ordinary central character is $\chi_\rho$, the infinitesimal character of the trivial representation. This is what "trivial infinitesimal character" means here; it is not the singular parameter $\chi_0$.

Write $\mathrm{HC}_\rho(\mathfrak{g}, K)$ for finitely generated Harish–Chandra modules on which the center acts by $\chi_\rho$, and $\mathrm{Coh}_K(\mathcal{D}_Y)$ for coherent strongly $K$-equivariant left $\mathcal{D}_Y$-modules.

**Theorem 55.1** (Beilinson–Bernstein localization: standard input)**.** *We have $R \simeq \Gamma(\mathcal{B}, \mathcal{D}_{\mathcal{B}})$, and*

$$\mathrm{Loc}_{\mathcal{B}}(X) = \mathcal{D}_{\mathcal{B}} \otimes_R X, \qquad \Gamma_{\mathcal{B}}(\mathcal{M}) = \Gamma(\mathcal{B}, \mathcal{M})$$

*are mutually inverse exact equivalences. On equivariant finitely generated objects this gives*

$$\mathrm{HC}(R, K) \simeq \mathrm{Coh}_K(\mathcal{D}_{\mathcal{B}}).$$

*There is also an equivalence between all $R$-modules and quasi-coherent left $\mathcal{D}_{\mathcal{B}}$-modules, which we use in the proofs below.*

This is the localization theorem of [BB81]. In particular, $\Gamma_{\mathcal{B}}$ is exact and detects zero objects. The algebraic $\tau$-invariant theorem stated in Subsection 55.3 is the other substantial representation-theoretic input. The geometric descent results below are deduced from these inputs.

55.2. **Untwisted localization on partial flag varieties.** For a smooth morphism $f : X \to Y$, use the unshifted inverse image

$$f^\circ \mathcal{N} = \mathcal{D}_{X \to Y} \otimes_{f^{-1}\mathcal{D}_Y} f^{-1}\mathcal{N}, \qquad \mathcal{D}_{X \to Y} = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{D}_Y.$$

Its underlying $\mathcal{O}_X$-module is $f^*\mathcal{N}$. For smooth $f$, this functor is exact. No cohomological shift is included in $f^\circ$.

**Proposition 55.2** (Partial-flag localization and compatibility)**.** *For ordinary untwisted differential operators,*

$$A_J \simeq U(\mathfrak{g})/I_J.$$

*The functors* $\Gamma_{\mathcal{P}_J}$ *and* $\mathrm{Loc}_{\mathcal{P}_J} = \mathcal{D}_{\mathcal{P}_J} \otimes_{A_J} (-)$ *are exact inverse equivalences, including on the corresponding coherent and equivariant categories. Moreover, for* $I_J X = 0$*,*

$$\mathrm{Loc}_{\mathcal{B}}(X) \simeq \pi_J^{\circ} \mathrm{Loc}_{\mathcal{P}_J}(X). \tag{55.1}$$

*The essential image of* $\pi_J^{\circ}$ *corresponds under full-flag localization precisely to* $R$*-modules annihilated by* $I_J/I_{\mathcal{B}}$*.*

*Proof.* Put $Y = \mathcal{P}_J$, $\pi = \pi_J$, and $\mathcal{E} = \mathcal{D}_{\mathcal{B}\to Y}$. The global section $1 \otimes 1$ generates $\mathcal{E}$ as a left $\mathcal{D}_{\mathcal{B}}$-module. Indeed, in smooth local coordinates, vector fields on the base have local lifts, and applying the lifted differential operators to $1 \otimes 1$ produces all base differential operators. Hence there is a surjection

$$\mathcal{D}_{\mathcal{B}} \twoheadrightarrow \mathcal{E}, \qquad \ker = \mathcal{D}_{\mathcal{B}}\, \mathcal{T}_{\mathcal{B}/Y}, \tag{55.2}$$

where $\mathcal{T}_{\mathcal{B}/Y}$ is the relative tangent sheaf and the product denotes the left ideal it generates. The kernel description is immediate in local product coordinates: quotienting kills the vertical derivatives.

Since $\pi_* \mathcal{O}_{\mathcal{B}} = \mathcal{O}_Y$, the projection formula gives

$$\Gamma(\mathcal{B}, \mathcal{E}) = \Gamma(Y, \mathcal{D}_Y) = A_J.$$

Apply the exact functor $\Gamma_{\mathcal{B}}$ to (55.2). This proves that $R \to A_J$ is surjective, hence $A_J = U(\mathfrak{g})/I_J$.

For any quasi-coherent left $\mathcal{D}_Y$-module $\mathcal{N}$, the same projection formula gives

$$\Gamma(\mathcal{B}, \pi^{\circ}\mathcal{N}) = \Gamma(Y, \mathcal{N}), \tag{55.3}$$

compatibly with the $U(\mathfrak{g})$-actions. Smooth surjective pullback is exact and faithful. Thus (55.3) and Theorem 55.1 show that $\Gamma_Y$ is exact and detects zero objects.

For completeness, this implies the claimed equivalence directly. If $V$ is an $A_J$-module, take a presentation by free $A_J$-modules. Exactness of $\Gamma_Y$ and its commutation with direct sums give

$$\Gamma_Y(\mathcal{D}_Y \otimes_{A_J} V) \simeq V.$$

For any $\mathcal{N}$, the evaluation map $\mathcal{D}_Y \otimes_{A_J} \Gamma_Y \mathcal{N} \to \mathcal{N}$ becomes an isomorphism after applying $\Gamma_Y$. Its kernel and cokernel therefore vanish. These constructions preserve equivariance and restrict to the finitely generated/coherent categories.

Finally, both sides of (55.1) have global sections $X$ as $R$-modules, so Theorem 55.1 identifies them. Equivalently,

$$\mathrm{Loc}_{\mathcal{B}}(A_J) \simeq \mathcal{D}_{\mathcal{B}\to Y},$$

and tensoring this identity with $X$ gives (55.1). The assertion about the essential image follows. In particular, this image is closed under subobjects and quotients, and pullback preserves simplicity here. □

**Remark 55.3** (Scope of the equivalence and a conflicting statement in the source)**.** Two assertions must be distinguished:

$$\mathrm{Mod}(A_J) \simeq \mathrm{Mod}(\mathcal{D}_{\mathcal{P}_J}), \qquad R_J = A_J.$$

The first is $\mathcal{D}$-affinity. The second concerns the image of $U(\mathfrak{g})$. If, for a twisted sheaf $\mathcal{D}^\lambda_{\mathcal{P}_J}$, its image $R^\lambda_J$ is smaller than $A^\lambda_J = \Gamma(\mathcal{D}^\lambda_{\mathcal{P}_J})$, extension of scalars from $R^\lambda_J$ cannot be identified with the localization equivalence over $A^\lambda_J$. Outside the appropriate localization range, global sections itself can also fail to be an equivalence.

For the *ordinary untwisted* sheaves used in this note, both assertions hold by Proposition 55.2. An independent published reference is [Yah17, Section 2.3.1, Definition 2.24, Proposition 2.30, and Lemma 4.1]: in that paper the ordinary sheaf is $\mathcal{D}^0$, and regularity and antidominance are tested on $\lambda - \rho$. Thus $\lambda = 0$ is regular and antidominant. Lemma 4.1 gives $R_J = A_J$, and Proposition 2.30 gives the abelian localization equivalence. The cyclic-transfer proof above also appears in [Bie90, Chapter I, Proposition 5.6].

The paragraph preceding Proposition 3.12 in [CNT12] states that $U(\mathfrak{g}) \to A_J$ need not be surjective, immediately after defining $\mathcal{D}_{\mathcal{P}_J}$ as ordinary differential operators. Taken literally with that untwisted definition, this conflicts with the preceding proof and the published localization results just cited. It should therefore not be used as a general counterstatement to Proposition 55.2. The failure it describes does occur for more general twisted operators; this does not mean that the quoted paragraph itself specifies a twist. Also, algebraic surjectivity must be distinguished from surjectivity in each degree of the PBW/order filtrations.

55.3. **Algebraic $\tau$-invariants and parabolic ideals.** For $w \in W$, let $L_w$ be the simple highest-weight module with *unshifted* highest weight $-w\rho - \rho$, and put $\mathfrak{I}_w = \operatorname{Ann}_{U(\mathfrak{g})} L_w$. Thus $\mathfrak{I}_e = I_{\mathcal{B}}$ and $\mathfrak{I}_{w_0}$ is the augmentation ideal. These conventions use the antidominant shifted parameter $-\rho$.

**Theorem 55.4** (Algebraic characterization of $\tau$: standard input)**.** *For primitive ideals with infinitesimal character $\rho$, the usual $\tau$-invariant has the following descriptions:*

$$\tau(\mathfrak{I}_w) = \{\alpha \in \Pi : w\alpha \in \Phi^-\}, \tag{55.4}$$

$$\tau(I) = \{\alpha \in \Pi : \mathfrak{I}_{s_\alpha} \subseteq I\}. \tag{55.5}$$

*For an irreducible Harish–Chandra module $X$, $\tau(X) = \tau(\operatorname{Ann} X)$. Equivalently, $\alpha \in \tau(X)$ precisely when translation from $\rho$ to the adjacent $\alpha$-wall kills $X$.*

Formula (55.4) is [BV85, Section 3, (3.4)–(3.5)]. The pen-minimal-ideal description (55.5) and its agreement with module descent sets are recalled in [Bin08, Section 6.3, Definition 6.8 and Theorem 6.9]. These are algebraic input, rather than definitions of geometric descent.

**Proposition 55.5.** *For $\mathcal{P}_J = G/P_J$,*

$$I_J = \operatorname{Ann} M_{\mathfrak{p}_J}(-2\rho_{\mathfrak{u}_J}) = \mathfrak{I}_{w_J}, \qquad \boxed{\tau(I_J) = J},$$

*where $w_J$ is the longest element of $W_J$ and $M_{\mathfrak{p}_J}(-2\rho_{\mathfrak{u}_J})$ is an irreducible scalar generalized Verma module. In particular, $I_\alpha = \mathfrak{I}_{s_\alpha}$ and $\tau(I_\alpha) = \{\alpha\}$.*

*Proof.* Write $\mathfrak{p}_J = \mathfrak{l}_J + \mathfrak{u}_J$ and $\rho = \rho_{\mathfrak{l}_J} + \rho_{\mathfrak{u}_J}$. Let $y = eP_J \in \mathcal{P}_J$, and consider the simple left $\mathcal{D}_{\mathcal{P}_J}$-module $\delta_y$ of distributions supported at $y$. A local-coordinate calculation of the isotropy action on its generating delta distribution, followed by PBW, gives

$$\Gamma(\mathcal{P}_J, \delta_y) \simeq U(\mathfrak{g}) \otimes_{U(\mathfrak{p}_J)} \mathbb{C}_{-2\rho_{\mathfrak{u}_J}} = M_{\mathfrak{p}_J}(-2\rho_{\mathfrak{u}_J}).$$

The character $-2\rho_{\mathfrak{u}_J}$ is the isotropy character of the left delta generator. This is also [Yah17, Propositions 2.32–2.33]. By Proposition 55.2, this module is simple.

Its annihilator is exactly $I_J$. One inclusion follows from the action through $A_J$. Conversely, a global differential operator that annihilates all distributions supported at $y$ has zero formal germ at $y$: distributions of all orders detect every Taylor coefficient of a differential operator. It is therefore zero on the smooth irreducible variety $\mathcal{P}_J$.

Since

$$w_J\rho = \rho - 2\rho_{\mathfrak{l}_J}, \qquad -2\rho_{\mathfrak{u}_J} = -w_J\rho - \rho,$$

this simple highest-weight module is $L_{w_J}$. Apply (55.4). The longest Levi element sends exactly the simple roots in $J$ to negative roots, so $\tau(I_J) = J$. □

**Corollary 55.6** (Order preservation)**.** *At infinitesimal character $\rho$,*

$$\boxed{I \subseteq I' \implies \tau(I) \subseteq \tau(I').}$$

*In particular, $I_J \subseteq \operatorname{Ann} X$ implies $J \subseteq \tau(X)$.*

*Proof.* Using $I_\alpha = \mathfrak{I}_{s_\alpha}$, formula (55.5) becomes

$$\tau(I) = \{\alpha \in \Pi : I_\alpha \subseteq I\}.$$

If $I_\alpha \subseteq I \subseteq I'$, then $I_\alpha \subseteq I'$. The final assertion follows from Proposition 55.5. □

55.4. **Combining simple-root directions.** There is a useful stronger statement: the parabolic ideal is generated by its simple-root ideals.

**Proposition 55.7** (Generation of the parabolic ideal)**.** *For every $J \subseteq \Pi$,*

$$\boxed{I_J = I_{\mathcal{B}} + \sum_{\alpha \in J} I_\alpha.} \tag{55.6}$$

*Equivalently, in $R = U(\mathfrak{g})/I_{\mathcal{B}}$,*

$$I_J/I_{\mathcal{B}} = \sum_{\alpha \in J} (I_\alpha/I_{\mathcal{B}}).$$

*The term $I_{\mathcal{B}}$ is redundant when $J \neq \varnothing$ but is needed for the empty-set convention.*

*Proof.* Set

$$\mathcal{T}_J = \mathcal{T}_{\mathcal{B}/\mathcal{P}_J}, \qquad \mathcal{T}_\alpha = \mathcal{T}_{\mathcal{B}/\mathcal{P}_\alpha}, \qquad \mathcal{J}_J = \mathcal{D}_{\mathcal{B}}\mathcal{T}_J.$$

We first prove an equality of left ideals:

$$\mathcal{J}_J = \sum_{\alpha \in J} \mathcal{D}_{\mathcal{B}}\mathcal{T}_\alpha. \tag{55.7}$$

Each $\mathcal{T}_\alpha$ lies in $\mathcal{T}_J$, giving one inclusion. For the other, work on the opposite big cell $N^- \subseteq \mathcal{B}$. For $f_\alpha \in \mathfrak{g}_{-\alpha}$, the curve

$$n \exp(t f_\alpha) B$$

lies in the $\pi_\alpha$-fiber through $nB$. Thus $\mathcal{T}_\alpha$ is generated there by the left-invariant field $f_\alpha$. The Lie algebra generated by the $f_\alpha$ for $\alpha \in J$ is $\mathfrak{n}^- \cap \mathfrak{l}_J$, whose left-invariant fields form a frame for $\mathcal{T}_J$ on this cell.

A left ideal containing vector fields $V, W$ contains their bracket, because $[V, W] = VW - WV$. Consequently the right side of (55.7) contains every vertical field on the big cell. Translate the big cell by $G$ to cover $\mathcal{B}$; all the distributions in question are $G$-invariant. This proves (55.7).

By (55.2) and exactness of $\Gamma_{\mathcal{B}}$,

$$\Gamma(\mathcal{B}, \mathcal{J}_J) = I_J/I_{\mathcal{B}}, \qquad \Gamma(\mathcal{B}, \mathcal{J}_\alpha) = I_\alpha/I_{\mathcal{B}}.$$

Apply the same exact functor to the surjection $\bigoplus_{\alpha\in J} \mathcal{J}_\alpha \twoheadrightarrow \mathcal{J}_J$. This gives the asserted sum. For $J = \varnothing$, both sides of the quotient identity are zero. □

**Theorem 55.8** (Parabolic descent criterion)**.** *Let $X \in \mathrm{HC}_\rho(\mathfrak{g}, K)$ be irreducible and put $\mathcal{M} = \mathrm{Loc}_{\mathcal{B}}(X)$. For $J \subseteq \Pi$, the following conditions are equivalent:*

(i) $J \subseteq \tau(X)$;
(ii) $I_J \subseteq \mathrm{Ann}(X)$;
(iii) $\mathcal{M} \simeq \pi_J^\circ \mathcal{N}$ *for a coherent strongly $K$-equivariant $\mathcal{D}_{\mathcal{P}_J}$-module $\mathcal{N}$.*

*When these conditions hold, $\mathcal{N}$ is unique up to isomorphism, is irreducible, and equals* $\mathrm{Loc}_{\mathcal{P}_J}(X)$.

*Proof.* The equivalence of (i) and (ii) follows from (55.5), Proposition 55.5, and (55.6). Proposition 55.2 proves (ii)⇒(iii). Conversely, (iii) gives

$$X = \Gamma_{\mathcal{B}}\mathcal{M} = \Gamma_{\mathcal{P}_J}\mathcal{N}$$

as $U(\mathfrak{g})$-modules, so $I_J$ annihilates $X$. The equivalences in Proposition 55.2 prove uniqueness and irreducibility. □

**Corollary 55.9.** *The coarsest standard partial flag variety to which $\mathcal{M}$ descends is $\mathcal{P}_{\tau(X)}$. More precisely, it descends to $\mathcal{P}_J$ exactly when $J \subseteq \tau(X)$.*

*Proof.* Apply Theorem 55.8. Enlarging $J$ forgets more of a full flag. □

55.5. **What descent means for the orbit connection.** A simple strongly $K$-equivariant $\mathcal{D}_{\mathcal{B}}$-module is determined by a pair $(Q', \mathcal{L})$, where $Q' \subseteq \mathcal{B}$ is a $K$-orbit and $\mathcal{L}$ is an irreducible $K$-equivariant flat connection on $Q'$. It is the intermediate extension

$$\mathcal{M} = \mathrm{IC}_{\mathcal{B}}(Q', \mathcal{L}) = j'_{!*}\mathcal{L}, \qquad j' : Q' \hookrightarrow \mathcal{B},$$

and has support $\overline{Q'}$. The notation $j'_{!*}$ includes extension by zero from the orbit closure to the ambient variety. Flat connections can equivalently be discussed through their local systems of horizontal sections.

**Proposition 55.10** (Orbit formulation of descent)**.** *Let $X$, $\mathcal{M}$, and $(Q', \mathcal{L})$ be as above. Put*

$$Q = \pi_J(Q'), \qquad S = \pi_J^{-1}(Q), \qquad \pi_Q = \pi_J|_S : S \longrightarrow Q.$$

*Then $J \subseteq \tau(X)$ if and only if both of the following hold:*

(a) *$Q'$ is the open dense $K$-orbit in $S$;*
(b) *$\mathcal{L}$ extends to a $K$-equivariant flat connection $\widehat{\mathcal{L}}$ on all of $S$ of the form $\widehat{\mathcal{L}} \simeq \pi_Q^*\mathcal{E}$, for an irreducible $K$-equivariant flat connection $\mathcal{E}$ on $Q$.*

*In (b), once the $K$-equivariant flat extension to $S$ exists, its descent to $Q$ is automatic and unique because the flag fibers are simply connected.*

*Proof.* Suppose first that $J \subseteq \tau(X)$. By Theorem 55.8, $\mathcal{M} = \pi_J^\circ\mathcal{N}$ for a simple equivariant module $\mathcal{N} = \mathrm{IC}_{\mathcal{P}_J}(Q_0, \mathcal{E})$. On $\pi_J^{-1}(Q_0)$, this is precisely the flat connection $\pi_{Q_0}^*\mathcal{E}$. The preimage of $Q_0$ has a unique open dense $K$-orbit: its connected components are irreducible,

$K$ permutes them transitively, and there are finitely many $K$-orbits. This orbit is $Q'$, and $Q_0 = Q$. This proves (a)–(b).

Conversely, suppose (a)–(b) hold. Define $\mathcal{N} = \mathrm{IC}_{\mathcal{P}_J}(Q, \mathcal{E})$. Smooth pullback commutes with intermediate extension, and pullback preserves simplicity in the present setting by Proposition 55.2. Thus $\pi_J^\circ \mathcal{N}$ is the simple intermediate extension of its restriction $\mathcal{L}$ to $Q'$. Uniqueness of that extension gives $\mathcal{M} = \pi_J^\circ \mathcal{N}$. Theorem 55.8 now gives $J \subseteq \tau(X)$.

Finally, on each connected component, the homotopy exact sequence of the locally trivial fibration $\pi_Q$, with connected simply connected fiber, gives

$$\pi_1(S) \simeq \pi_1(Q).$$

Consequently every local system on $S$ is pulled back uniquely from $Q$. Applied to the horizontal sections of the equivariant flat extension, this constructs the descended local system, with equivariance descending by uniqueness. Equivariant local systems on the homogeneous space $Q$ arise from representations of the finite component group of a stabilizer, and give algebraic regular connections. Thus the descended object is an algebraic $K$-equivariant flat connection, as claimed. □

**Corollary 55.11** (The simple-root statement)**.** *For an irreducible $X$, with orbit connection $(Q', \mathcal{L})$,*

$$\boxed{\alpha \in \tau(X) \quad \Longleftrightarrow \quad \mathcal{L} \text{ descends through } \pi_\alpha \text{ in the sense of Proposition 55.10.}}$$

The familiar phrase "$\mathcal{L}$ fibers over a flat connection on $\mathcal{P}_J$" means exactly this extension-and-pullback statement. If $q \in Q$ and $F_q = \pi_Q^{-1}(q) \simeq \mathcal{B}_{\mathfrak{l}_J}$, then

$$\widehat{\mathcal{L}}|_{F_q} \simeq (\mathcal{O}_{F_q}, d) \otimes_{\mathbb{C}} \mathcal{E}_q.$$

Thus the restriction is a trivial connection of rank $\dim_{\mathbb{C}} \mathcal{E}_q$. The flag variety $F_q$ is the fiber of the geometric projection; $\mathcal{E}_q$ is the vector-space fiber of the connection. These are two different uses of "fiber."

The density and extension conditions cannot be replaced merely by $\mathcal{L} \simeq (\pi_J|_{Q'})^* \mathcal{E}$. That identity on a smaller orbit does not control singularities on the remaining orbits in $S$.

For $J = \{\alpha\}$, the extending dense-orbit cases in [LV83, Lemma 3.5] are (a), (b2), (c2), and (d2). The intersection with a fiber is respectively $\mathbb{P}^1$, $\mathbb{A}^1$, or $\mathbb{C}^\times$, and the connection extends across the missing points. Case (e) has a dense orbit but fails extension. This gives the explicit rank-one calculation behind the geometric interpretation.

## 55.6. **Examples and consequences.**

**Example 55.12** (Rank one: support alone does not determine $\tau$)**.** Take $G = SL_2(\mathbb{C})$ and $K \simeq SO_2(\mathbb{C}) \simeq \mathbb{C}^\times$. After choosing coordinates, its orbits on $\mathcal{B} = \mathbb{P}^1$ are

$$\{0\}, \qquad \{\infty\}, \qquad \mathbb{C}^\times.$$

There is one simple root $\alpha$, and $\mathcal{P}_\alpha$ is a point. The trivial local system on the open orbit extends to the trivial connection $(\mathcal{O}_{\mathbb{P}^1}, d)$. Its global sections form the trivial representation, and $\tau = \{\alpha\}$.

The other irreducible equivariant local system on $\mathbb{C}^\times$ has monodromy $-1$. It does not extend across 0 and $\infty$, so its simple intermediate extension has $\tau = \varnothing$. Both modules have support all of $\mathbb{P}^1$, but different $\tau$-invariants.

The delta-module at either closed orbit also has $\tau = \varnothing$. Although a connection on that point is tautologically pulled back from a point, its orbit is not dense in the full $\pi_\alpha$-fiber. This explains why the density qualification is indispensable.

**Example 55.13** (Flags in $\mathbb{C}^3$)**.** For $G = SL_3(\mathbb{C})$, a full flag is a line contained in a plane. With the standard numbering, $\pi_{\alpha_1}$ forgets the line and retains the plane. Its fiber is the $\mathbb{P}^1$ of lines in that plane. Thus $\alpha_1 \in \tau(X)$ precisely when the orbit connection extends to the full inverse image of its orbit of planes and is constant along the choices of a line. The projection $\pi_{\alpha_2}$ has the analogous interpretation with the line retained and the plane forgotten.

**Proposition 55.14** (Extreme cases)**.** *For irreducible $X \in \mathrm{HC}_\rho(\mathfrak{g}, K)$:*

(i) $\tau(X) = \varnothing$ *if and only if* $\mathrm{Loc}_{\mathcal{B}}(X)$ *descends to no* $\mathcal{P}_J$ *with* $J \neq \varnothing$.

(ii) $\tau(X) = \Pi$ *if and only if* $\mathfrak{g}$ *acts trivially on* $X$. *If* $K$ *is connected, this is the trivial* $(\mathfrak{g}, K)$*-module. In general,* $X$ *is an irreducible representation of* $K/K^\circ$ *with trivial* $\mathfrak{g}$*-action.*

*Proof.* Part (i) is Theorem 55.8. For (ii), use $\mathcal{P}_\Pi = \mathrm{pt}$ and $I_\Pi = \ker(U(\mathfrak{g}) \to \mathbb{C})$, the augmentation ideal. Compatibility of the two actions then makes $K^\circ$ act trivially as well. □

For arbitrary primitive ideals, inclusion of $\tau$-invariants does not recover inclusion of ideals. Theorem 55.8 provides a useful special converse when the smaller ideal is parabolic:

$$\boxed{I_J \subseteq I \quad \Longleftrightarrow \quad J \subseteq \tau(I).}$$

In particular, $I_J$ is the smallest primitive ideal at $\chi_\rho$ whose $\tau$-invariant contains $J$. This does not make the general map $I \mapsto \tau(I)$ an order embedding.

All these statements have been made at the ordinary regular character $\chi_\rho$. Arbitrary twists and singular infinitesimal characters require the corresponding twisted localization and translation results; the untwisted formulas above should not be transferred to that setting without changing the hypotheses.

## 56. Ample cones and Parabolic Subgroups

2026-10-22

Let $X$ be a conical simplectic singularity, it has Namikawa space:

$$\mathfrak{P}^X = \mathfrak{P}_0^X \oplus \bigoplus_{i=1}^{r} (\mathfrak{h}_i^*)^{\pi_1(\mathcal{L}_i)}.$$

The Namikawa Weyl group has a canonical fundamental chamber:

$$\mathfrak{P}_{\mathbb{R}}^{\geq 0} = \mathfrak{P}_{0,\mathbb{R}}^X \oplus \bigoplus_{i=1}^{r} \mathfrak{P}_{i,\mathbb{R}}^{\geq 0}.$$

Then there is a bijection:

$$\begin{array}{rccc} \mathrm{Amp}\colon & \{\mathbb{Q}\text{-factorial terminalization}\}/\sim & \longrightarrow & \{\text{ample cones}\} \\ & (Y \to X) & \longmapsto & \mathrm{Amp}(Y/X) \end{array}$$

When $X = \mathrm{Spec}(\mathbb{C}[\tilde{\mathbb{O}}])$ for some nilpotent cover $\tilde{\mathbb{O}}$, suppose $\tilde{\mathbb{O}}_L$ is a birationally rigid orbit of a Levi $L$ such that $\mathrm{Bind}_L^G \tilde{\mathbb{O}}_L = \tilde{\mathbb{O}}$.

We have a surjective map:

$$\begin{array}{rccl}\varphi: & \{\text{parabolics containing } L \text{ as a Levi}\} & \longrightarrow & \{\mathbb{Q}\text{-factorial terminalization}\}/\sim \\ & P & \longmapsto & Y_P := G\times^P (\tilde{X}_L \times \mathfrak{n}_P)\end{array}$$

For each $P$ we have an identification:

$$\eta_P : \mathfrak{X}(\mathfrak{l}) \longrightarrow \mathfrak{P}^X,$$

and a dominant cone $C_P \subseteq \mathfrak{X}(\mathfrak{l})$. They satisfy $\eta_P(C_P) = \mathrm{Amp}(Y_P/X)$.

We have an equality:

$$|W_X| \cdot |\{\text{ample cones}\}| = |\{\text{parabolics containing } L \text{ as a Levi}\}|.$$

**Example 56.1.** $G = \mathrm{SL}_n(\mathbb{C})$, $X = \mathrm{Spec}(\mathbb{C}[\mathbb{O}_{[p,q]^t}])$. Then $L = S(\mathrm{GL}_p(\mathbb{C}) \times \mathrm{GL}_q(\mathbb{C}))$. $|\{\text{parabolics containing } L \text{ as a Levi}\}| = 2$.

- If $p \neq q$, $|\{\text{ample cones}\}| = 2$, so $W_X = \textit{trivial}$;
- If $p = q$, $|\{\text{ample cones}\}| = 1$, so $W_X = \mathbb{Z}/2\mathbb{Z}$.

## 57. Computation of Birational Induction

2026-10-23

Let $G(n) = \mathrm{Sp}_n(\mathbb{C}), \mathrm{SO}_n(\mathbb{C})$. $L = \mathrm{GL}_m(\mathbb{C}) \times G(a)$ is a Levi of a maximal parabolic subgroup.

A birational rational nilpotent orbit cover of $L$ is of the form:

$$\{0\} \times \tilde{\mathbb{O}}_p \in \mathfrak{gl}_m \times \mathfrak{g}(a).$$

Let $\pi : G\times^P (\tilde{X}_L \times \mathfrak{n}_P) \to \bar{\mathbb{O}}$. Then $\tilde{\mathbb{O}} = \mathrm{BInd}_L^G(\{0\} \times \tilde{\mathbb{O}}_p) := \pi^{-1}(\mathbb{O})$.

Next we determine the degree of $\tilde{\mathbb{O}}$ and the Namikawa Weyl group $W_{\tilde{X}}$ of $\tilde{X} = \mathrm{Spec}(\mathbb{C}[\tilde{\mathbb{O}}])$.

57.1. $\tilde{\mathbb{O}}_{\mathbf{p}} = \mathbb{O}_{\mathbf{p}}$ **is a nilpotent orbit.** Suppose $\mathbf{p} = [p_1, p_2, \cdots, p_m, \cdots]$ (add 0 at the end). Form

$$\mathbf{r} = \mathbf{p} + (2^m) = (p_1 + 2, \ldots, p_m + 2, p_{m+1}, \ldots).$$

The ordinary induced orbit has partition

$$\mathbf{q} = \begin{cases}\mathbf{r}_{B/D}, & G(n) = \mathrm{SO}_n, \\ \mathbf{r}_C, & G(n) = \mathrm{Sp}_n,\end{cases}$$

where the subscript denotes the appropriate collapse; see [MMY25, Proposition 5.9(ii)].

In this situation, the collapse can be described precisely:

Let $d = p_m$, $t = |\{j \in \{1, 2, \cdots, m\} \,|\, p_j = d\}|$, collapse is needed if and only if

$$\begin{cases}B, D, & d \text{ is even, } t \text{ is odd;} \\ C, & d \text{ is odd, } t \text{ is odd.}\end{cases}$$

When collapse is needed,

$$\mathbf{q} = [p_1 + 2, \cdots, p_{m-1} + 2, p_m + 1, p_{m+1} + 1, p_{m+2}, \cdots].$$

**Proposition 57.1.** *In type $B, C$:*

$$\deg(\tilde{\mathbb{O}}) = \begin{cases}1, & \textit{no collapse is needed;} \\ 2, & \textit{collapse is needed.}\end{cases}$$

*Proof.* The degree statements are Namikawa's type I and type II induction calculations: see [Nam22a, Claims 2.3.1–2.3.2] in type $C$ and [Nam22a, Claims 3.6.1–3.6.2] in type $B$. We give the corresponding fiber calculation. Write the Jordan blocks of $x \in \mathbb{O}$ in decreasing order as $\mathbf{q} = (q_1, q_2, \ldots)$. Choose a Jordan basis

$$e_{j,1}, \ldots, e_{j,q_j}, \qquad xe_{j,s} = e_{j,s-1}, \quad xe_{j,1} = 0,$$

compatible with the bilinear form. Different block lengths are orthogonal, and pairings within equal-length blocks occur only between opposite positions:

$$\langle e_{j,s}, e_{j',t} \rangle = 0 \quad \text{unless} \quad q_j = q_{j'}, \quad s + t = q_j + 1.$$

Note that

$$\pi^{-1}(x) = \{F \subseteq V \mid \dim F = m, \quad \langle F, F \rangle = 0, \quad F \subseteq \ker x, \quad \mathrm{Jord}(\bar{x} \text{ on } F^{\perp}/F) = \mathbf{p}\}$$

- When collapse is not needed, we have
$$\mathbf{q} = [p_1 + 2, \cdots, p_m + 2, p_{m+1}, \cdots],$$
then $F \subseteq \mathrm{im}(x^d) \cap \ker(x) = \mathrm{span}\{e_{1,1}, e_{2,1}, \cdots, e_{m,1}\}$. But $\dim F = m$, this implies that $F = \mathrm{im}(x^d) \cap \ker(x)$. So, $\deg(\tilde{\mathbb{O}}) = 1$.
- When collapse is needed, we have
$$\mathbf{q} = [p_1 + 2, \cdots, p_{m-1} + 2, p_m + 1, p_{m+1} + 1, p_{m+2}, \cdots],$$
then $F \subseteq \mathrm{im}(x^d) \cap \ker(x) =: K$, note that
$$K = \mathrm{span}\{e_{1,1}, \cdots, e_{m-1,1}\} \oplus \mathrm{span}\{e_{m,1}, e_{m+1,1}\},$$
Write
$$U = \mathrm{span}\{e_{1,1}, \ldots, e_{m-1,1}\}.$$
Then
$$K = U \oplus \mathrm{span}\{e_{m,1}, e_{m+1,1}\}, \qquad \dim U = m - 1.$$
Define a bilinear form on $K$ by
$$B(v, w) = \langle \widetilde{v}, w \rangle, \qquad x^d \widetilde{v} = v.$$
This is well defined because
$$\ker x^d = (\mathrm{im}\, x^d)^{\perp}.$$
Let $\varepsilon = 1$ in orthogonal type and $\varepsilon = -1$ in symplectic type. Skew-adjointness of $x$ gives
$$B(w, v) = \varepsilon(-1)^d B(v, w).$$
In the collapse case, $d$ is even in orthogonal type and odd in symplectic type. Hence $B$ is symmetric.

For a bottom vector $e_{j,1} \in K$, we may choose $e_{j,d+1}$ as its preimage under $x^d$. Therefore
$$B(e_{j,1}, e_{j',1}) = \langle e_{j,d+1}, e_{j',1} \rangle.$$
By the pairing condition on the Jordan basis, this can be nonzero only when $q_j = q_{j'} = d + 1 = h$. The pairing between the top and bottom vectors of the two length-$h$ blocks is nondegenerate. It follows that
$$\mathrm{rad}\, B = U,$$
so $B$ induces a nondegenerate symmetric form on the two-dimensional space $K/U$.

Since $F \subseteq x^d(F^\perp)$, every $v \in F$ has a preimage $\widetilde{v} \in F^\perp$. Thus

$$B(v, w) = 0 \qquad (v, w \in F).$$

The image of $F$ in $K/U$ is consequently isotropic and has dimension at most one. Hence

$$\dim(F \cap U) \geq \dim F - 1 = m - 1 = \dim U,$$

which proves $U \subseteq F$. Thus $F/U$ must be an isotropic line in $K/U$.

Choose a hyperbolic basis of $K/U$ and apply the corresponding change of basis at every level of the two length-$h$ Jordan chains. Denote the resulting chains by

$$u_h \mapsto \cdots \mapsto u_1 \mapsto 0, \qquad v_h \mapsto \cdots \mapsto v_1 \mapsto 0,$$

with

$$B(u_1, u_1) = B(v_1, v_1) = 0, \qquad B(u_1, v_1) = 1.$$

A line spanned by $\alpha u_1 + \beta v_1$ is isotropic precisely when

$$B(\alpha u_1 + \beta v_1, \alpha u_1 + \beta v_1) = 2\alpha\beta = 0.$$

Therefore the only possible subspaces are

$$F_+ = U \oplus \mathbb{C}u_1, \qquad F_- = U \oplus \mathbb{C}v_1.$$

So $\deg(\tilde{\mathbb{O}}) = 2$.

□

**Proposition 57.2.** *In type D:*

$$\deg(\tilde{\mathbb{O}}) = \begin{cases} 1, \textit{ no collapse needed;} \\ 2, \textit{ collapse needed, } a \neq 0; \\ 1, \textit{ collapse needed, } a = 0. \end{cases}$$

*The last case occurs precisely when $m = n$ is odd.*

*Proof.* See [Nam22a, Claims 3.6.1–3.6.2], including the degree-one exception when the residual orthogonal factor is absent. First suppose $m < n$. Then $G/P$ parametrizes all isotropic $m$-dimensional subspaces of $V$. Since $\mathbb{O}_{\mathbf{p}}$ is birationally rigid and $\mathbf{p}$ is nonempty, $\mathbf{p}$ is not very even. Thus the additional very-even-source exception in [Nam22a, Claim 3.6.2] does not occur here. Consequently, the Jordan partition $\mathbf{p}$ determines the source orbit uniquely, and the fiber description in the preceding proof remains valid.

The same Jordan-basis calculation therefore applies. If no collapse is needed, there is exactly one subspace in the fiber. If collapse is needed, the two subspaces

$$F_+ = U \oplus \mathbb{C}u_1, \qquad F_- = U \oplus \mathbb{C}v_1$$

both belong to $G/P$ and give quotient Jordan type $\mathbf{p}$. Hence the degree is respectively 1 or 2.

Now suppose $m = n$. Then $\mathbf{p} = \varnothing$, and $G/P$ is one fixed connected component of the maximal isotropic Grassmannian

$$\mathrm{OGr}(n, 2n) = \mathrm{OGr}^+(n, 2n) \sqcup \mathrm{OGr}^-(n, 2n).$$

If $n$ is even, no collapse is needed and $\mathbf{q} = (2^n)$. For $x$ in the induced orbit,

$$x^2 = 0, \qquad \operatorname{im} x = \ker x, \qquad \dim \ker x = n.$$

Thus the unique possible subspace is $F = \ker x$, and the degree is 1. Here $(2^n)$ is very even; the choice of $P$ determines which of the two corresponding nilpotent orbits is induced.

If $n$ is odd, collapse is needed and

$$\mathbf{q} = (2^{n-1}, 1^2).$$

The preceding calculation gives precisely two maximal isotropic subspaces satisfying the Jordan-partition conditions:

$$F_+ = U \oplus \mathbb{C}u_1, \qquad F_- = U \oplus \mathbb{C}v_1, \qquad U = \operatorname{im} x.$$

In particular,

$$F_+ \cap F_- = U, \qquad \dim U = n - 1.$$

Two maximal isotropic subspaces $F, F'$ belong to the same connected component of $\mathrm{OGr}(n, 2n)$ if and only if

$$\dim(F \cap F') \equiv n \pmod 2.$$

Hence $F_+$ and $F_-$ lie in opposite components. Exactly one belongs to the chosen $G/P$, so the fiber consists of one point and the degree is 1. □

Next, we determine the Namikawa Weyl groups:

**Proposition 57.3.** *We have*

$$W_{\tilde{X}} \cong \begin{cases} \mathbb{Z}/2\mathbb{Z}, & \textit{no collapse is needed;} \\ 1, & \textit{collapse is needed.} \end{cases}$$

*Proof.* By [Los22a, Proposition 4.7(3)], there is a short exact sequence

$$1 \longrightarrow W_{\tilde{X}} \longrightarrow N_G(L, \mathbb{O}_L)/L \longrightarrow \mathrm{Aut}_G(\tilde{X}) \longrightarrow 1.$$

Since $\tilde{X} = \operatorname{Spec} \mathbb{C}[\tilde{\mathbb{O}}]$, restriction to the dense open orbit induces an isomorphism

$$\mathrm{Aut}_G(\tilde{X}) \cong \mathrm{Deck}(\tilde{\mathbb{O}}/\mathbb{O}).$$

Indeed, a $G$-equivariant Poisson automorphism preserves the moment map, by its uniqueness for semisimple $G$, and hence restricts to a deck transformation. Conversely, every deck transformation is $G$-equivariant and Poisson, and its action on $\mathbb{C}[\tilde{\mathbb{O}}]$ extends it uniquely to $\tilde{X}$.

Consequently,

$$|W_{\tilde{X}}| = \frac{|N_G(L, \mathbb{O}_L)/L|}{|\operatorname{Deck}(\tilde{\mathbb{O}}/\mathbb{O})|}.$$

We now compute $N_G(L, \mathbb{O}_L)/L$. We will show that it has order two, except when $G = \mathrm{SO}_{2m}$, $a = 0$, and $m$ is odd, in which case it is trivial.

Choose a decomposition

$$V = E \oplus V_0 \oplus E^*, \qquad \dim E = m, \quad \dim V_0 = a,$$

compatible with the bilinear form. Then

$$L = \left\{ \mathrm{diag}(A, H, (A^{-1})^{\mathsf{t}}) : A \in \mathrm{GL}(E),\ H \in G(V_0) \right\}.$$

Since $L$ is a maximal Levi subgroup, its connected center is the one-dimensional torus

$$S = Z(L)^\circ = \left\{ \mathrm{diag}(tI_E, I_{V_0}, t^{-1} I_{E^*}) : t \in \mathbb{C}^\times \right\}.$$

Every element of $N_G(L)$ normalizes $S$, and $C_G(S) = L$. Hence conjugation induces an injection

$$N_G(L)/L \hookrightarrow \mathrm{Aut}_{\mathrm{alg.grp}}(S) \cong \mathbb{Z}/2\mathbb{Z}.$$

In particular,

$$|N_G(L, \mathbb{O}_L)/L| \leq 2.$$

We now construct an element of $N_G(L, \mathbb{O}_L)$ acting on $S$ by inversion. Choose dual bases $e_1, \ldots, e_m$ of $E$ and $f_1, \ldots, f_m$ of $E^*$, and fix $x_0 \in \mathbb{O}_{\mathbf{p}}$.

In symplectic type, define $w$ by

$$we_i = f_i, \qquad wf_i = -e_i, \qquad w|_{V_0} = \mathrm{id}\,.$$

This preserves the symplectic form, normalizes $L$, and fixes the element $(0, x_0) \in \mathbb{O}_L$. Moreover,

$$wsw^{-1} = s^{-1} \qquad (s \in S).$$

Thus $w \in N_G(L, \mathbb{O}_L) \setminus L$.

In orthogonal type, first define $w_0$ by

$$w_0 e_i = f_i, \qquad w_0 f_i = e_i, \qquad w_0|_{V_0} = \mathrm{id}\,.$$

This is an isometry with

$$\det(w_0) = (-1)^m.$$

If $m$ is even, take $w = w_0$.

Suppose that $m$ is odd and $a > 0$. Since $\mathbb{O}_{\mathbf{p}}$ is an ordinary birationally rigid orbit, $\mathbf{p}$ contains a part 1. The sum of its length-one Jordan blocks is nondegenerate, so it contains a nondegenerate line $D \subseteq V_0$ on which $x_0$ vanishes. Let

$$r = -\,\mathrm{id}_D \oplus \mathrm{id}_{D^\perp} \in O(V_0).$$

Then $\det(r) = -1$ and $rx_0 = x_0 r$. Extend $r$ by the identity on $E \oplus E^*$ and set $w = w_0 r$. Thus $\det(w) = 1$, so $w \in G$. Again, $w$ normalizes $L$, fixes $(0, x_0)$, and acts on $S$ by inversion.

Consequently, except when $G = \mathrm{SO}_{2m}$, $a = 0$, and $m$ is odd, we have

$$N_G(L, \mathbb{O}_L)/L \cong \mathbb{Z}/2\mathbb{Z}.$$

By the preceding degree computation,

$$\deg(\tilde{\mathbb{O}}/\mathbb{O}) = \begin{cases} 1, & \text{no collapse is needed;} \\ 2, & \text{collapse is needed.} \end{cases}$$

A connected cover of degree one or two is Galois. Therefore

$$\left|\mathrm{Deck}(\tilde{\mathbb{O}}/\mathbb{O})\right| = \deg(\tilde{\mathbb{O}}/\mathbb{O}),$$

and the exact sequence gives

$$|W_{\tilde{X}}| = \frac{2}{\deg(\tilde{\mathbb{O}}/\mathbb{O})}.$$

This proves the asserted formula in these cases.

Finally, suppose $G = \mathrm{SO}_{2m}$, $a = 0$, and $m$ is odd. Any isometry acting on $S$ by inversion exchanges $E$ and $E^*$ and has determinant $(-1)^m = -1$. Hence no such element belongs to $G$, and

$$N_G(L)/L = 1.$$

The exact sequence therefore gives $W_{\tilde{X}} = 1$. Since the partition $(2^m)$ requires orthogonal collapse when $m$ is odd, this agrees with the asserted formula. □

57.2. **General Case.** Let

$$L = \mathrm{GL}_m \times G(a)$$

be a Levi subgroup of a maximal parabolic subgroup $P \subseteq G$, and put

$$\mathbb{O}_L = \{0\} \times \mathbb{O}_{\mathbf{p}}.$$

Let

$$\nu_L : \tilde{\mathbb{O}}_L \longrightarrow \mathbb{O}_L$$

be a connected, birationally rigid $L$-equivariant cover. Write

$$D = \deg(\nu_L), \qquad \tilde{\mathbb{O}} = \mathrm{Bind}_L^G(\tilde{\mathbb{O}}_L), \qquad \mathbb{O} = \mathrm{Ind}_L^G(\mathbb{O}_L), \qquad \tilde{X} = \operatorname{Spec} \mathbb{C}[\tilde{\mathbb{O}}].$$

When $a = 0$, we take $\mathbf{p} = \varnothing$. In this case $\mathbb{O}_L$ is a point, so its only connected equivariant cover is trivial and $D = 1$.

The underlying orbit $\mathbb{O}$ has partition

$$\mathbf{q} = \mathrm{collapse}_{B/C/D}\big(\mathbf{p} + (2^m)\big).$$

If $\mathbf{q}$ is very even, the choice of $P$ and the source orbit specifies which of the two nilpotent orbits is induced.

Notice that $\mathbb{O}_L$ need not itself be birationally rigid. Nevertheless, in orthogonal type with $a > 0$, the partition $\mathbf{p}$ contains a part 1. Indeed, a birationally rigid cover has no codimension-two symplectic leaves in its affinization [LMM26, Corollary 7.28(iv)]. The criterion for an orthogonal orbit to admit such an $\mathrm{SO}_a$-equivariant cover is [LMM26, Proposition 7.34(i)]; it gives

$$p_i - p_{i+1} \le 2, \qquad p_i - p_{i+1} < 2 \quad \text{if } p_i \text{ is even.}$$

Applying this to the last positive part, with zero padding, shows that the smallest part is 1. In particular, $\mathbf{p}$ is not very even.

**Proposition 57.4.** *We have*

$$\deg(\tilde{\mathbb{O}}/\mathbb{O}) = \begin{cases} D, & \textit{no collapse is needed;} \\ 2D, & \textit{collapse is needed and } a > 0; \\ 1, & \textit{collapse is needed and } a = 0. \end{cases}$$

*The last case occurs precisely when $G = \mathrm{SO}_{2m}$ and $m$ is odd.*

*Proof.* Let $\mathfrak{n}_P$ be the nilradical of $P$. Consider

$$Z = G \times^P (\mathbb{O}_L + \mathfrak{n}_P)$$

with its generalized Springer map

$$\mu : Z \longrightarrow \overline{\mathbb{O}}.$$

Pulling back $\nu_L$ along the Levi projection $\mathbb{O}_L + \mathfrak{n}_P \to \mathbb{O}_L$ gives

$$\tilde{Z} = G \times^P \big((\mathbb{O}_L + \mathfrak{n}_P) \times_{\mathbb{O}_L} \tilde{\mathbb{O}}_L\big).$$

Forgetting the lift to $\tilde{\mathbb{O}}_L$ defines a finite étale morphism

$$\rho : \tilde{Z} \longrightarrow Z$$

of degree $D$.

By the construction of birational induction, the restriction of $\tilde{\mu} := \mu \circ \rho$ over $\mathbb{O}$ is the cover $\tilde{\mathbb{O}} \to \mathbb{O}$. Multiplicativity of generic degrees gives

$$\deg(\tilde{\mathbb{O}}/\mathbb{O}) = \deg(\tilde{\mu}) = \deg(\rho)\deg(\mu) = D\deg(\mu).$$

The ordinary generalized Springer degrees are computed in [Nam22a, Claims 2.3.1–2.3.2 and 3.6.1–3.6.2]. These calculations apply even when the underlying source orbit is not birationally rigid. We spell out the cases.

In symplectic type, the preceding Jordan-basis computation gives

$$\deg(\mu) = \begin{cases} 1, & \text{no collapse is needed;} \\ 2, & \text{collapse is needed.} \end{cases}$$

This includes $a = 0$, when $\mathbf{q} = (2^m)$ and no collapse is needed.

In orthogonal type with $a > 0$, the same computation applies. Since $\mathbf{p}$ contains a part 1, it is not very even, so the quotient Jordan partition determines the source orbit uniquely. Thus the degree is again 1 without collapse and 2 with collapse.

It remains to consider $G = \mathrm{SO}_{2m}$ and $a = 0$. Here $D = 1$, and $G/P$ is one fixed connected component of $\mathrm{OGr}(m, 2m)$.

If $m$ is even, no collapse is needed and $\mathbf{q} = (2^m)$. The preceding type $D$ computation shows that the fiber over $x \in \mathbb{O}$ consists of the unique subspace $F = \ker x$, which belongs to the chosen component because $x$ lies in the orbit induced using $P$. Hence $\deg(\mu) = 1$.

If $m$ is odd, collapse is needed and

$$\mathbf{q} = (2^{m-1}, 1^2).$$

The two candidate maximal isotropic subspaces have intersection of dimension $m - 1$ and therefore lie in opposite components of $\mathrm{OGr}(m, 2m)$. Exactly one belongs to $G/P$, so $\deg(\mu) = 1$ in this case as well. □

We next compute the Namikawa Weyl group. For a nontrivial source cover, we use the extended normalizer, which records a normalizer element together with a lift of its action to the cover, as in [MMY25, Definition 3.33].

Put

$$\Gamma_L = \mathrm{Deck}(\tilde{\mathbb{O}}_L/\mathbb{O}_L), \qquad \Gamma = \mathrm{Deck}(\tilde{\mathbb{O}}/\mathbb{O}).$$

Let

$$\tilde{X}_L = \operatorname{Spec} \mathbb{C}[\tilde{\mathbb{O}}_L],$$

and let

$$\mu_L : \tilde{X}_L \longrightarrow \mathfrak{l}$$

be the moment map extending $\nu_L$, using an invariant bilinear form to identify $\mathfrak{l}^*$ with $\mathfrak{l}$.

Define

$$R = \left\{ g \in N_G(L, \mathbb{O}_L) : \mathrm{Ad}(g)|_{\mathbb{O}_L} \text{ lifts to } \tilde{\mathbb{O}}_L \right\} / L$$

and

$$\widetilde{W}^{\tilde{X}} = \left\{ (g, \varphi) \in N_G(L) \times \mathrm{Aut}_{\mathrm{Pois}}(\tilde{X}_L) : \mu_L \circ \varphi = \mathrm{Ad}(g) \circ \mu_L \right\} / L.$$

Here $L$ is embedded diagonally via its natural action on $\tilde{X}_L$. Projection to the normalizer gives an exact sequence

$$1 \longrightarrow \Gamma_L \longrightarrow \widetilde{W}^{\tilde{X}} \longrightarrow R \longrightarrow 1$$

and [MMY25, Theorem 3.34(iii)] gives

$$1 \longrightarrow W_{\tilde{X}} \longrightarrow \widetilde{W}^{\tilde{X}} \longrightarrow \mathrm{Aut}_G(\tilde{X}) \longrightarrow 1,$$

where $\mathrm{Aut}_G$ denotes $G$-equivariant Poisson automorphisms. See also [LMM26, Proposition 7.5(ii)] for this kernel description of the Namikawa Weyl group.

**Proposition 57.5.** *In all cases, including $a = 0$, we have*

$$W_{\tilde{X}} \cong \begin{cases} \mathbb{Z}/2\mathbb{Z}, & \textit{no collapse is needed;} \\ 1, & \textit{collapse is needed.} \end{cases}$$

*Proof.* For the linear classical groups under consideration, nilpotent centralizer component groups are abelian. Consequently, every connected equivariant nilpotent cover is Galois. In particular,

$$|\Gamma_L| = D, \qquad |\Gamma| = \deg(\tilde{\mathbb{O}}/\mathbb{O}).$$

Restriction to the dense open orbit identifies

$$\mathrm{Aut}_G(\tilde{X}) \cong \Gamma.$$

Indeed, a $G$-equivariant Poisson automorphism preserves the moment map and hence restricts to a deck transformation. Conversely, deck transformations are $G$-equivariant and Poisson, and extend uniquely to the affinization. The two exact sequences therefore give

$$|W_{\tilde{X}}| = \frac{|\Gamma_L||R|}{|\Gamma|} = \frac{D|R|}{\deg(\tilde{\mathbb{O}}/\mathbb{O})}.$$

First consider the exceptional case

$$G = \mathrm{SO}_{2m}, \qquad a = 0, \qquad m \text{ odd}.$$

Here $D = 1$. Any isometry acting by inversion on $Z(L)^\circ$ exchanges $E$ and $E^*$ and has determinant $(-1)^m = -1$. Hence

$$N_G(L)/L = 1, \qquad R = 1.$$

The preceding proposition gives $\deg(\tilde{\mathbb{O}}/\mathbb{O}) = 1$, so $W_{\tilde{X}} = 1$. Since $(2^m)$ requires orthogonal collapse, this agrees with the asserted formula.

For the remainder of the proof, exclude this exceptional case. We claim that

$$R = N_G(L, \mathbb{O}_L)/L \cong \mathbb{Z}/2\mathbb{Z}.$$

As before, the action on the one-dimensional torus $Z(L)^\circ$ gives

$$|N_G(L)/L| \leq 2.$$

It remains to exhibit a normalizer element acting by inversion whose action on the source orbit lifts to the source cover.

Choose a decomposition

$$V = E \oplus V_0 \oplus E^*$$

compatible with the bilinear form. In symplectic type, choose the normalizer element exchanging $E$ and $E^*$ to act trivially on $V_0$. Its action on $\mathbb{O}_L$ is the identity, so the identity map on $\tilde{\mathbb{O}}_L$ supplies a lift.

In orthogonal type with $m$ even, the same argument applies, including when $a = 0$.

Suppose now that we are in orthogonal type and $m$ is odd. Since the exceptional case has been excluded, $a > 0$. Fix $x_0 \in \mathbb{O}_{\mathbf{p}}$. Since $\mathbf{p}$ contains a part 1, choose a determinant-minus-one isometry

$$r \in O(V_0)_{x_0}$$

by reflecting in a nondegenerate length-one Jordan summand. Combining $r$ with the exchange of $E$ and $E^*$ gives an element of $\mathrm{SO}(V)$ normalizing $L$ and acting by inversion on $Z(L)^\circ$.

We check that conjugation by $r$ lifts to the source cover. Write

$$A = \mathrm{SO}(V_0)_{x_0}/\mathrm{SO}(V_0)^\circ_{x_0},$$

and let $H \subseteq A$ be the subgroup corresponding to $\tilde{\mathbb{O}}_L$. Since

$$\mathrm{SO}(V_0)^\circ_{x_0} = O(V_0)^\circ_{x_0},$$

the inclusion of centralizers identifies $A$ with a subgroup of

$$O(V_0)_{x_0}/O(V_0)^\circ_{x_0}.$$

The latter group is abelian. Hence conjugation by $r$ acts trivially on $A$ and preserves $H$. It therefore lifts to the source cover.

Thus $R \cong \mathbb{Z}/2\mathbb{Z}$ in every nonexceptional case. Using the preceding degree computation, we obtain

$$|W_{\tilde{X}}| = \frac{2D}{\deg(\tilde{\mathbb{O}}/\mathbb{O})} = \begin{cases} 2, & \text{no collapse is needed;} \\ 1, & \text{collapse is needed.} \end{cases}$$

This completes the proof. □

For universal target covers, [Nam22b, Corollary 18] describes the Weyl group as generated by reflections whose corresponding generalized Springer maps have degree one. This gives the same rank-one dichotomy. The extended-normalizer argument above applies to arbitrary connected birationally rigid source covers.

## 58. Strict GK-Dimension Drop for Harish–Chandra Cells

2026-10-25

This section supplies the strict-drop input used in Section 50. The argument combines the comparison with highest-weight double cells in [BMSZ22] with positive projective functors. Its additional input is the positive-idempotent theorem of Kildetoft–Mazorchuk [KM16, Prop. 18]. We assemble the proof from these results, rather than attributing a strict-drop theorem for Harish–Chandra cells to BMSZ.

58.1. **Projective functors and their positive action.** Let $G$ be a real reductive group in Harish–Chandra's class, with complexified semisimple Lie algebra $\mathfrak{g}$. Put $U = U(\mathfrak{g})$ and $Z = Z(U)$. We use the Harish–Chandra parameter convention: $\chi_\lambda : Z \to \mathbb{C}$ is the central character of the highest-weight module $L(\lambda - \rho)$, where $\rho$ is the half-sum of the positive roots. We first give the definitions at a regular integral character. The component needed at a regular nonintegral character is specified at the end of this subsection.

58.1.1. *Central-character projection and natural summands.* For a module $M$ on which $Z$ acts locally finitely, define

$$\mathrm{pr}_\lambda M = \{m \in M : (\ker \chi_\lambda)^N m = 0 \text{ for some } N \geq 1\}.$$

The exponent may depend on $m$. The decomposition into generalized central characters is

$$M = \bigoplus_\chi \mathrm{pr}_\chi M,$$

so each $\mathrm{pr}_\lambda$ is an exact projection functor. Let $\mathcal{M}_\lambda$ be the ambient category of modules with generalized central character $\chi_\lambda$, and let $\mathcal{HC}_\lambda$ be its Harish–Chandra subcategory.

For a finite-dimensional representation $E$, tensoring uses the diagonal $\mathfrak{g}$-action

$$x(e \otimes m) = xe \otimes m + e \otimes xm, \qquad x \in \mathfrak{g}.$$

Tensoring can change the infinitesimal character. To prescribe source and target, use

$$F_E^{\mu,\lambda} : \mathcal{M}_\lambda \longrightarrow \mathcal{M}_\mu, \qquad F_E^{\mu,\lambda}(M) = \mathrm{pr}_\mu(E \otimes M).$$

For the Harish–Chandra restriction, $E$ is understood to be compatible with the chosen $(\mathfrak{g}, K)$-category.

**Definition 58.1** (Projective functor)**.** A *projective functor* from $\lambda$ to $\mu$ is a natural direct summand of $F_E^{\mu,\lambda}$ for some finite-dimensional $E$. Thus there are a functor $F'$ and a natural isomorphism

$$F \oplus F' \cong F_E^{\mu,\lambda}.$$

The splitting must commute with every module homomorphism. Choosing a summand separately for each object does not define a projective functor. The definition is made on the ambient categories and then restricted to Harish–Chandra modules; see [MMM24, §2.2].

These functors are exact: tensoring over $\mathbb{C}$, projection to a generalized central character, and taking a natural direct summand all preserve exactness. They also have exact biadjoint projective functors, obtained from the dual tensor representation and the corresponding projections. The word projective does not mean that $F(M)$ is a projective module for every $M$. For example, the identity functor is projective, by taking $E = \mathbb{C}$. In category $\mathcal{O}$, projective functors preserve projective objects because they have exact right adjoints.

58.1.2. *Translation functors and the meaning of $\theta_w$.* For compatible dominant parameters $\lambda, \mu$ with integral difference, choose the usual finite-dimensional irreducible representation $E$ having extremal weight $\mu - \lambda$. Translation is the projective functor

$$T_\lambda^\mu(M) = \mathrm{pr}_\mu(E \otimes M).$$

If $\lambda$ is regular and $\mu$ lies on the $s$-wall of its closed chamber and on no other wall, translation through that wall is

$$\theta_s = T_\mu^\lambda \circ T_\lambda^\mu. \tag{58.1}$$

It translates onto the wall and then back to the regular character.

To specify $\theta_w$ precisely, choose $\lambda$ dominant, regular and integral, and write $W$ for the Weyl group. In the category-$\mathcal{O}$ block at $\chi_\lambda$, put

$$\Delta_w = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_{w\lambda - \rho}.$$

Let $L_w$ be its simple quotient and $P_w \twoheadrightarrow L_w$ its indecomposable projective cover. In particular, $\Delta_e = P_e$ is the dominant Verma module.

**Notation 58.2.** The functor $\theta_w$ is the unique, up to natural isomorphism, indecomposable projective endofunctor characterized by

$$\boxed{\theta_w(\Delta_e) \cong P_w.} \tag{58.2}$$

Here indecomposable means that the functor has no decomposition into two nonzero functors. Existence and uniqueness, and the fact that these are all the indecomposable projective endofunctors, are classification theorems; see [MMM24, §2.2]. Every projective endofunctor is a finite direct sum of the $\theta_w$.

Thus $\theta_e = \mathrm{Id}$, and $\theta_s$ for a simple reflection is the wall-crossing functor in (58.1). More generally, for a reduced expression $w = s_1 \cdots s_k$, with the above indexing convention, the composition

$$\Theta = \theta_{s_k} \circ \cdots \circ \theta_{s_1}$$

has a decomposition

$$\Theta \cong \theta_w \oplus \bigoplus_{x<w} \theta_x^{\oplus m_x}, \qquad m_x \in \mathbb{Z}_{\geq 0}, \tag{58.3}$$

where $<$ is Bruhat order. The distinguished summand $\theta_w$ occurs once and is independent of the reduced expression. For example, in type $A_2$ with simple reflections $s, t$,

$$\theta_s \circ \theta_t \circ \theta_s \cong \theta_{sts} \oplus \theta_s.$$

The whole composition of wall crossings need not be indecomposable.

Category $\mathcal{O}$ supplies the labels and the characterization of these ambient functors. The same functor restricts to Harish–Chandra modules, so $\theta_w X$ means applying it to $X$. The projective cover $P_w$ above belongs to category $\mathcal{O}$, not to the Harish–Chandra category.

58.1.3. *The two Grothendieck groups and positivity.* Let $\mathcal{P}_\lambda$ be the additive category of ambient projective endofunctors at $\chi_\lambda$, with natural transformations as morphisms. Composition makes it a monoidal category. Its real split Grothendieck algebra is

$$\mathcal{A} = \mathbb{R} \otimes_{\mathbb{Z}} K_0^{\mathrm{split}}(\mathcal{P}_\lambda). \tag{58.4}$$

The defining relations and multiplication are

$$[F \oplus H] = [F] + [H], \qquad [F][H] = [F \circ H].$$

Thus $b_w = [\theta_w]$, $w \in W$, is a basis. A decomposition of functors gives the multiplication constants:

$$\theta_x \circ \theta_y \cong \bigoplus_z \theta_z^{\oplus c_{xy}^z} \quad \Longrightarrow \quad b_x b_y = \sum_z c_{xy}^z b_z, \qquad c_{xy}^z \in \mathbb{Z}_{\geq 0}.$$

This is a *positive basis*: every multiplication structure constant is nonnegative. Kazhdan–Lusztig theory identifies the algebra with $\mathbb{R}[W]$, using the KL basis specialized at Hecke parameter $v = 1$. An inverse $w \mapsto w^{-1}$ can occur in this identification according to the convention for indexing functors. In either convention,

$$b_e \longleftrightarrow 1, \qquad b_s \longleftrightarrow 1 + s.$$

For instance,

$$(1+s)^2 = 2(1+s), \qquad \theta_s^2 \cong \theta_s \oplus \theta_s.$$

The wall-crossing functor corresponds to $1+s$, rather than to the group element $s$ alone.

There is a second, ordinary Grothendieck group,

$$M_\lambda = \mathbb{R} \otimes_{\mathbb{Z}} K_0(\mathcal{HC}_\lambda).$$

It imposes $[M] = [M'] + [M'']$ for every short exact sequence $0 \to M' \to M \to M'' \to 0$. Its basis consists of the classes $[X]$ of irreducible Harish–Chandra modules. Exactness defines an action by $[F][X] = [F(X)]$, and a composition series gives

$$b_w[X] = \sum_Y [\theta_w X : Y]\,[Y], \qquad [\theta_w X : Y] \in \mathbb{Z}_{\geq 0}. \tag{58.5}$$

Here $[\theta_w X : Y]$ is the Jordan–Hölder multiplicity of $Y$ in $\theta_w X$. The split group records direct sums of functors; the ordinary group records composition factors of modules.

Concretely, define the module preorder by

(58.6)

$$Y \leq X \quad \Longleftrightarrow \quad Y \text{ is a composition factor of } F(X) \text{ for some projective endofunctor } F.$$

It is reflexive because the identity is projective, and transitive because projective functors are exact and closed under composition. Mutual reachability defines the Harish–Chandra cells for this action. At a regular integral character this is the usual coherent-continuation cell relation. A composition factor may occur anywhere in a composition series; it need not be a submodule or quotient of the whole module $F(X)$.

58.1.4. *The component at a regular nonintegral character.* For general regular $\lambda$, the integral Weyl group is

$$W_\lambda = \langle s_\alpha : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \rangle.$$

It is not the stabilizer of $\lambda$. To use a basis indexed by $W_\lambda$, specify the component of projective functors. Let $\Xi = \mathbb{Z}\Delta(\mathfrak{g}, \mathfrak{h})$ be the root lattice and $P_{\mathrm{wt}}$ the weight lattice. Harish–Chandra bimodules are graded by $P_{\mathrm{wt}}/\Xi$ through their adjoint weights. The degree-zero, or $\Xi$-component, consists of bimodules whose adjoint weights lie in $\Xi$. Let $\mathcal{P}_\lambda^\Xi$ be the corresponding component of ambient projective endofunctors at $\chi_\lambda$.

This component contains the identity and is closed under direct sums, direct summands, composition and biadjunction. On the corresponding indecomposable block of category $\mathcal{O}$, Soergel's description identifies it with the projective-functor category for a regular integral block with Weyl group $W_\lambda$; see [MMM24, §§2.3–2.4]. Its indecomposable functors are therefore labeled $\theta_w$, $w \in W_\lambda$, and

$$\mathbb{R} \otimes_{\mathbb{Z}} K_0^{\mathrm{split}}(\mathcal{P}_\lambda^\Xi) \cong \mathbb{R}[W_\lambda]$$

with the same positive KL basis and the same action by composition multiplicities. At a regular integral character, this recovers the projective endofunctors described above.

For the rest of this section, put $\mathcal{P} = \mathcal{P}_\lambda^\Xi$ and use $\mathcal{A}$ for its real split Grothendieck algebra. At a nonintegral character, the projective endofunctors in the cell argument refer to this component. No claim is made that all ambient projective endofunctors at $\chi_\lambda$ are indexed by $W_\lambda$. This also specifies the translation action used in (58.6) and in the definition of the cells below.

58.2. **Statement and reduction to one double cell.** Fix a regular infinitesimal character $\lambda$, which need not be integral, and use the projective-functor component just specified. Write

$$C' \le C$$

when the Harish–Chandra cell $C'$ belongs to the downward closure of $C$: its simple modules occur as composition factors under the projective-functor action generated from $C$. Write $C' < C$ if also $C' \neq C$. Tensoring with finite-dimensional modules and taking subquotients cannot increase GK dimension, so

$$C' \le C \quad \Longrightarrow \quad d(C') \le d(C), \qquad d(C) := \operatorname{GKdim} C.$$

In particular, $d(C)$ is well defined.

**Theorem 58.3** (Strict GK-dimension drop)**.** *For Harish–Chandra cells at a regular infinitesimal character,*

$$\boxed{C' < C \quad \Longrightarrow \quad d(C') < d(C).}$$

**Definition 58.4** (The annihilator double cell)**.** Use the coherent-family notation of Section 50, and choose a regular dominant parameter $\nu \in \Lambda$. Regular evaluation identifies the basal coherent families in $C$ with its simple modules. For $\Psi \in C$, put

$$X = \Psi(\nu), \qquad I_X = \operatorname{Ann}_{U(\mathfrak{g})} X.$$

The primitive ideal $I_X$ determines a unique highest-weight left cell

$$\mathcal{L}_\Psi = \mathcal{L}_{\nu, I_X} \subseteq \operatorname{Coh}_\Lambda(\mathcal{K}(\mathfrak{g}, \mathfrak{b})).$$

Define $D(C)$ to be the highest-weight double cell containing $\mathcal{L}_\Psi$. It is independent of the choice of $\Psi \in C$ by [BMSZ22, Lem. 4.12]; this is the attached double cell of [BMSZ22, Def. 4.13]. Thus its construction is

$$\Psi \in C \ \longmapsto \ \operatorname{Ann} \Psi(\nu) \ \longmapsto \ \mathcal{L}_\Psi \ \longmapsto \ D(C).$$

The annihilator ideals and their left cells can vary within $C$, but the containing double cell is fixed.

Let $\sigma_{D(C)}$ be the special $W_\lambda$-representation attached to $D(C)$. Put $N = |\Delta^+(\mathfrak{g})|$. For an irreducible representation $\sigma$ of $W_\lambda$, write

$$b(\sigma) = \min\{j : \sigma \text{ occurs in } \operatorname{Sym}^j(\mathfrak{h})\}.$$

This is the least occurrence degree denoted by $a(\sigma)$ in [BMSZ22, §3.6]. The annihilator-variety formula gives

$$d(C) = \frac{1}{2} \dim \operatorname{AV}(\operatorname{Ann} X) = N - b(\sigma_{D(C)}), \qquad X \in C. \tag{58.7}$$

Here we use the half-dimension property for irreducible Harish–Chandra modules, together with the description of annihilator varieties by special representations; see [Vog78] and [BMSZ22, Prop. 3.26].

**Lemma 58.5.** *If $C' \le C$ and $d(C') = d(C)$, then $D(C') = D(C)$.*

*Proof.* By [BMSZ22, Lem. 4.8], the attached double-cell cones satisfy

$$\langle D(C') \rangle_{LR} \subseteq \langle D(C) \rangle_{LR}.$$

If the inclusion is strict, $\sigma_{D(C')}$ occurs in the latter cone and is different from $\sigma_{D(C)}$. Proposition 3.22(a),(b) of [BMSZ22] then gives

$$b(\sigma_{D(C')}) > b(\sigma_{D(C)}).$$

Equation (58.7) contradicts equality of the GK dimensions. The cones must therefore coincide, and the same proposition identifies their double cells. □

The lemma concerns highest-weight double cells. It does not by itself identify the Harish–Chandra cells $C'$ and $C$.

58.3. **The projective-functor apex.** Use $\mathcal{P}$, $\mathcal{A}$ and $b_w = [\theta_w]$ from Subsection 58.1, with the component specified in Subsubsection 58.1.4 when $\lambda$ is nonintegral. Equation (58.5) makes the ordinary Grothendieck group a positive based module; its cells are the Harish–Chandra cells defined by (58.6).

We distinguish the order on acting functors from the order on module cells. For indecomposable projective functors, use the Kildetoft–Mazorchuk convention

$$F \preceq G \quad \Longleftrightarrow \quad G \text{ is a direct summand of } H \circ F \circ K$$

for suitable projective functors $H, K$. Mutual comparability defines the two-sided cells of $\mathcal{P}$. Equivalently, $\mathcal{J}_1 \preceq \mathcal{J}_2$ if a basis element in $\mathcal{J}_2$ occurs with positive coefficient in $abc$, for some basis element $b$ in $\mathcal{J}_1$ and basis elements $a, c$. Thus multiplication moves upward for $\preceq$, whereas applying a functor moves downward for the module-cell order $\leq$.

Write the based cell quotient as

$$V_C = V_{\leq C}/V_{<C},$$

where the numerator is spanned by the cells $\leq C$ and the denominator by the cells $< C$. For an exact projective functor $F$, its induced action is

$$[F][X] = \sum_{Y \in C} [FX : Y]\,[Y], \qquad X \in C. \tag{58.8}$$

In particular, $F$ acts nontrivially on $V_C$ exactly when $[FX : Y] > 0$ for some $X, Y \in C$. The weaker condition $FX \neq 0$ alone does not suffice: all its constituents could lie in strictly lower cells.

**Definition 58.6** (The action apex)**.** The *apex* $\mathcal{J}(C)$ is the greatest, for $\preceq$, two-sided cell of projective functors acting nontrivially on $V_C$.

The term apex refers to this greatest acting cell. Its existence and uniqueness follow from positivity and transitivity, as the next lemma shows; they do not follow from finiteness alone. This definition uses the action on $V_C$, independently of the annihilators used to define $D(C)$.

**Lemma 58.7** (Existence and uniqueness of the apex)**.** *Let $A$ be a finite-dimensional unital algebra with a positive basis $B$ containing $1$, and let $V$ be a nonzero unital based $A$-module on which the elements of $B$ act by nonnegative matrices. Assume transitivity: for every pair of basis vectors $v_i, v_j$, some $b \in B$ has a positive coefficient of $v_j$ in $bv_i$. Then the two-sided cells acting nontrivially on $V$ have a greatest element.*

*Proof.* We show that any two acting cells have an acting common upper bound. Choose $a, b \in B$ acting nontrivially, and choose basis vectors such that $bv_i$ contains $v_j$ with positive

coefficient and $av_k \neq 0$. Transitivity supplies $x \in B$ such that $xv_j$ contains $v_k$ with positive coefficient. Positivity gives

$$axbv_i \neq 0.$$

Expand the product in the positive basis:

$$axb = \sum_{c \in B} \gamma_c c, \qquad \gamma_c \geq 0.$$

Some $c$ with $\gamma_c > 0$ acts nontrivially on $v_i$. Since $c$ occurs in $axb$, its cell is above both the cell of $a$ and the cell of $b$ in the two-sided order. This is the required common upper bound.

The set of acting cells is nonempty, since 1 acts as the identity, and finite, so it has a maximal element. Any two maximal elements coincide by the common-upper-bound property. The same property, applied to the unique maximal element and any acting cell, shows that this maximal element is greatest. □

This is the based-module apex result of [KM16, Prop. 14 and §9.2]. It applies to $V_C$: positivity comes from composition multiplicities, and transitivity comes from mutual reachability within a cell. A path between two elements of $C$ stays in $C$, since leaving and returning would put every intermediate element in the same cell. Thus passing to $V_C$ preserves the required reachability. No identification with annihilator cells is used here.

58.4. **Identifying the apex with the annihilator cell.** Let $\iota$ denote the standard correspondence from highest-weight double cells to two-sided cells of projective functors. For a simple $X \in C$, form the primitive ideal

$$I_X = \operatorname{Ann}_U X = \{u \in U : uX = 0\}.$$

In the regular integral notation above, Duflo's theorem gives a $w \in W$ such that $I_X = \operatorname{Ann}_U L_w = \operatorname{Ann}_U L(w\lambda - \rho)$. Explicitly, its attached projective-functor two-sided cell is

$$\mathcal{J}_X = \{\theta_v : v \sim_{LR} w\}. \tag{58.9}$$

The primitive-ideal/KL correspondence makes this independent of the choice of $w$. Equivalently, the same ideal that determines the highest-weight left cell $\mathcal{L}_\Psi$ determines a right cell $\mathcal{R}_X$ of projective functors, and $\mathcal{J}_X$ is the two-sided cell containing it. At a regular nonintegral character, use the corresponding $W_\lambda$-block and the component $\mathcal{P}_\lambda^\Xi$. Thus $\mathcal{J}_X$ is a cell of acting functors, whereas $C$ is a cell of simple Harish–Chandra modules. The cell correspondence gives

$$\mathcal{J}_X = \iota(D(C)). \tag{58.10}$$

See [MMM24, §§2.6, 3.2]. This identifies two constructions from the primitive ideal; identifying either of them with the action apex $\mathcal{J}(C)$ still requires a proof.

58.4.1. *Vanishing depends only on the annihilator.*

**Lemma 58.8.** *Let $X$ and $L$ be simple $\mathfrak{g}$-modules with the same annihilator, and let $F$ be a projective functor with domain their central-character block. Then*

$$\operatorname{Ann} X = \operatorname{Ann} L \quad \Longrightarrow \quad \big(FX = 0 \iff FL = 0\big). \tag{58.11}$$

*Proof.* Let $\chi$ be their central character and put

$$U_\chi = U(\mathfrak{g})/U(\mathfrak{g})\ker\chi.$$

On modules annihilated by $\ker\chi$, the Harish–Chandra bimodule representing $F$ is a direct summand $B_F$ of $E \otimes U_\chi$ for some finite-dimensional $\mathfrak{g}$-module $E$; see [MMM24, §2.3]. As a right $U_\chi$-module, $E\otimes U_\chi$ is finite free. Choose a basis of $E$ and write the bimodule projection onto $B_F$ as an idempotent matrix

$$p = (p_{ij}), \qquad p_{ij} \in U_\chi, \qquad p^2 = p.$$

For any left $U_\chi$-module $M$, tensoring this split projection with $M$ gives

$$FM \cong B_F \otimes_{U_\chi} M = \operatorname{im}\big(p_M : E \otimes M \longrightarrow E \otimes M\big).$$

Consequently,

$$FM = 0 \quad\Longleftrightarrow\quad p_M = 0 \quad\Longleftrightarrow\quad p_{ij} \in \operatorname{Ann}_{U_\chi} M \text{ for every } i, j.$$

Applying this criterion to $X$ and $L$ proves the claim. Only their scalar central character is used in this step; no assertion about a common power of $\ker\chi$ on general objects of the block is needed. □

58.4.2. *The upper bound from adjunction in category $\mathcal{O}$.*

**Lemma 58.9** (The annihilator upper bound)**.** *Let $X$ be simple at a regular infinitesimal character. For an indecomposable projective endofunctor $F$,*

$$FX \neq 0 \quad\Longrightarrow\quad \operatorname{cell}(F) \preceq \mathcal{J}_X. \tag{58.12}$$

*Proof.* First work in a regular integral block. Choose a simple highest-weight module $L$ with

$$\operatorname{Ann} L = \operatorname{Ann} X = I.$$

Such an $L$ exists by Duflo's theorem, recalled in [MMM24, §2.1]. Let $P_L$ be its projective cover in $\mathcal{O}$, and let $\Delta_{\mathrm{dom}}$ be the dominant Verma module. There is an indecomposable projective functor $G$ such that

$$G\Delta_{\mathrm{dom}} \cong P_L.$$

Under the annihilator–cell correspondence, $\operatorname{cell}(G) = \mathcal{J}_X$. The classification of projective functors by their values on the dominant Verma module is recalled in [MMM24, §2.2].

Suppose $FX \neq 0$. By Lemma 58.8, $FL \neq 0$. Since projectives generate the block of $\mathcal{O}$, some indecomposable projective $P$ satisfies

$$\operatorname{Hom}_{\mathcal{O}}(P, FL) \neq 0.$$

Write $F^*$ for the biadjoint of $F$. Adjunction gives

$$\operatorname{Hom}_{\mathcal{O}}(F^*P, L) \cong \operatorname{Hom}_{\mathcal{O}}(P, FL) \neq 0.$$

The module $F^*P$ is projective, since $F^*$ has the exact right adjoint $F$. Its decomposition into indecomposable projectives must therefore contain $P_L$. Here and below, $\mid$ denotes being a direct summand:

$$P_L \mid F^*P. \tag{58.13}$$

Write $P \cong H\Delta_{\mathrm{dom}}$ for an indecomposable projective functor $H$. Decompose $F^*H$ into indecomposable projective functors and evaluate on $\Delta_{\mathrm{dom}}$. Their values are the corresponding pairwise nonisomorphic indecomposable projectives, so (58.13) implies

$$G \mid F^*H.$$

By the definition of the two-sided preorder, this gives

$$\mathrm{cell}(F^*) \preceq \mathrm{cell}(G).$$

Finally, the triangle identities for adjunction make $F$ a retract of $FF^*F$ and $F^*$ a retract of $F^*FF^*$. Thus

$$F \mid FF^*F, \qquad F^* \mid F^*FF^*,$$

which gives both inequalities between their two-sided cells. Hence

$$\mathrm{cell}(F) = \mathrm{cell}(F^*) \preceq \mathrm{cell}(G) = \mathcal{J}_X.$$

At a regular nonintegral character, use the corresponding projective-functor blocks and their dominant Verma modules. The same classification and adjunction argument applies, with the integral Weyl group $W_\lambda$ as in [MMM24, §2.4]. This proves the upper bound without using any strict GK-dimension statement. □

This proves the annihilator criterion stated in [MMM24, §3.2] by reduction to category $\mathcal{O}$. In particular, its contrapositive says that functors outside the downward $\preceq$-closure of $\mathcal{J}_X$ annihilate $X$.

58.4.3. *Detection and identification of the apex.* Let $\theta_{d_X}$ be the Duflo functor in the right cell $\mathcal{R}_X$. The counit $\theta_{d_X} \to \mathrm{Id}$ evaluates nontrivially on $X$, by [MMM24, Rem. 12]. Since $X$ is simple, this gives a surjection

$$\theta_{d_X} X \twoheadrightarrow X, \qquad d_X \in \mathcal{J}_X. \tag{58.14}$$

The functors and their natural transformations are realized by Harish–Chandra bimodules, so the detection argument applies to Harish–Chandra modules as well.

**Proposition 58.10.** *The action apex and the annihilator double cell are related by*

$$\boxed{\mathcal{J}(C) = \iota(D(C)).}$$

*Consequently,*

$$D(C') = D(C) \quad \Longrightarrow \quad \mathcal{J}(C') = \mathcal{J}(C). \tag{58.15}$$

*Proof.* If $F$ acts nontrivially on $V_C$, then (58.8) gives $X, Y \in C$ with $[FX : Y] > 0$. In particular, $FX \neq 0$. Lemma 58.9 and (58.10) therefore give

$$\mathrm{cell}(F) \preceq \mathcal{J}_X = \iota(D(C)).$$

This holds for every acting functor and every relevant choice of $X \in C$.

For detection, fix $X \in C$. The surjection (58.14) gives

$$[\theta_{d_X} X : X] \geq 1.$$

Since $X$ belongs to $C$, this coefficient survives in $V_C$. Hence $\mathcal{J}_X = \iota(D(C))$ itself acts nontrivially on $V_C$. It is thus the greatest acting cell, which proves the asserted identification with $\mathcal{J}(C)$. In particular, equality of $D(C')$ and $D(C)$ gives equality of their action apices.

This uses the upper bound and detection, rather than following from the definitions alone. □

58.5. **Positive idempotents and two-cell subquotients.** Suppose $C' < C$ have a common apex $\mathcal{J}$ and no cell lies strictly between them. Let $V_{\leq C}$ be the based submodule spanned by the cells $\leq C$. The union of those cells other than $C'$ and $C$ is downward closed, by adjacency. Quotienting by its span gives a based module $M$ whose basis is exactly $C' \sqcup C$. With the $C'$-basis first, write

$$[b_w]_M = \begin{pmatrix} A_w & T_w \\ 0 & B_w \end{pmatrix}, \qquad A_w, B_w, T_w \geq 0. \tag{58.16}$$

The strict relation and adjacency imply that some $T_w$ is nonzero.

The annihilator criterion above shows that the based ideal

$$\mathcal{K}_{\mathcal{J}} = \operatorname{span}_{\mathbb{R}}\{b_w : \operatorname{cell}(w) \not\preceq \mathcal{J}\}$$

annihilates $M$: each such functor annihilates every simple module in its basis. Set

$$\mathcal{A}_{\mathcal{J}} = \mathcal{A}/\mathcal{K}_{\mathcal{J}}, \qquad I = \operatorname{span}_{\mathbb{R}}\{b_w : w \in \mathcal{J}\} \subseteq \mathcal{A}_{\mathcal{J}}.$$

We use the same symbols for the surviving basis elements. In this quotient, $I$ is a two-sided ideal. Since $\mathcal{J}$ is an idempotent two-sided cell, [KM16, Prop. 18(i)] supplies

$$e = \sum_{w \in \mathcal{J}} c_w b_w, \qquad c_w > 0, \qquad e^2 = e. \tag{58.17}$$

For applicability to an idempotent two-sided cell, use also [KM16, Prop. 17].

**Lemma 58.11.** *The matrix of $e$ on either cell quotient $V_{C'}$ or $V_C$ is entrywise strictly positive.*

*Proof.* Let $V$ be either transitive cell module, and let $v$ be a basis vector. The support of $Iv$ means the union of the supports of $b_w v$ for $w \in \mathcal{J}$. This support is nonempty. Indeed, if $Iv = 0$, the right ideal property gives $I\mathcal{A}_{\mathcal{J}} v = 0$. Positivity then forces $I$ to kill every basis vector reachable from $v$. Transitivity would imply $IV = 0$, contrary to the definition of the apex.

The left ideal property makes this nonempty support closed under reachability: if a basis vector occurs in $b_w v$ and another occurs after applying $b_u$ to it, positivity makes the latter occur in $b_u b_w v \in Iv$. Transitivity therefore makes the support equal to the whole cell basis. Since all coefficients $c_w$ in (58.17) are positive, every entry in the column $ev$ is positive. This holds for every basis vector $v$. □

**Lemma 58.12** (Vanishing of the off-diagonal action)**.** *In* (58.16), $T_w = 0$ *for every* $w$.

*Proof.* By Lemma 58.11,

$$[e]_M = \begin{pmatrix} P & R \\ 0 & Q \end{pmatrix}, \qquad P, Q > 0, \quad R \geq 0, \tag{58.18}$$

where all inequalities are entrywise. Idempotency gives

$$P^2 = P, \qquad Q^2 = Q, \qquad PR + RQ = R.$$

Multiplying the last identity on the left by $P$ yields

$$PR + PRQ = PR, \qquad \text{hence} \qquad PRQ = 0.$$

Strict positivity of $P, Q$ and nonnegativity of $R$ imply $R = 0$. This is the matrix mechanism in [MMM24, Lem. 2].

It remains to eliminate the off-diagonal blocks of all basis elements. By (58.17),

$$0 = R = \sum_{w \in \mathcal{J}} c_w T_w.$$

There is no cancellation among nonnegative matrices, so $T_w = 0$ for $w \in \mathcal{J}$. Thus every element of $I$ acts block diagonally.

For an arbitrary surviving basis element $b_w$, the element $eb_we$ belongs to $I$. Since $[e]_M$ is block diagonal, its off-diagonal block is

$$[eb_we]_{12} = PT_wQ.$$

This vanishes because $eb_we \in I$. Positivity again forces $T_w = 0$. Basis elements killed in $\mathcal{A}_{\mathcal{J}}$ already act by zero. □

Thus adjacent distinct cells cannot have a common apex. The calculation uses ordinary composition multiplicities in a based module; it does not require identifying these with the split Grothendieck basis used in the birepresentation formulation of [MMM24, Lem. 2].

58.6. **Completion of the strict-drop proof.**

*Proof of Theorem 58.3.* Suppose $C' < C$ but $d(C') = d(C)$. The cell poset of the fixed block is finite, so choose a saturated chain from $C'$ to $C$. Monotonicity forces every cell on the chain to have the same GK dimension. By Lemma 58.5 all have the same attached double cell, and by Proposition 58.10 all have the same apex.

Choose two adjacent cells on the chain. Their based subquotient $M$ has some nonzero off-diagonal block $T_w$, as in (58.16). This contradicts Lemma 58.12. Hence

$$C' \leq C, \quad d(C') = d(C) \quad \Longrightarrow \quad C' = C.$$

Combining this with monotonicity proves the strict inequality. □

In particular, this supplies Assumption 50.1 in the regular-character setting of Section 50.

58.7. **Coherent families and singular characters.** For coherent families in the BMSZ setting, GK dimension is defined by evaluation at a regular dominant parameter. The regular-character theorem therefore gives the strict-drop statement for those cells directly.

At a singular infinitesimal character, use the translation principle in a compatible integral block [Vog81, §7.3]. Let $T$ translate from a regular character onto the wall and let $U$ translate out. For a simple module $X$ at the singular character, choose its simple regular lift $\widetilde{X}$ with

$$T\widetilde{X} \cong X, \qquad \widetilde{X} \text{ a subquotient of } UX.$$

Since both translations are summands of tensoring with finite-dimensional modules, these two relations give

$$\operatorname{GKdim} X \leq \operatorname{GKdim} \widetilde{X} \leq \operatorname{GKdim} X.$$

Thus $\operatorname{GKdim} \widetilde{X} = \operatorname{GKdim} X$.

The preorder transfers in both directions. If $Y$ is a subquotient of $\theta X$ at the singular character, exactness implies

$$\widetilde{Y} \text{ is a subquotient of } UY, \text{ hence of } U\theta T\widetilde{X}.$$

Conversely, if $\widetilde{Y}$ is a subquotient of $\psi\widetilde{X}$ at the regular character, then

$$Y = T\widetilde{Y} \text{ is a subquotient of } T\psi\widetilde{X}, \text{ hence of } T\psi UX.$$

The composites $U\theta T$ and $T\psi U$ are projective functors on the respective blocks. Applying these observations to chains shows that comparability, and equality of cells, transfer between simple modules and their regular lifts. Distinct comparable singular cells consequently lift to distinct comparable regular cells with the same respective GK dimensions. The regular theorem proves strict drop at the singular character as well.

## References


[ABV92] J. Adams, D. Barbasch, and D. A. Vogan Jr. *The Langlands classification and irreducible characters for real reductive groups*. Vol. 104. Progress in Mathematics. Boston, MA: Birkhäuser Boston, 1992. xii+318 pp. DOI: 10.1007/978-1-4612-0383-4 (↑ 26).

[AC09] J. Adams and F. du Cloux. "Algorithms for representation theory of real reductive groups". *J. Inst. Math. Jussieu* 8.2 (2009), pp. 209–259. DOI: 10.1017/S1474748008000352 (↑ 4).

[Ach21] P. N. Achar. *Perverse sheaves and applications to representation theory*. Vol. 258. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society, 2021. xii+562 pp. DOI: 10.1090/surv/258 (↑ 44).

[BB81] A. Beĭlinson and J. Bernstein. "Localisation de $g$-modules". *C. R. Acad. Sci. Paris Sér. I Math.* 292.1 (1981), pp. 15–18 (↑ 75).

[BB82] W. Borho and J.-L. Brylinski. "Differential operators on homogeneous spaces. I. Irreducibility of the associated variety for annihilators of induced modules". *Invent. Math.* 69.3 (1982), pp. 437–476. DOI: 10.1007/BF01389364 (↑ 10).

[BB93] A. Beĭlinson and J. Bernstein. "A proof of Jantzen conjectures". In: *I. M. Gel'fand Seminar*. Vol. 16, Part 1. Adv. Soviet Math. Providence, RI: American Mathematical Society, 1993, pp. 1–50 (↑ 34).

[Bie90] F. V. Bien. *$\mathcal{D}$-modules and spherical representations*. Vol. 39. Mathematical Notes. Princeton, NJ: Princeton University Press, 1990. x+131 pp. DOI: 10.1515/9781400862078 (↑ 77).

[Bin08] B. Binegar. *HC-cells, nilpotent orbits, primitive ideals, and Weyl group representations*. Lecture notes from the conference Representation Theory of Lie Groups and Applications, Paris, 15–18 December 2008. Institut Henri Poincaré. 2008. URL: https://math.okstate.edu/people/binegar/Research/Paris-2008.pdf (↑ 77).

[BK04] R. Bezrukavnikov and D. Kaledin. "Fedosov quantization in algebraic context". *Mosc. Math. J.* 4.3 (2004), pp. 559–592, 782. DOI: 10.17323/1609-4514-2004-4-3-559-592 (↑ 55).

[BMSZ22] D. Barbasch, J.-J. Ma, B. Sun, and C.-B. Zhu. *Special unipotent representations of real classical groups: counting and reduction*. Version 4. Revised 2 January 2025. 2022. arXiv: 2205.05266v4 [math.RT] (↑ 7, 62, 64, 65, 90, 94, 95).

[BV80] D. Barbasch and D. A. Vogan Jr. "The local structure of characters". *J. Funct. Anal.* 37.1 (1980), pp. 27–55. DOI: 10.1016/0022-1236(80)90026-9 (↑ 14).

[BV82] D. Barbasch and D. A. Vogan Jr. "Primitive ideals and orbital integrals in complex classical groups". *Math. Ann.* 259.2 (1982), pp. 153–199. DOI: 10.1007/BF01457308 (↑ 31).

[BV83] D. Barbasch and D. A. Vogan Jr. "Primitive ideals and orbital integrals in complex exceptional groups". *J. Algebra* 80.2 (1983), pp. 350–382. DOI: 10.1016/0021-8693(83)90006-6 (↑ 64, 66).

[BV85] D. Barbasch and D. A. Vogan Jr. "Unipotent representations of complex semisimple groups". *Ann. of Math. (2)* 121.1 (1985), pp. 41–110. DOI: 10.2307/1971193 (↑ 77).

[CNT12] D. Ciubotaru, K. Nishiyama, and P. E. Trapa. "Regular orbits of symmetric subgroups on partial flag varieties". In: *Representation theory, complex analysis, and integral geometry*. New York: Birkhäuser/Springer, 2012, pp. 61–86. DOI: 10.1007/978-0-8176-4817-6_4 (↑ 77).

[GGS21] R. Gomez, D. Gourevitch, and S. Sahi. "Whittaker supports for representations of reductive groups". *Ann. Inst. Fourier (Grenoble)* 71.1 (2021), pp. 239–286. DOI: 10.5802/aif.3372 (↑ 13).

[GZ14] R. Gomez and C.-B. Zhu. "Local theta lifting of generalized Whittaker models associated to nilpotent orbits". *Geom. Funct. Anal.* 24.3 (2014), pp. 796–853. DOI: 10.1007/s00039-014-0276-5 (↑ 14).

[Har77] R. Hartshorne. *Algebraic geometry*. Vol. 52. Graduate Texts in Mathematics. New York and Heidelberg: Springer-Verlag, 1977. xvi+496 pp. (↑ 40).

[HTT08] R. Hotta, K. Takeuchi, and T. Tanisaki. *D-modules, perverse sheaves, and representation theory*. Vol. 236. Progress in Mathematics. Boston, MA: Birkhäuser Boston, 2008. xii+407 pp. DOI: 10.1007/978-0-8176-4523-6 (↑ 22).

[Jos80] A. Joseph. "Goldie rank in the enveloping algebra of a semisimple Lie algebra. II". *J. Algebra* 65.2 (1980), pp. 284–306. DOI: 10.1016/0021-8693(80)90218-5 (↑ 63).

[Jos85] A. Joseph. "On the associated variety of a primitive ideal". *J. Algebra* 93.2 (1985), pp. 509–523. DOI: 10.1016/0021-8693(85)90172-3 (↑ 10, 30).

[Kin81] D. R. King. "The character polynomial of the annihilator of an irreducible Harish-Chandra module". *Amer. J. Math.* 103.6 (1981), pp. 1195–1240. DOI: 10.2307/2374231 (↑ 63).

[KL80] D. Kazhdan and G. Lusztig. "A topological approach to Springer's representations". *Adv. Math.* 38.2 (1980), pp. 222–228. DOI: 10.1016/0001-8708(80)90005-5 (↑ 17).

[KM16] T. Kildetoft and V. Mazorchuk. "Special modules over positively based algebras". *Doc. Math.* 21 (2016), pp. 1171–1192. DOI: 10.4171/DM/555 (↑ 64, 65, 90, 96, 99).

[KT84] M. Kashiwara and T. Tanisaki. "The characteristic cycles of holonomic systems on a flag manifold related to the Weyl group algebra". *Invent. Math.* 77.1 (1984), pp. 185–198. DOI: 10.1007/BF01389142 (↑ 17, 18).

[KV95] A. W. Knapp and D. A. Vogan Jr. *Cohomological induction and unitary representations*. Vol. 45. Princeton Mathematical Series. Princeton, NJ: Princeton University Press, 1995. xx+948 pp. DOI: 10.1515/9781400883936 (↑ 11).

[LMM26] I. Losev, L. Mason-Brown, and D. Matvieievskyi. *Unipotent ideals and Harish-Chandra bimodules*. Version 4. Revised 22 March 2026. 2026. arXiv: 2108.03453v4 [math.RT] (↑ 87, 89).

[Los22a] I. Losev. "Deformations of symplectic singularities and orbit method for semisimple Lie algebras". *Selecta Math. (N.S.)* 28.2, Art. 30 (2022). DOI: 10.1007/s00029-021-00754-y (↑ 85).

[Los22b] I. Losev. *MATH720: Topics in representation theory – geometry and representation theory of singular symplectic varieties*. Lecture notes, spring semester. Yale University. 2022. URL: https://ivanloseu.github.io/MATH720_F22.html (↑ 36, 40, 42, 46, 54).

[LV83] G. Lusztig and D. A. Vogan Jr. "Singularities of closures of $K$-orbits on flag manifolds". *Invent. Math.* 71.2 (1983), pp. 365–379. DOI: 10.1007/BF01389103 (↑ 68, 80).

[LY21] N. C. Leung and S. Yu. "Equivariant deformation quantization and coadjoint orbit method". *Duke Math. J.* 170.8 (2021), pp. 1781–1850. DOI: 10.1215/00127094-2020-0066 (↑ 55).

[LY23] I. Losev and S. Yu. *On Harish-Chandra modules over quantizations of nilpotent orbits*. Version 2. Revised 4 September 2025. 2023. arXiv: 2309.11191v2 [math.RT] (↑ 54, 61).

[MMM24] M. Mackaay, V. Mazorchuk, and V. Miemietz. "Applying projective functors to arbitrary holonomic simple modules". *J. Lond. Math. Soc. (2)* 110.2, Art. e12965 (2024). DOI: 10.1112/jlms.12965 (↑ 91–93, 96–98, 100).

[MMY25] L. Mason-Brown, D. Matvieievskyi, and S. Yu. "Unipotent representations of complex groups and extended Sommers duality". *Proc. Lond. Math. Soc. (3)* 130.3, Art. e70035 (2025). DOI: 10.1112/plms.70035 (↑ 82, 88, 89).

[MW87] C. Mœglin and J.-L. Waldspurger. "Modèles de Whittaker dégénérés pour des groupes $p$-adiques". *Math. Z.* 196.3 (1987), pp. 427–452. DOI: 10.1007/BF01200363 (↑ 13).

[Nam22a] Y. Namikawa. "Birational geometry for the covering of a nilpotent orbit closure". *Selecta Math. (N.S.)* 28.4, Art. 75 (2022). DOI: 10.1007/s00029-022-00790-2 (↑ 83, 84, 88).

[Nam22b] Y. Namikawa. "Birational geometry for the covering of a nilpotent orbit closure II". *J. Algebra* 600 (2022), pp. 152–194. DOI: 10.1016/j.jalgebra.2022.01.036 (↑ 90).

[Nev99] M. Nevins. "Admissible nilpotent coadjoint orbits of $p$-adic reductive Lie groups". *Represent. Theory* 3 (1999), pp. 105–126. DOI: 10.1090/S1088-4165-99-00072-2 (↑ 21).

[Sch87] J. O. Schwartz. "The determination of the admissible nilpotent orbits in real classical groups". PhD thesis. Massachusetts Institute of Technology, 1987. URL: http://gateway.proquest.com/openurl?url_ver=Z39.88-2004&rft_val_fmt=info:ofi/fmt:kev:mtx:dissertation&res_dat=xri:pqdiss&rft_dat=xri:pqdiss:0376526 (↑ 21).

[SGA3] M. Demazure and A. Grothendieck. *Schémas en groupes (SGA 3)*. French. Ed. by P. Gille and P. Polo. Documents Mathématiques 7, 8, 28. Réédition de 1962/64. Société Mathématique de France, 2011. URL: https://webusers.imj-prg.fr/~patrick.polo/SGA3/ (↑ 11).

[Sun15] B. Sun. "Almost linear Nash groups". *Chinese Ann. Math. Ser. B* 36.3 (2015), pp. 355–400. DOI: 10.1007/s11401-015-0915-7 (↑ 13).

[Vog78] D. A. Vogan Jr. "Gelfand-Kirillov dimension for Harish-Chandra modules". *Invent. Math.* 48.1 (1978), pp. 75–98. DOI: 10.1007/BF01390063 (↑ 62, 94).

[Vog81] D. A. Vogan Jr. *Representations of real reductive Lie groups*. Vol. 15. Progress in Mathematics. Boston, MA: Birkhäuser, 1981. xvii+754 pp. (↑ 64, 70, 100).

[Vog82] D. A. Vogan Jr. "Irreducible characters of semisimple Lie groups. IV. Character-multiplicity duality". *Duke Math. J.* 49.4 (1982), pp. 943–1073. DOI: 10.1215/S0012-7094-82-04946-8 (↑ 63, 68).

[Vog83] D. A. Vogan Jr. "Irreducible characters of semisimple Lie groups. III. Proof of Kazhdan-Lusztig conjecture in the integral case". *Invent. Math.* 71.2 (1983), pp. 381–417. DOI: 10.1007/BF01389104 (↑ 72).

[Wil15] G. Williamson. *Illustrated guide to perverse sheaves*. Course given in Pisa, 19–30 January 2015. University of Sydney. 2015. URL: https://www.maths.usyd.edu.au/u/geordie/perverse_course/ (visited on 2026-04-16) (↑ 33).

[Yah17] K. Yahiro. "Radon transforms of twisted D-modules on partial flag varieties". *Publ. Res. Inst. Math. Sci.* 53.4 (2017), pp. 597–627. DOI: 10.4171/PRIMS/53-4-5 (↑ 77).